

- (vii) The sum of interior angle of an n -sided polygon is $(n-1) \times 180^\circ$
- (viii) The sum of the interior angles of any quadrilateral is always 180° .
- (ix) The set of integers is finite.

The following section will discuss various standard methods for combining statements to create new statements.

8.1.1 Logical Operators

The letters p, q etc., will use to donate the statements. A brief list of the symbols which will be used is given below:

Symbols	How to be read	Symbolic expression	How to be read
$\sim$	Not	$\sim p$	Not p , negation of p
$\wedge$	And	$p \wedge q$	p and q
$\vee$	Or	$p \vee q$	p or q
$\rightarrow$	If . . . then, implies	$p \rightarrow q$	If p then q , p implies q
$\leftrightarrow$	Is equivalent to, if and only if	$p \leftrightarrow q$	p if and only if q , p is equivalent to q

8.1.2 Explanation of the Use of the Symbols

1. Negation

If p is any statement, its negation is denoted by $\sim p$, read ‘not p ’. It follows from this definition that if p is true, $\sim p$ is false, and if p is false, $\sim p$ is true. The possible truth values of p and $\sim p$ are given in table:1, which is called a truth table, where the true value is denoted by T and the false value is denoted by F.

Table 1

p	$\sim p$
T	F
F	T

2. Conjunction

The conjunction of two statements p and q is symbolically written as $p \wedge q$ (p and q). A conjunction is considered to be true only if both statements are true. So, the truth table of $p \wedge q$ is given in Table: 2.

Table 2

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example 1: Whether the following statements are true or false.

- (i) Lahore is the capital of the Punjab and Quetta is the capital of Balochistan.
 (ii) $4 < 5 \wedge 8 < 10$ (iii) $2 + 2 = 3 \wedge 6 + 6 = 10$

Solution:

Clearly conjunctions (i) and (ii) are true whereas (iii) is false.

3. Disjunction

The disjunction of p and q is symbolically written as $p \vee q$ (p or q). The disjunction $p \vee q$ is considered to be true when at least one of the statements is true. It is false when both of them are false. The truth table $p \vee q$ is given in Table: 3.

Example 2: 10 is a positive integer or 0 is a rational number. Find truth value of this disjunction.

Solution: Since both statements are true, the disjunction is true.

Example 3: Triangle can have two right angles or Lahore is the capital of Sindh. Find the truth value of this disjunction.

Solution: Both statements are false, the disjunction is false.

4. Implication or conditional

A compound statement of the form if p then q ($p \rightarrow q$) also written as p **implies** q is called a **conditional** or an **implication**. p is called the **antecedent** or **hypothesis** and q is called the **consequent** or the **conclusion**.

A conditional is regarded as false only when the antecedent is true and the consequent is false. In all other cases conditional is considered to be true. So, the truth table of $p \rightarrow q$ is given in Table: 4.

Table 3

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Table 4

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

We attempt to clear the position with the help of an example. Consider the conditional:

If person A lives in Lahore, then he lives in Pakistan.

If the antecedent is false, i.e., A does not live in Lahore, he may still be living in Pakistan. We have no reason to say that he does not live in Pakistan.

We cannot, therefore, say that the conditional is false. So we must regard it as true. Similarly, when both the antecedent and consequent of the conditional under consideration are false, then is no justification for quarrelling with the statement.

5. Biconditional $p \leftrightarrow q$

The statement $p \rightarrow q \wedge q \rightarrow p$ is shortly written as $p \leftrightarrow q$ and is called the **biconditional** or **equivalence**. It is read p iff q (iff stands for “if and only if”)

We draw up its truth table.

From the Table 5 it appears that

$p \leftrightarrow q$ is true only when both statements p and q are true or both statements p and q are false.

Table 5				
p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

6. Conditionals related with a given conditional.

Let p and q be the statements and $p \rightarrow q$ be a given conditional, then

- (i) $q \rightarrow p$ is called the **converse** of $p \rightarrow q$;
- (ii) $\sim p \rightarrow \sim q$ is called the **inverse** of $p \rightarrow q$;
- (iii) $\sim q \rightarrow \sim p$ is called the **contrapositive** of $p \rightarrow q$.

The truth values of these new conditionals are given below in Table 6.

Table 6

				Given conditional	Converse	Inverse	Contrapositive
p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$q \rightarrow p$	$\sim p \rightarrow \sim q$	$\sim q \rightarrow \sim p$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

From the table 6, it appears that

- (i) Any conditional and its contrapositive are equivalent; therefore, any theorem may be proved by proving its contrapositive.
- (ii) The converse and inverse are equivalent to each other.

Example 4: Prove that in any universal set, the empty set ϕ is a subset of any set A .

Solution: Let U be the universal set. Consider the conditional:

$$\forall x \in U, x \in \phi \rightarrow x \in A \quad \dots(i)$$

The antecedent of this conditional is false because no $x \in U$, is a member of ϕ .

Hence, the conditional is true.

Example 5: Construct the truth table of $[(p \rightarrow q) \wedge p]$ and $[(p \rightarrow q) \wedge p] \rightarrow q$

Solution:

The desired truth Table 7 is given below:

Table 7

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$[(p \rightarrow q) \wedge p] \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

8.1.3 Mathematical Proof

Suppose Fayyaz is a student in Grade 9. One day, he arrived home late due to heavy traffic in a city. His father, however, suspected that Fayyaz had not gone to school and instead spent the day elsewhere. To address his concerns, his father asked, “Tell me the truth, did you go to school today? Fayyaz responded, saying, “Yes, I did.” Still doubtful, his father asked, “What proof do you have that you attended school? To satisfy his father's concern, Fayaz says that my classmate Ahmad went to school with me and could confirm with him. But his father was still not convinced by his words. Now, how will he prove his father's claim that he went to school or not? To prove his father's claim, Fayyaz would need to present some evidence, like his attendance for that day, which was recorded in the school attendance register, or CCTV footage from the school to prove that he was indeed present that day.

Consider another situation, you have bought a mobile phone with a warranty of about one year. After using the mobile phone for a few days, your mobile phone breaks down, so you take it to the mobile company or service provider. The customer support representative will ask you for proof if you want to claim your mobile phone's warranty. To claim the warranty on the mobile phone, you must present the warranty card as documented proof to the customer service representative. Generally, we have to prove and disprove many claims and statements in our daily routine. In mathematics, proofs provides the evidence that a statement is correct, demonstrating a logical sequence of steps that lead to the final conclusion.

Example 6: Prove the following mathematical statements.

- (a) If x is an odd integer, then x^2 is also an odd integer
- (b) The sum of two odd numbers is an even number

Solution:

- (a) Let x be an odd integer. Then by definition of an odd integer, we can express x as:

$$x = 2k + 1 \text{ for some } k \in \mathbb{Z}$$

Now
$$\begin{aligned} x^2 &= (2k + 1)^2 = 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2m + 1, \text{ where } m = 2k^2 + 2k \in \mathbb{Z} \end{aligned}$$

Thus, $x^2 = 2m + 1$ for some $m \in \mathbb{Z}$

Therefore, x^2 is an odd integer, by definition of an odd integer.

- (b) Let x and y be odd integers. Then by definition of an odd integer, we can express x and y as:

$$x = 2k + 1 \text{ and } y = 2n + 1 \text{ for some } k \text{ and } n \in \mathbb{Z}.$$

Thus,
$$\begin{aligned} x + y &= (2k + 1) + (2n + 1) \\ &= 2k + 2n + 1 + 1 \\ &= 2(k + n + 1) = 2m, \text{ where } k + n + 1 = m \in \mathbb{Z} \end{aligned}$$

So, $x + y = 2m$ for some $m \in \mathbb{Z}$.

Therefore, $x + y$ is an even integer, by definition of an even integer.

Note:

If x is odd, then x can be expressed in the form: $x = 2k + 1$ for some $k \in \mathbb{Z}$

Note:

If x is an even integer, then x can be expressed in the form: $x = 2k$ for some $k \in \mathbb{Z}$

Example 7: Prove that for any two non-empty sets A and B , $(A \cup B)' = A' \cap B'$.

Proof: Let $x \in (A \cup B)'$

$$\begin{aligned} &\Rightarrow x \notin (A \cup B) \\ &\Rightarrow x \notin A \text{ and } x \notin B \\ &\Rightarrow x \in A' \text{ and } x \in B' \\ &\Rightarrow x \in A' \cap B' \end{aligned}$$

But $x \in (A \cup B)'$ is an arbitrary element

$$\text{Therefore, } (A \cup B)' \subseteq A' \cap B' \quad \dots (i)$$

Now, suppose that $y \in A' \cap B'$

$$\begin{aligned} &\Rightarrow y \in A' \text{ and } y \in B' \\ &\Rightarrow y \notin A \text{ and } y \notin B \\ &\Rightarrow y \notin (A \cup B) \\ &\Rightarrow y \in (A \cup B)' \end{aligned}$$

$$\text{Thus } A' \cap B' \subseteq (A \cup B)' \quad \dots (ii)$$

From equations (i) and (ii) we conclude that

$$(A \cup B)' = A' \cap B', \text{ hence proved.}$$

Note:

A set B is a subset of a set A if every element of set B is also an element of a set A .

Mathematically, we write it as:

$$B \subseteq A \text{ if } \forall x \in B \Rightarrow x \in A$$

8.1.4 Theorem, Conjecture and Axiom

In previous sections, we have explored mathematical statements and their corresponding proofs. We will now move on to a more advanced concept known as theorems. A **theorem** is a mathematical statement that has been proved true based on previously known facts. For example, the following statements are theorem:

- (i) **Theorem:** The sum of the interior angles of a quadrilateral is 360 degrees.
- (ii) **The Fundamental Theorem of Arithmetic:** Every integer greater than 1 can be uniquely expressed as a product of prime numbers up to the order of the factors.
- (iii) **Fermat's Last Theorem:** There are no three positive integers a, b, c , which satisfy the equation $a^n + b^n = c^n$, where $n \in \mathbb{N}$ and $n > 2$

One of the famous theorems was named after the 17th-century French mathematician Pierre Fermat. Let's examine Fermat's Last Theorem for specific values of n and see how they apply. For $n = 2$, the statement simplifies to $a^2 + b^2 = c^2$ which does have solutions. This is the well-known Pythagorean theorem. For instance, $3^2 + 4^2 = 5^2$ holds true because $9 + 16 = 25$.

Now, let's examine the statement for $n = 3$. The statement becomes $a^3 + b^3 = c^3$.

After centuries of searching, no such integer solution has been found, and Wiles' proof confirmed that no such numbers exist. For example, $3^3 + 4^3 \neq 5^3$ because $91 \neq 125$.

Fermat claimed he could prove this theorem but noted that the margin of his book was too small for such a meaningful explanation. Despite his assertion, many mathematicians found it challenging to prove the theorem for centuries. The theorem remained unproven for over 350 years and became one of the most famous problems in mathematics. In 1993, Andrew Wiles from Princeton University announced a proof after working on it for over seven years, spanning hundreds of pages. This illustrates that some factual statements are not immediately evident.

Conjecture: A **conjecture** is a mathematical statement or hypothesis that is believed to be true based on observations but has not yet been proved. In mathematics, conjectures often serve as hypotheses, and if a conjecture is proven to be true, it becomes a theorem. Conversely, if evidence is found that disproves it, the conjecture is shown to be false. Here, is another well-known statement that has gained enough recognition to be named. First proposed in the 18th century by the German mathematician Christian Goldbach, it is known as the Goldbach Conjecture. The Goldbach Conjecture states that:

Statement: Every even integer greater than 2 is a sum of two prime numbers.

We must agree that the conjecture is either true or false. It appears to be true based on empirical evidence, as many even numbers greater than 2 can indeed be written as the sum of two prime numbers: for example, $4 = 2 + 2$, $6 = 3 + 3$, $12 = 5 + 7$, among others. However, this does not preclude the possibility that some large even number may exist that cannot be expressed as the sum of two primes. The conjecture would be proven false if such a number is found. Despite extensive efforts since Goldbach first posed the problem over 260 years ago, no proof has been found to determine whether the conjecture is true or false. Nevertheless, conjecture is a valid mathematical statement, as it must be either true or false.

In mathematics, we frequently encounter situations where it is necessary to determine the truth of a given statement without proving it. Next, we will study the same statement, which is known as axiom.

An **axiom** is a mathematical statement that we believe to be true without any evidence or requiring any proof. In other words, these statements are basic facts that form the starting point for further ideas and are based on everyday experiences. Moreover, there is no evidence contradicting these statements. For example, the following are the statements of axioms.

Axiom: Through a given point, infinitely many lines can pass.

Euclid Axioms: A straight line can be drawn between any two points.

Peano Axioms: Every natural number has a successor, which is also a natural number.

Axiom of Extensionality: Two sets are equal if they have the same elements.

Axiom of Power Set: Any set has a set of all its subsets.

Considering the above example, we will find that there is no need to prove these statements. For example, our intuition recognizes that infinitely many lines can pass through a point, so there is no need to prove it.

Axioms are sometimes referred to as postulates. Both Axioms and Postulates describe statements that are accepted as true without requiring proof. However, postulates are associated explicitly with geometry, while axioms can pertain to broader mathematical contexts.

Next, we are going to prove the statement of a theorem.

Example 8: Prove that $\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$ where a, b, c and d are non-zero real numbers.

Solution: L.H.S = $\frac{a}{b} + \frac{c}{d} = \frac{a}{b} \times 1 + \frac{c}{d} \times 1$ ($\because$ Multiplicative identity)

$$= \frac{a}{b} \times \left(d \times \frac{1}{d} \right) + \frac{c}{d} \times \left(b \times \frac{1}{b} \right) \quad (\because \text{Multiplicative inverse})$$

$$= \frac{a}{b} \times \frac{d}{d} + \frac{c}{d} \times \frac{b}{b} \quad \left(\because a \times \frac{1}{b} = \frac{a}{b} \right)$$

$$= \frac{ad}{bd} + \frac{cb}{db} \quad \left(\because \text{Rule of production of fraction } \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \right)$$

$$\begin{aligned}
 &= \frac{ad}{bd} + \frac{bc}{bd} & (\because \text{Commutative law of multiplication } ab = ba) \\
 &= ad \times \frac{1}{bd} + bc \times \frac{1}{bd} & \left(\because a \times \frac{1}{b} = \frac{a}{b} \right) \\
 &= (ad + bc) \cdot \frac{1}{bd} & (\because \text{Distributive property}) \\
 &= \frac{(ad + bc)}{bd} = \text{R.H.S}
 \end{aligned}$$

$$\Rightarrow \text{L.H.S} = \text{R.H.S}$$

$$\text{Thus, } \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

Hence proved.

8.1.5 Deductive Proof

As discussed earlier, deductive reasoning is a way of drawing conclusions from premises believed to be true. If the premises are true, then the conclusion must also be true. For example: All human beings need to breathe to live. Ahmad is a human. Therefore, Ahmad is also breathing to live.

Similarly, in mathematics, deductive proof in an algebraic expression is a technique to show the validity of a mathematical statement through logical reasoning based on known rules, theorems, axioms, or previously proven statements. Deductive reasoning is broadly used in algebra to validate identities and solve equations.

Example 9: Prove that: $(x + 1)^2 + 7 = x^2 + 2x + 8$

Solution: **Proof:** L.H.S = $(x + 1)^2 + 7$

$$\begin{aligned}
 &= (x + 1)(x + 1) + 7 & (\because x^m \cdot x^n = x^{m+n}) \\
 &= x \cdot (x + 1) + 1 \cdot (x + 1) + 7 & (\because \text{Right distributive law}) \\
 &= x \cdot x + x \cdot 1 + 1 \cdot x + 1 \cdot 1 + 7 & (\because \text{Right distributive law}) \\
 &= x^2 + 1 \cdot x + 1 \cdot x + 1 + 7 & (\because \text{Commutative law \& } x^m \cdot x^n = x^{m+n}) \\
 &= x^2 + (1 + 1)x + 8 & (\because \text{Left distributive law}) \\
 &= x^2 + 2x + 8 = \text{R.H.S}
 \end{aligned}$$

$$\Rightarrow \text{L.H.S} = \text{R.H.S}$$

Thus, $(x + 1)^2 + 7 = x^2 + 2x + 8$. Hence proved

Example 10: Prove that $\frac{45x + 15}{15} = 3x + 1$ by justifying each step.

Solution: **Proof:** $L.H.S = \frac{45x+15}{15}$

$$= \frac{1}{15} \times (45x+15) \quad \left(\because \frac{a}{b} = \frac{1}{b} \times a \right)$$

$$= \frac{1}{15} \times (15 \times 3x + 15 \times 1) \quad (\because \text{Multiplicative Identity})$$

$$= \frac{1}{15} \times 15(3x + 1) \quad (\because \text{Distributive Law})$$

$$= \left(\frac{1}{15} \times 15 \right) \cdot (3x+1) \quad (\because \text{Associative Law})$$

$$= 1 \cdot (3x + 1) \quad (\because \text{Multiplicative Inverse})$$

$$= 3x + 1 = R.H.S \quad (\because \text{Multiplicative Identity})$$

$$\Rightarrow L.H.S = R.H.S$$

Thus, $\frac{45x+15}{15} = 3x + 1$ hence proved.

EXERCISE 8

1. Four options are given against each statement. Encircle the correct option.

- (i) Which of the following expressions is often related to inductive reasoning?
 - (a) based on repeated experiments
 - (b) if and only if statements
 - (c) Statement is proven by a theorem
 - (d) based on general principles
- (ii) Which of the following sentences describe deductive reasoning?
 - (a) general conclusions from a limited number of observations
 - (b) based on repeated experiments
 - (c) based on units of information that are accurate
 - (d) draw conclusion from well-known facts
- (iii) Which one of the following statements is true?
 - (a) The set of integers is finite
 - (b) The sum of the interior angles of any quadrilateral is always 180°
 - (c) $\frac{22}{7} \notin \mathbb{Q}'$
 - (d) All isosceles triangles are equilateral triangles
- (iv) Which of the following statements is the best to represent the negation of the statement "The stove is burning"?
 - (a) the stove is not burning.

- (b) the stove is dim
 - (c) the stove is turned to low heat
 - (d) it is both burning and not burning.
 - (v) The conjunction of two statements p and q is true when:
 - (a) both p and q are false.
 - (b) both p and q are true.
 - (c) only q is true.
 - (d) only p is true
 - (vi) A conditional is regarded as false only when:
 - (a) antecedent is true and consequent is false.
 - (b) consequent is true and antecedent is false.
 - (c) antecedent is true only.
 - (d) consequent is false only.
 - (vii) Contrapositive of $q \rightarrow p$ is
 - (a) $q \rightarrow \sim p$
 - (b) $\sim q \rightarrow p$
 - (c) $\sim p \rightarrow \sim q$
 - (d) $\sim q \rightarrow \sim p$
 - (viii) The statement "Every integer greater than 2 is a sum of two prime numbers" is:
 - (a) theorem
 - (b) conjecture
 - (c) axiom
 - (d) postulates
 - (ix) The statement "A straight line can be drawn between any two points" is :
 - (a) theorem
 - (b) conjecture
 - (c) axiom
 - (d) logic
 - (x) The statement "The sum of the interior angle of a triangle is 180° " is:
 - (a) converse
 - (b) theorem
 - (c) axiom
 - (d) conditional
2. Write the converse, inverse and contrapositive of the following conditionals:
 - (i) $\sim p \rightarrow q$
 - (ii) $q \rightarrow p$
 - (iii) $\sim p \rightarrow \sim q$
 - (iv) $\sim q \rightarrow \sim p$
 3. Write the truth table of the following
 - (i) $\sim(p \vee q) \vee (\sim q)$
 - (ii) $\sim(\sim q \vee \sim p)$
 - (iii) $(p \vee q) \leftrightarrow (p \wedge q)$
 4. Differentiate between a mathematical statement and its proof. Give two examples.
 5. What is the difference between an axiom and a theorem? Give examples of each.
 6. What is the importance of logical reasoning in mathematical proofs? Give an example to illustrate your point.
 7. Indicate whether it is an axiom, conjecture or theorem and explain your reasoning.
 - (i) There is exactly one straight line through any two points.
 - (ii) Every even number greater than 2 can be written as the sum of two prime numbers.

- (iii) The sum of the angles in a triangle is 180° .
8. Formulate simple deductive proofs for each of the following algebraic expressions, prove that the L.H.S is equal to the R.H.S:
- (i) prove that $(x - 4)^2 + 9 = x^2 - 8x + 25$
- (ii) prove that $(x + 1)^2 - (x - 1)^2 = 4x$
- (iii) prove that $(x + 5)^2 - (x - 5)^2 = 20x$
9. Prove the following by justifying each step:
- (i) $\frac{4+16x}{4} = 1+4x$ (ii) $\frac{6x^2+18x}{3x^2-27} = \frac{2x}{x-3}$
- (iii) $\frac{x^2+7x+10}{x^2-3x-10} = \frac{x+5}{x-5}$
10. Suppose x is an integer. If x is odd, then $9x + 4$ is odd.
11. Suppose x is an integer. If x is odd, then $7x + 5$ is even.
12. Prove the following statements
- (i) If x is an odd integer, then show that $x^2 - 4x + 6$ is odd.
- (ii) If x is an even integer then show that $x^2 + 2x + 4$ is even.
13. Prove that for any two non-empty sets A and B , $(A \cap B)' = A' \cup B'$.
14. If x and y are positive real numbers and $x^2 < y^2$ then $x < y$.
15. Prove that the sum of the interior angles of a triangle is 180° .
16. If a , b and c are non-zero real numbers, prove that:
- (i) $\frac{a}{b} = \frac{c}{d} \Leftrightarrow ad = bc$ (ii) $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$ (iii) $\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}$

Unit 9

Similar Figures

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Identify similarity of polygons. Area and volume of similar figures.
- Solve problems using the relationship between areas of similar figures and volume of similar solids.
- Solve real life problems that involve the properties of regular polygons, triangles and parallelograms (such as building architectural structures, fencing, tiling, painting and carpeting a room).

INTRODUCTION

The concept of similarity dates back to ancient Greece, where Greek mathematicians, particularly Euclid, developed the fundamental principles of geometry. In his creative work, "The Elements", Euclid established the foundations of plane geometry, including the theory of similar triangles and polygons. Euclid's further work laid the groundwork for modern geometry and the concept of similarity remains central in many branches of mathematics, including trigonometry and algebra.

9.1 Similarity of Polygons

Similar figures have same shape but not necessarily of same size. Two polygons are similar if their corresponding angles are equal and the corresponding

Remember!

Three or more than three-sided closed figure is called polygon.

sides are proportional (i.e., the ratios of the lengths of corresponding sides are equal). This means that if two polygons are similar, one is a scaled version of the other. For example, all equilateral triangles are similar to each other because they have the same angles and the measure of the sides are proportional.

9.1.1 Identification of Similar Triangles

- (i) If two angles in one triangle are congruent to two corresponding angles in another triangle, the third angle in each triangle must be congruent. Since the angles are the same, the triangles are similar. Similarity symbol is ' $\sim$ '.

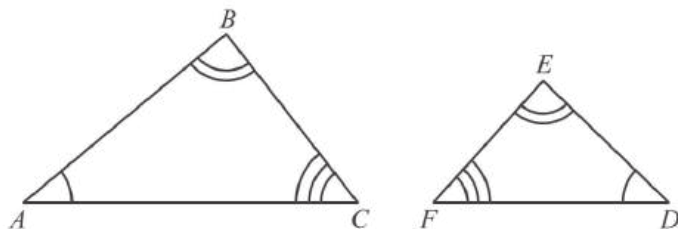
i.e., In the correspondence of the triangles ABC and DEF .

$$m\angle A = m\angle D$$

$$m\angle B = m\angle E$$

$$m\angle C = m\angle F$$

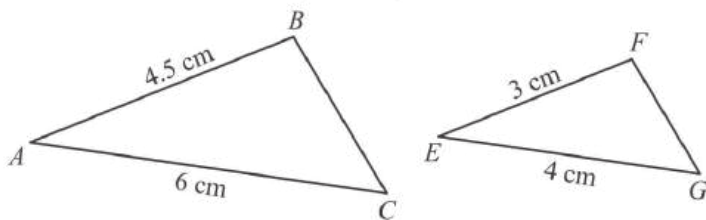
Hence, $\triangle ABC \sim \triangle DEF$



- (ii) If the ratio of two corresponding sides and their included angle are equal, then the triangles are similar. In the correspondence of the triangles ABC and EFG , $m\angle ABC = m\angle EFG$ and the ratio of the corresponding sides are

$$\frac{m\overline{AB}}{m\overline{EF}} = \frac{4.5}{3} = \frac{3}{2}$$

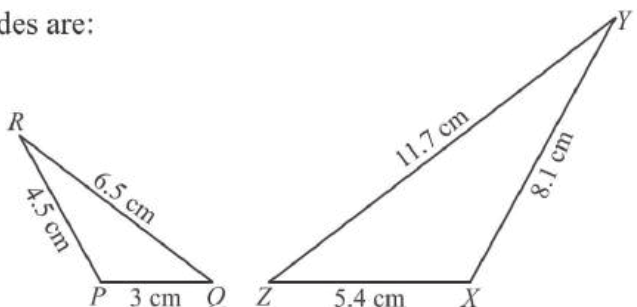
and $\frac{m\overline{AC}}{m\overline{EG}} = \frac{6}{4} = \frac{3}{2}$ Hence



triangles ABC and EFG are similar.

- (iii) If the ratio of all the corresponding sides are equal, then the triangles are similar. In the corresponding of $\triangle PQR$ and $\triangle XZY$, the ratio of corresponding sides are:

$$\begin{aligned} \frac{m\overline{PQ}}{m\overline{XZ}} &= \frac{m\overline{QR}}{m\overline{YZ}} = \frac{m\overline{PR}}{m\overline{XY}} \\ \frac{3}{5.4} &= \frac{6.5}{11.7} = \frac{4.5}{8.1} \\ \frac{5}{9} &= \frac{5}{9} = \frac{5}{9} \end{aligned}$$



Hence, the $\triangle PQR$ and $\triangle XZY$ are similar.

Example 1: If one pair of corresponding sides are parallel to each other, then the triangles so formed as shown in the figure are similar. i.e.,

In the figure, $\overline{AB}$ is parallel to $\overline{CD}$ and

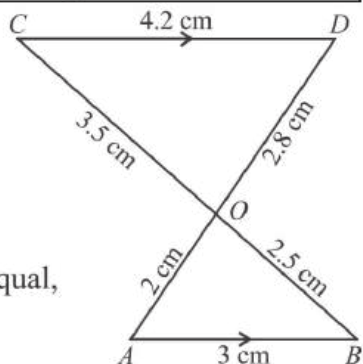
$$m\angle AOB = m\angle DOC \quad (\text{Vertically opposite angles})$$

$$m\angle A = m\angle D \quad (\text{Alternate angles of parallel lines})$$

$$m\angle B = m\angle C \quad (\text{Alternate angles of parallel lines})$$

Solution: Since all three corresponding angles are equal, so $\triangle OAB \sim \triangle ODC$

Need to Know! Proportionality of sides means one side is k times of its corresponding side.



The ratio of corresponding sides are equal i.e.,

$$\frac{m\overline{OA}}{m\overline{OD}} = \frac{m\overline{AB}}{m\overline{DC}} = \frac{m\overline{OB}}{m\overline{OC}}$$

$$\frac{2}{2.8} = \frac{3}{4.2} = \frac{2.5}{3.5}$$

$$\frac{5}{7} = \frac{5}{7} = \frac{5}{7}$$

So, the triangles OAB and ODC are similar.

Example 2:

In the triangles XBC and XDE , find the value of x and y .

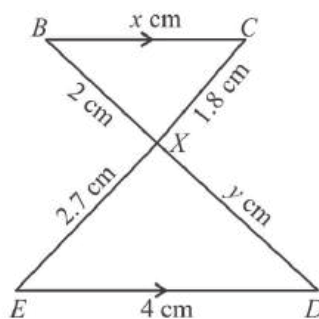
Solution: Since $\overline{BC}$ is parallel to $\overline{ED}$, so the triangles XBC and XDE are similar, so, the ratio of the corresponding sides are:

$$\frac{m\overline{XB}}{m\overline{XD}} = \frac{m\overline{BC}}{m\overline{DE}} = \frac{m\overline{XC}}{m\overline{XE}}$$

$$\frac{2}{y} = \frac{x}{4} = \frac{1.8}{2.7}$$

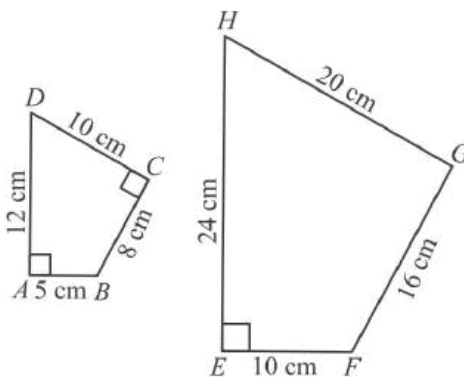
$$\frac{x}{4} = \frac{1.8}{2.7} \Rightarrow x = \frac{1.8}{2.7} \times 4 = 2.67 \text{ cm}$$

$$\frac{2}{y} = \frac{1.8}{2.7} \Rightarrow y = \frac{2.7}{1.8} \times 2 = 3 \text{ cm}$$



9.1.2 Similarity of Quadrilaterals

Example 3: The Quadrilateral $ABCD$ has side lengths $m\overline{AB} = 5\text{cm}$, $m\overline{BC} = 8$, $m\overline{CD} = 10\text{ cm}$, $m\overline{AD} = 12\text{cm}$, and its angles are $m\angle A = 90^\circ$, $m\angle B = 120^\circ$ and $m\angle C = 90^\circ$. Quadrilateral $EFGH$ has side lengths $m\overline{EF} = 10\text{ cm}$, $m\overline{FG} = 16\text{ cm}$, $m\overline{GH} = 20\text{cm}$, $m\overline{EH} = 24\text{ cm}$ and its angles are $m\angle E = 90^\circ$, $m\angle F = 120^\circ$ and $m\angle H = 60^\circ$. Prove that the quadrilateral $ABCD$ is similar to the quadrilateral $EFGH$. (Diagrams are not drawn to scale).



Solution: We see that in the quadrilateral $ABCD$:

$$m\angle D = 360^\circ - (90^\circ + 120^\circ + 90^\circ) = 60^\circ.$$

In the quadrilateral $EFGH$, $m\angle G = 360^\circ - (90^\circ + 120^\circ + 60^\circ) = 90^\circ$.

Now, check if the corresponding angles of the quadrilaterals are congruent:

$m\angle A = m\angle E = 90^\circ$, $m\angle B = m\angle F = 120^\circ$, $m\angle C = m\angle G = 90^\circ$ and $m\angle D = m\angle H = 60^\circ$.

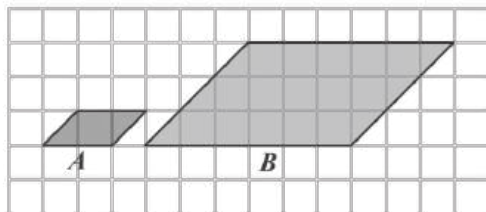
Next, check the ratios of the corresponding sides:

Ratio of $\overline{AB}$ to $\overline{EF}$: $\frac{m\overline{AB}}{m\overline{EF}} = \frac{5}{10} = \frac{1}{2}$, Ratio of $\overline{BC}$ to $\overline{FG}$: $\frac{m\overline{BC}}{m\overline{FG}} = \frac{8}{16} = \frac{1}{2}$

Ratio of $\overline{CD}$ to $\overline{GH}$: $\frac{m\overline{CD}}{m\overline{GH}} = \frac{10}{20} = \frac{1}{2}$ Ratio of $\overline{AD}$ to $\overline{EH}$: $\frac{m\overline{AD}}{m\overline{EH}} = \frac{12}{24} = \frac{1}{2}$

Since the corresponding angles are congruent and the corresponding sides are proportional (with a ratio of $\frac{1}{2}$), so the quadrilateral $ABCD$ is similar to the quadrilateral $EFGH$.

Example 4: Find whether the parallelograms are similar given that one of the angle between sides is 45° in both the parallelograms.



Solution:

Since opposite angles in a parallelogram are equal and adjacent angles are supplementary, so the corresponding angles (45° , 135° , 45° , and 135°) in both parallelograms are equal. So, the parallelograms are similar.

Measure of the base of smaller parallelogram, $b_1 = 2$ units

Measure of the base of larger parallelogram, $b_2 = 6$ units.

Measure of the height of smaller parallelogram, $h_1 = 1$ unit

Measure of the height of larger parallelogram, $h_2 = 3$ units.

Ratio of corresponding lengths are equal. i.e., $\frac{b_1}{b_2} = \frac{2}{6} = \frac{1}{3}$ and $\frac{h_1}{h_2} = \frac{1}{3}$

Therefore, $\frac{b_1}{b_2} = \frac{h_1}{h_2}$

Example 5 The perimeter of a regular octagon is 48 cm. Another octagon has sides that are 1.2 times the sides of the first octagon. What is the length of side of the second octagon?

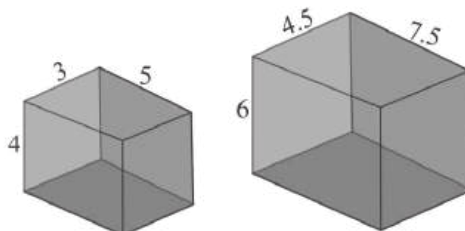
Solution: Perimeter of first regular octagon = 48 cm

$$\text{Side length of first regular octagon} = \frac{48}{8} = 6 \text{ cm.}$$

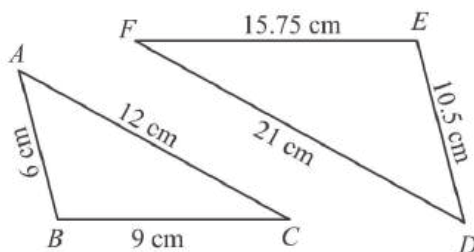
$$\text{Side length of second regular octagon} = 6 \times 1.2 = 7.2 \text{ cm.}$$

EXERCISE 9.1

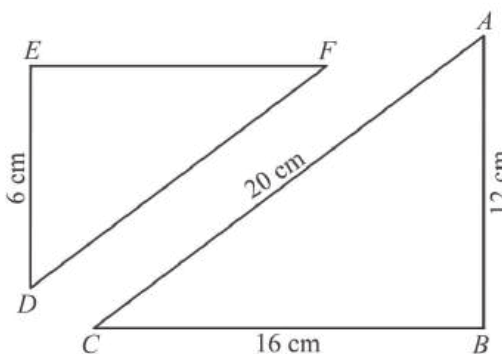
1. Find whether the solids are similar. All lengths are in cm.



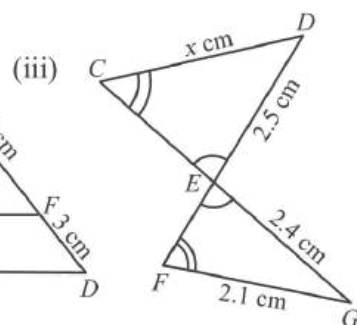
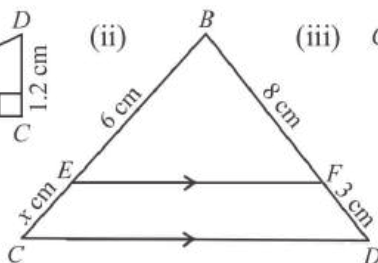
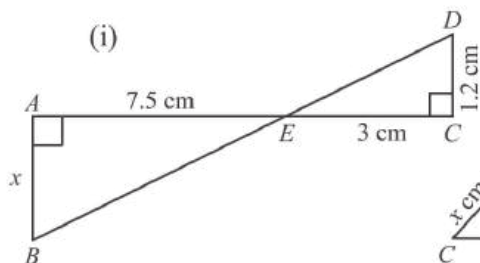
2. In triangle ABC , the sides are given as $\overline{AB} = 6$ cm, $\overline{BC} = 9$ cm and $\overline{CA} = 12$ cm. In triangle DEF , the sides are given as $\overline{DE} = 10.5$ cm, $\overline{EF} = 15.75$ cm, and $\overline{FD} = 21$ cm. Prove that the triangles are similar.



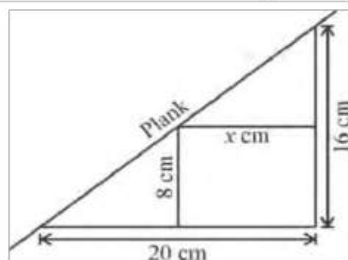
3. In the given figure, $\triangle ABC \sim \triangle DEF$, $\overline{AB} = 12$ cm, $\overline{AC} = 20$ cm and $\overline{BC} = 16$ cm. In $\triangle DEF$, $\overline{DE} = 6$ cm. Find $\overline{DF}$ and $\overline{EF}$.



4. Find the value of x in each of the following:

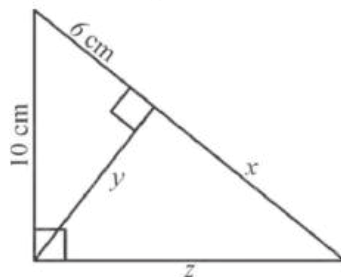


5. A plank is placed straight upstairs that 20 cm wide and 16 cm deep. A rectangular box of height 8 cm and width x cm is placed on a stair under the plank. Find the value of x .



6. A man who is 1.8 m tall casts a shadow of a 0.76 m in length. If at the same time a telephone pole casts a 3 m shadow, find the height of the pole.

7. Find the values of x , y and z in the given figure.

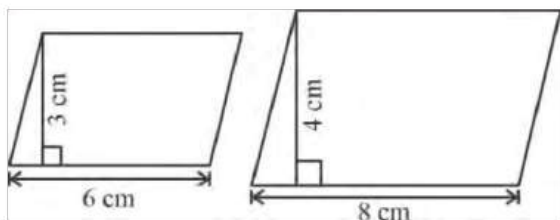


8. Draw an isosceles trapezoid $ABCD$ where $\overline{AB} \parallel \overline{CD}$ and $m\overline{AB} > m\overline{CD}$. Draw diagonals $\overline{AC}$ and $\overline{BD}$, intersecting at E . Prove that $\triangle ABE$ is similar to $\triangle CDE$. If $m\overline{AB} = 8$ cm, $m\overline{CD} = 4$ cm, and $m\overline{AE} = 3$ cm, find the length of $\overline{CE}$.
9. A regular dodecagon has its side lengths decreased by a factor of $\frac{1}{\sqrt{2}}$. If the perimeter of the original dodecagon is 72 cm. What is the side length of scaled dodecagon?

9.2 Area of Similar Figures

There are two parallelograms with corresponding bases 6 cm and 8 cm and corresponding altitudes 3 cm and 4 cm respectively. The ratio between their lengths is 3 : 4 written as:

$$\frac{\ell_1}{\ell_2} = \frac{3}{4}$$



The area of smaller parallelogram is: $A_1 = \text{base} \times \text{altitude}$
 $= 6 \times 3 = 18 \text{ cm}^2$

The area of larger parallelogram is: $A_2 = \text{base} \times \text{altitude}$
 $= 8 \times 4 = 32 \text{ cm}^2$

The ratio of their areas is:

$$\frac{A_1}{A_2} = \frac{9}{16} = \left(\frac{3}{4}\right)^2$$

$$\frac{A_1}{A_2} = \left(\frac{\ell_1}{\ell_2}\right)^2$$

Where A_1 and A_2 are areas and ℓ_1 and ℓ_2 are any two corresponding lengths of similar figures.

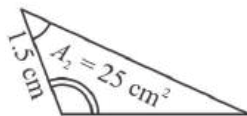
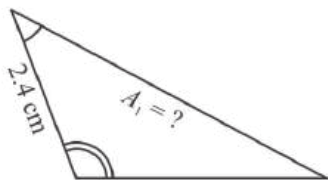
Hence the ratio of the areas of any two similar figures is equal to the square of the ratio of any two corresponding lengths of the figures.

$$\frac{A_1}{A_2} = \left(\frac{\ell_1}{\ell_2}\right)^2$$

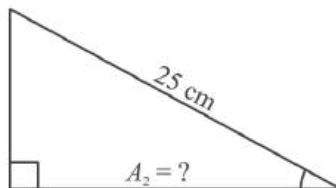
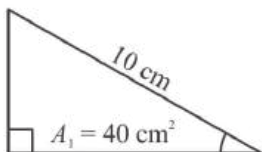
Since each length is k times of the other, we take $\frac{\ell_1}{\ell_2} = k$, then $\frac{A_1}{A_2} = k^2$. i.e. Area A_1 is k^2 times the area A_2 . k is called scale factor.

Example 6: Find the unknown value in the following:

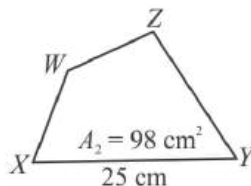
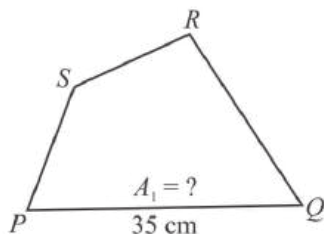
(i)



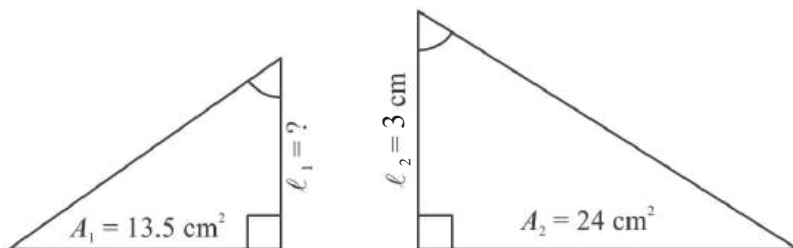
(ii)



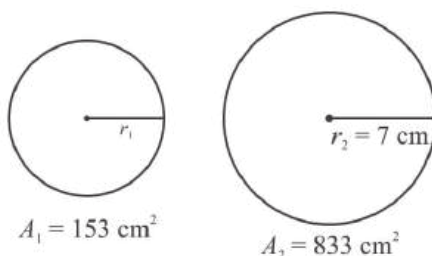
(iii) The quadrilaterals $PQRS$ and $XYZW$ are similar where $m\overline{PQ} = 35 \text{ cm}$ and $m\overline{XY} = 25 \text{ cm}$.



(iv)



(v)



Solution: (i) Since two pairs of corresponding angles are equal i.e., triangles are similar. We use the formula for ratio of areas of similar figures.

$$\frac{A_1}{A_2} = \left(\frac{\ell_1}{\ell_2} \right)^2$$

Here $\ell_1 = 2.4 \text{ cm}$, $\ell_2 = 1.5 \text{ cm}$, $A_2 = 25 \text{ cm}^2$, $A_1 = ?$

$$\frac{A_1}{25} = \left(\frac{2.4}{1.5} \right)^2$$

$$\frac{A_1}{25} = \left(\frac{8}{5} \right)^2$$

$$A_1 = \frac{64}{25} \times 25 = 64 \text{ cm}^2$$

(ii) Apply formula:

$$\frac{A_1}{A_2} = \left(\frac{\ell_1}{\ell_2} \right)^2$$

Here $\ell_1 = 10 \text{ cm}$, $\ell_2 = 25 \text{ cm}$, $A_1 = 40 \text{ cm}^2$, $A_2 = ?$

$$\frac{40}{A_2} = \left(\frac{10}{25} \right)^2$$

$$\frac{40}{A_2} = \left(\frac{2}{5} \right)^2$$

$$\frac{40}{A_2} = \frac{4}{25}$$

$$A_2 = 40 \times \frac{25}{4} = 250 \text{ cm}^2$$

- (iii) It is given that the quadrilateral $PQRS$ is similar to quadrilateral $XYZW$.

$$\frac{A_1}{A_2} = \left(\frac{\ell_1}{\ell_2} \right)^2$$

Here $\ell_1 = 35 \text{ cm}$, $\ell_2 = 25 \text{ cm}$, $A_1 = ?$, $A_2 = 98 \text{ cm}^2$

$$\frac{A_1}{98} = \left(\frac{35}{25} \right)^2$$

$$\frac{A_1}{98} = \left(\frac{7}{5} \right)^2$$

$$A_1 = \frac{49}{25} \times 98 = 192.08 \text{ cm}^2$$

- (iv) Since two pairs of corresponding angles in both triangles are equal, so triangles are similar.

$$\therefore \frac{A_1}{A_2} = \left(\frac{\ell_1}{\ell_2} \right)^2$$

Here $\ell_1 = ?$, $\ell_2 = 3 \text{ cm}$, $A_1 = 13.5 \text{ cm}^2$, $A_2 = 24 \text{ cm}^2$

$$\frac{13.5}{24} = \left(\frac{\ell_1}{3} \right)^2$$

$$\frac{135}{240} = \left(\frac{\ell_1}{3} \right)^2$$

$$\frac{9}{16} = \left(\frac{\ell_1}{3} \right)^2$$

$$\sqrt{\left(\frac{\ell_1}{3} \right)^2} = \sqrt{\frac{9}{16}} \quad (\text{Taking square root})$$

$$\frac{\ell_1}{3} = \frac{3}{4}$$

$$\ell_1 = \frac{9}{4}$$

$$= 2.25 \text{ cm}$$

(v) For similar spheres

$$\frac{A_1}{A_2} = \left(\frac{\ell_1}{\ell_2} \right)^2$$

Here $r_1 = ?$, $r_2 = 7$ cm, $A_1 = 153$ cm², $A_2 = 833$ cm²

$$\frac{153}{833} = \left(\frac{r_1}{7} \right)^2$$

$$\frac{9}{49} = \left(\frac{r_1}{7} \right)^2$$

$$\sqrt{\left(\frac{r_1}{7} \right)^2} = \sqrt{\frac{9}{49}} \quad (\text{Taking square root})$$

$$\frac{r_1}{7} = \frac{3}{7} \quad \Rightarrow \quad r_1 = 3 \text{ cm}$$

Example 7: Two polygons are similar with a ratio of corresponding sides being $\frac{3}{5}$. If

the area of the smaller polygon is 54 cm², find the area of the larger polygon.

Solution: The ratio of the areas of two similar polygons is the square of the ratio of

corresponding sides. So, $\frac{\text{Area of larger polygon}}{\text{Area of smaller polygon}} = \left(\frac{5}{3} \right)^2 = \frac{25}{9}$

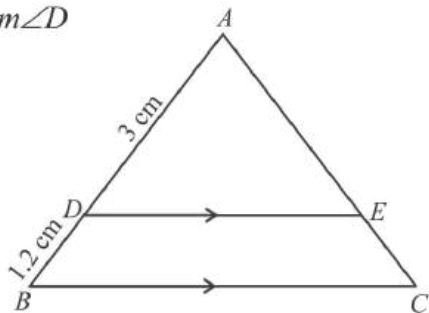
Therefore, Area of larger polygon = $\frac{25}{9} \times 54 = 150$ cm²

Example 8 Given that $\overline{BC} \parallel \overline{DE}$, prove that the triangles ABC and ADE are similar.

- If $m\overline{AB} = 3$ cm and $m\overline{BD} = 1.2$ cm, find the ratio of area of $\triangle ABC$ to the area of $\triangle ADE$.
- If area of $\triangle ADE$ is 125 cm², find the area of $\triangle ABC$ and area of trapezium $BCED$.

Solution: Since $m\angle A = m\angle A$ (common), $m\angle B = m\angle D$ and $m\angle C = m\angle E$ (Corresponding angles of parallel lines $\overline{BC}$ and $\overline{DE}$). Hence $\triangle ABC$ is similar to $\triangle ADE$.

$$(i) \quad \text{Ratio of sides} = \frac{m\overline{AB}}{m\overline{AD}} = \frac{3 + 1.2}{3} = \frac{4.2}{3} = \frac{7}{5}$$



$$\therefore \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle ADE} = \left(\frac{\ell_1}{\ell_2}\right)^2 = \left(\frac{7}{5}\right)^2 = \frac{49}{25}$$

(ii) Area of $\triangle ADE = 125 \text{ cm}^2$

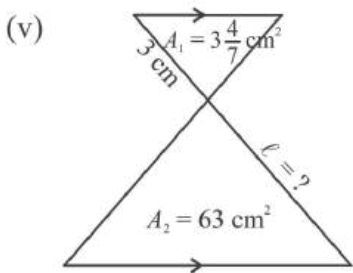
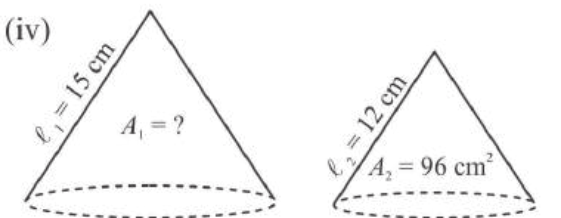
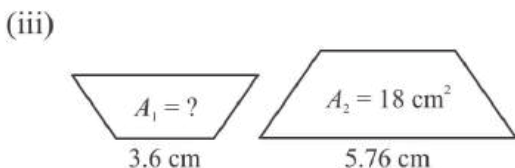
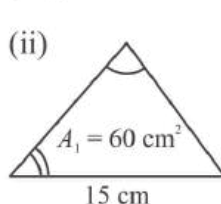
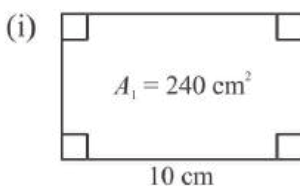
$$\frac{\text{Area of } \triangle ABC}{125} = \frac{49}{25}$$

$$\Rightarrow \text{Area of } \triangle ABC = \frac{49}{25} \times 125 = 245 \text{ cm}^2$$

$$\begin{aligned} \text{Area of trapezium } BCED &= \text{Area of } \triangle ABC - \text{Area of } \triangle ADE \\ &= 245 - 125 = 120 \text{ cm}^2 \end{aligned}$$

EXERCISE 9.2

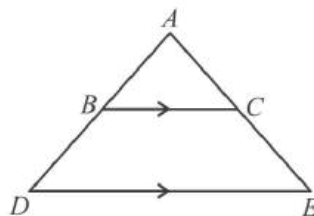
- Find the ratio of the areas of similar figures if the ratio of their corresponding lengths are: (i) 1:3 (ii) 3:4 (iii) 2:7 (iv) 8:9 (v) 6:5
- Find the unknowns in the following figures:



3. Given that area of $\triangle ABC = 36 \text{ cm}^2$ and $\overline{AB} = 6 \text{ cm}$,

$\overline{BD} = 4 \text{ cm}$. Find

- the area of $\triangle ADE$
- the area of trapezium $BCED$



4. Given that $\triangle ABC$ and $\triangle DEF$ are similar, with a scale factor of $k = 3$. If the area of $\triangle ABC$ is 50 cm^2 , find the area of triangle $\triangle DEF$?

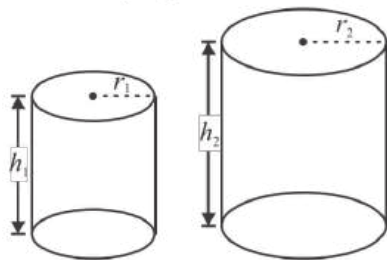
5. Quadrilaterals $ABCD$ and $EFGH$ are similar, with a scale factor of $k = \frac{1}{4}$. If the area of quadrilateral $ABCD$ is 64 cm^2 , find the area of quadrilateral $EFGH$.
6. The areas of two similar triangles are 16 cm^2 and 25 cm^2 . What is the ratio of a pair of corresponding sides?
7. The areas of two similar triangles are 144 cm^2 and 81 cm^2 . If the base of the large triangle is 30 cm , find the corresponding base of the smaller triangle.
8. A regular heptagon is inscribed in a larger regular heptagon and each side of the larger heptagon is 1.7 times the side of the smaller heptagon. If the area of the smaller heptagon is known to be 100 cm^2 , find the area of the larger heptagon.

9.3 Volume of Similar Solids

Two solids are said to be similar if they have same shape but possibly different sizes. Two solids are similar if lengths of the corresponding sides are proportional i.e., the ratio of the corresponding lengths are equal. e.g.,

The two cylinders are similar if $\frac{r_1}{r_2} = \frac{h_1}{h_2}$

If $r_1 = 4 \text{ cm}$, $r_2 = 5 \text{ cm}$, $h_1 = 8 \text{ cm}$ and $h_2 = 10 \text{ cm}$, then we note that:



$$\frac{r_1}{r_2} = \frac{4}{5} \text{ and } \frac{h_1}{h_2} = \frac{8}{10} = \frac{4}{5}.$$

$$\Rightarrow \frac{r_1}{r_2} = \frac{h_1}{h_2}$$

Volume of smaller cylinder,

$$\begin{aligned} V_1 &= \pi r_1^2 h_1 \\ &= \pi \times 4^2 \times 8 \\ &= 128\pi \text{ cm}^2 \end{aligned}$$

Volume of larger cylinder,

$$\begin{aligned} V_2 &= \pi r_2^2 h_2 \\ &= \pi \times 5^2 \times 10 \\ &= 250\pi \text{ cm}^2 \end{aligned}$$

$$\text{Ratio of volumes: } \frac{V_1}{V_2} = \frac{128\pi}{250\pi} = \frac{64}{125} = \left(\frac{4}{5}\right)^3$$

$$\text{So, } \frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 \text{ or } \frac{V_1}{V_2} = \left(\frac{h_1}{h_2}\right)^3$$

Hence the ratio of the volume of any two similar solids is equal to the cube of the ratio of any two corresponding lengths of the solids.

$$\frac{V_1}{V_2} = \left(\frac{\ell_1}{\ell_2} \right)^3$$

Since each length is k times of the other, we take $\frac{\ell_1}{\ell_2} = k$, then $\frac{V_1}{V_2} = k^3$. i.e., Volume V_1 is k^3 times the volume V_2 and k is called scale factor.

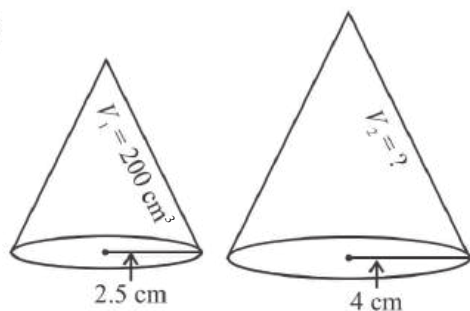
Since mass of a substance is proportional to its volume, the ratio of the mass of two similar solids is equal of to the ratio of their volumes. If the masses of two similar solids are w_1 and w_2 and volumes are V_1 and V_2 , then

$$\frac{V_1}{V_2} = \frac{w_1}{w_2}$$

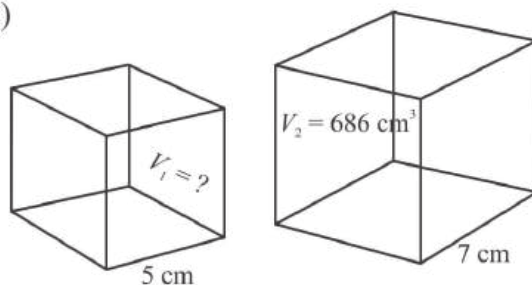
Therefore,
$$\frac{w_1}{w_2} = \left(\frac{\ell_1}{\ell_2} \right)^3$$

Example 9: Find the unknown volume in the following similar solids:

(i)



(ii)



Solution: (i) $\ell_1 = 2.5$ cm, $\ell_2 = 4$ cm

$$\frac{V_1}{V_2} = \left(\frac{\ell_1}{\ell_2} \right)^3$$

$$\frac{200}{V_2} = \left(\frac{2.5}{4} \right)^3$$

$$\frac{200}{V_2} = \left(\frac{5}{8} \right)^3$$

$$V_2 = 200 \times \frac{512}{125}$$

$$V_2 = 819.2 \text{ cm}^3$$

(ii) Using formula $\frac{V_1}{V_2} = \left(\frac{\ell_1}{\ell_2}\right)^3$

$$\frac{V_1}{686} = \left(\frac{5}{7}\right)^3$$

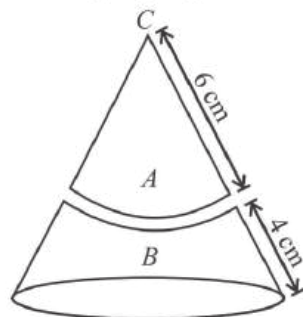
$$\left[\begin{array}{l} \ell_1 = 5 \text{ cm}, \ell_2 = 7 \text{ cm} \\ V_1 = ?, V_2 = 686 \text{ cm}^3 \end{array} \right]$$

$$\frac{V_1}{686} = \frac{125}{343}$$

$$\begin{aligned} V_1 &= \frac{125}{343} \times 686 \\ &= 250 \text{ cm}^3 \end{aligned}$$

Example 10: A solid cone C is cut into two pieces A and B with sloping edges 6 cm and 4 cm. Find the ratio of:

- the diameters of the bases of the cones A and C .
- the area of the bases of the cones A and C .
- the volumes of the cones A and C .
- If volume of cone A is 72 cm^3 , find the volume of solid B .



Solution: Let diameter of cone $A = d_1$

Diameter of cone $C = d_2$

- (i) The ratios of the corresponding lengths are equal because of similarity of the cones.

$$\therefore \frac{d_1}{d_2} = \frac{\ell_1}{\ell_2} = \frac{6}{10}$$

$$= \frac{3}{5}$$

$$\text{i.e., } \frac{\ell_1}{\ell_2} = \frac{3}{5}$$

(ii)
$$\frac{\text{Area of cone } A}{\text{Area of cone } C} = \left(\frac{\ell_1}{\ell_2}\right)^2$$

$$= \left(\frac{3}{5}\right)^2 = \frac{9}{25}$$

(iii)
$$\frac{\text{Volume of cone } A}{\text{Volume of cone } C} = \left(\frac{\ell_1}{\ell_2}\right)^3$$

$$= \left(\frac{3}{5}\right)^3 = \frac{27}{125}$$

(iv) $V_1 = \text{Volume of cone } A = 72 \text{ cm}^3$

$V_2 = \text{Volume of cone } C = ?$

$$\therefore \frac{V_1}{V_2} = \left(\frac{\ell_1}{\ell_2} \right)^3$$

$$\frac{72}{V_2} = \frac{27}{125}$$

$$V_2 = \frac{72 \times 125}{27} = 333\frac{1}{3} \text{ cm}^3$$

Volume of solid $B = \text{Volume of cone } C - \text{Volume of cone } A$

$$= 333\frac{1}{3} - 72 = 261\frac{1}{3} \text{ cm}^3$$

Example 11: The mass of sack of rice is 50 kg and height 60 cm. Find the mass of the similar sack of rice with height of 90 cm.

Solution: Mass of the smaller sack of rice $w_1 = 50 \text{ kg}$

Height of smaller sack of rice $h_1 = 60 \text{ cm}$

Mass of larger sack of rice $w_2 = ?$

Height of smaller sack of rice $h_2 = 90 \text{ cm}$

Using formula $\frac{w_1}{w_2} = \left(\frac{h_1}{h_2} \right)^3$

$$\frac{50}{w_2} = \left(\frac{60}{90} \right)^3 = \left(\frac{2}{3} \right)^3$$

$$\frac{50}{w_2} = \frac{8}{27}$$

$$w_2 = \frac{27 \times 50}{8} = 168.75 \text{ kg}$$

Example 12: The ratio of the corresponding lengths of two similar cylindrical cans is 3 : 2.

- The larger cylindrical can has surface area of 67.5 square metres. Find the surface area of the smaller cylindrical can.
- The smaller cylindrical can has a volume of 132 cubic metres. Find the volume of larger tin can.

Solution: (i) Surface area of larger can $= A_1 = 67.5 \text{ m}^2$
Surface area of smaller can $= A_2 = ?$

Ratio of corresponding lengths is $\frac{\ell_1}{\ell_2} = \frac{3}{2}$

Using formula for areas of the similar figures:

$$\frac{A_1}{A_2} = \left(\frac{\ell_1}{\ell_2} \right)^2$$

$$\frac{67.5}{A_2} = \left(\frac{3}{2} \right)^2 \Rightarrow A_2 = 67.5 \times \frac{4}{9} = 30 \text{ m}^2$$

(ii) Volume of smaller can = $V_2 = 132 \text{ m}^3$

Volume of larger can = $V_1 = ?$

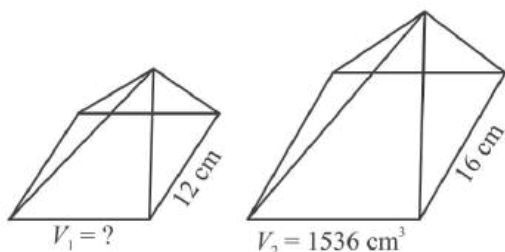
Using formula for volume of similar figures: $\frac{V_1}{V_2} = \left(\frac{\ell_1}{\ell_2} \right)^3$

$$\frac{V_1}{132} = \left(\frac{3}{2} \right)^3 \Rightarrow V_1 = 132 \times \frac{27}{8} = 445.5 \text{ m}^3$$

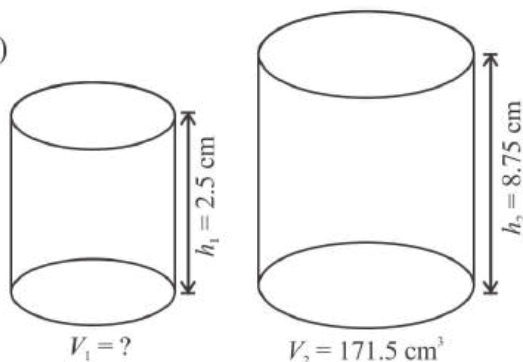
EXERCISE 9.3

- The radii of two spheres are in the ratio 3 : 4. What is the ratio of their volumes?
- Two regular tetrahedrons have volumes in the ratio 8 : 27. What is the ratio of their sides?
- Two right cones have volumes in the ratio 64 : 125. What is the ratio of:
 - their heights
 - their base areas?
- Find the missing value in the following similar solids.

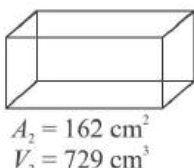
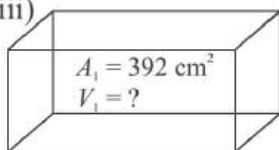
(i)



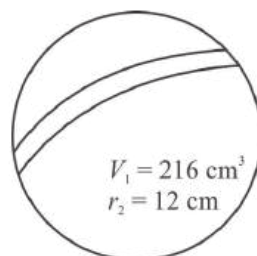
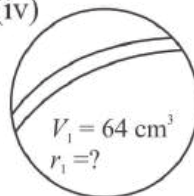
(ii)



(iii)



(iv)



5. The ratio of the corresponding lengths of two similar canonical cans is 3 : 2.
- The larger canonical can have surface area of 96 m^2 . Find the surface area of the smaller canonical can.
 - The smaller canonical can have a volume of 240 m^3 . Find the volume of larger canonical can.
6. The ratio of the heights of two similar cylindrical water tanks is 5 : 3.
- If the surface area of the larger tank is 250 square metres, find the surface area of the smaller tank.
 - If the volume of the smaller tank is 270 cubic metres, find the volume of the larger tank.

9.4 Geometrical Properties of Polygon and their Applications

9.4.1 Geometrical Properties of Regular Polygon

A regular polygon has all sides and all angles equal. Some of the common regular polygons are equilateral triangles, squares, regular pentagons, regular hexagons, etc.

Sum of Interior Angles: The formula for sum of interior angles of n -sided polygon is $(n - 2) \times 180^\circ$.

Interior Angle: For a regular n -sided polygon:

$$\text{Size of each Interior Angle} = \frac{(n - 2) \times 180^\circ}{n}$$

For instance, a regular hexagon has $n = 6$, so each interior angle is

$$\frac{(6 - 2) \times 180^\circ}{6} = \frac{720^\circ}{6} = 120^\circ$$

Exterior Angle: The sum of all exterior angles of any polygon is always 360° regardless of the number of sides. The exterior angle of a regular n -sided polygon is:

$$\text{Exterior Angle} = \frac{360^\circ}{n}$$

The interior and exterior angles are supplementary at a vertex i.e.,

$$\text{Interior} + \text{exterior angle} = 180^\circ$$

Diagonals: The total number of diagonals in a regular polygon with n sides is $\frac{n(n-3)}{2}$

Symmetry: A regular n -sided polygon has rotational symmetry and reflexive symmetry both of order n . e.g., a regular hexagon has six lines of symmetry and has rotational symmetry of order 6. A regular n -sided polygon can be rotated by $\frac{360^\circ}{n}$ and will look the same.

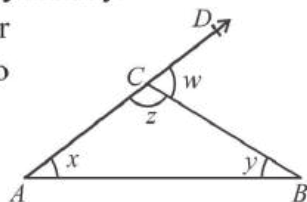
9.4.2 Geometrical Properties of Triangle

A triangle is a polygon with three sides and three angles. Triangles come in various types based on side length and angle measure.

Angle sum: The sum of the interior angles in any triangle is always 180° . In equilateral triangle, all sides are equal, and each angle is 60° . It has three lines of symmetry and rotational symmetry of order 3. In isosceles triangle, two sides are equal, and the angles opposite to the equal sides are also equal. It has one line of symmetry.

Exterior angle of a triangle: The measure of an exterior angle in a triangle is equal to sum of the measures of two opposite interior angles i.e.,

In $\triangle ABC$, $m\angle A + m\angle B = m\angle BCD$
i.e., $x + y = w$



9.4.3 Geometrical Properties of Parallelogram

A parallelogram is a quadrilateral whose opposite sides are parallel and equal in length and opposite angles are equal. Its adjacent angles are supplementary. The diagonals of a parallelogram bisect each other (they cross each other at the midpoint). They are not equal in length.

Recall:

Rectangle: All angles are 90° and diagonals are equal.

Rhombus: All sides are equal, and diagonals bisect each other at right angles.

Square: All sides are equal, all angles are 90° and diagonals are equal and bisect each other at right angles.

Example 13: Find the measure of each interior angle of a regular pentagon.

Solution: Interior angle = $\frac{(n-2) \times 180^\circ}{n}$

$$= \frac{(5-2) \times 180^\circ}{5} = \frac{540^\circ}{5} = 108^\circ$$

Each exterior angle is: $\frac{360^\circ}{5} = 72^\circ$

9.4.4 Applications of Polygons

Architects use polygons in building designs, while engineers rely on them to make strong structures like bridges. In art and design, polygons help create beautiful patterns and 3-D models. On maps, polygons show areas like cities or land boundaries. Polygons are also used in video games and animations to build characters and scenes. In science, they appear in molecular shapes, natural patterns like honeycombs and even in the design of telescope mirrors. Their simple, versatile shapes make polygons essential in many fields.

Tessellation

A tessellation is a pattern of shapes that fit together perfectly, without any gaps or overlaps, covering a plane. These shapes can be repeated infinitely to create a repeating pattern. Tessellations can be created using a single shape or a combination of shapes. They can be regular or irregular and they can exhibit various symmetries and patterns.

Only three regular polygons can tessellate the plane on their own: equilateral triangles, squares, and regular hexagons. They have symmetries. Hexagons (interior angle 120°) can tessellate perfectly because three hexagons meet at each vertex to form a 360° angle with no space creating a natural look inspired by honeycombs.

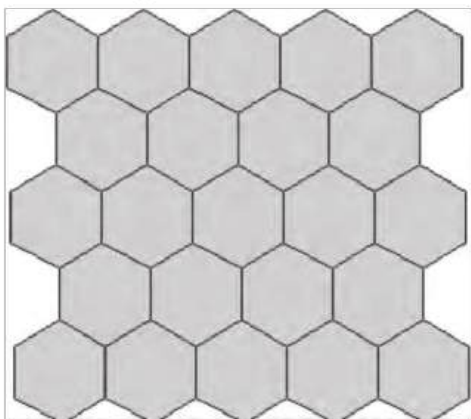
Remember:

Equilateral triangles can tessellate perfectly because the internal angle of each equilateral triangle is 60° , and six of these triangles meet at a point to form a 360° angle, allowing them to fill space seamlessly. Squares can tessellate perfectly because each square has an internal angle of 90° and four squares meet at a point to form a 360° angle.

Remember:

Regular pentagons and other polygons with angles that don't add up to 360° at each vertex cannot form gap-free patterns. i.e., Tessellation is not possible.

Regular Tessellation



Irregular Tessellation



Example 14: A tessellation is created using a combination of regular pentagons and decagons. Find the sum of the angles at a vertex where a pentagon and a decagon meet.

Solution

$$\begin{aligned}\text{Interior angle of regular decagon} &= \frac{(n-2) \times 180^\circ}{n} \\ &= \frac{(10-2) \times 180^\circ}{10} = \frac{1440^\circ}{10} = 144^\circ\end{aligned}$$

Interior angle of regular pentagon = 108°

Sum of angles = $144^\circ + 108^\circ = 252^\circ$. Since, angle sum $\neq 360^\circ$. Tessellation cannot be done.

Example 15: A parallelogram-shaped room has a base of 10 metres and a height of 8m. Babar wants to carpet the room using rolls that cover 20 m^2 each. How many rolls of carpet do he need?

Solution: The area of the parallelogram = $A = \text{base} \times \text{height} = 10 \times 8 = 80 \text{ m}^2$

$$\text{Number of rolls needed: } \frac{80}{20} = 4 \text{ rolls}$$

Example 16: Find the area of the equilateral triangle ABC of side length s .

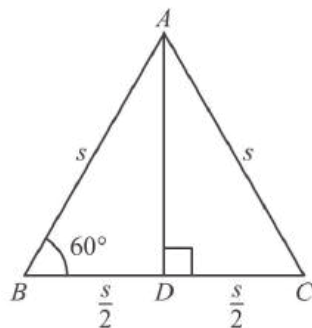
Solution: Draw perpendicular from A to side BC at point D . In the right angled triangle ABD :

Using trigonometric ratios: $\sin 60^\circ = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$

$$\frac{\sqrt{3}}{2} = \frac{m\overline{AD}}{s} \Rightarrow m\overline{AD} = \frac{\sqrt{3}}{2} s$$

$$\text{Area of triangle } ABC = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times s \times \frac{\sqrt{3}}{2} s$$

$$\text{Area of triangle } ABC = \frac{\sqrt{3}}{4} s^2$$



Example 17: Ali wants to create a floor design that uses regular hexagons (each with a side length of 1 metre) and equilateral triangles (each with a side length of 1 metre) to cover a rectangular area measuring 10 m by 5 m. Find how many hexagons and triangles Ali will need to complete the tessellation.

Solution: To find the area of an equilateral triangle with side length s , we can use the formula:

$$\text{Area of a triangle} = \frac{\sqrt{3}}{4} \cdot s^2$$

Multiply by 6 (since there are 6 triangles)

$$\text{Area of a hexagon} = \frac{6\sqrt{3}}{4} \cdot s^2 = \frac{3\sqrt{3}}{2} \cdot s^2$$

$$\text{Area of a hexagon} = \frac{3\sqrt{3}}{2} \times s^2 \approx \frac{3\sqrt{3}}{2} \times (1 \text{ m})^2 \approx 2.598 \text{ m}^2$$

$$\text{Area of an equilateral triangle} = \frac{\sqrt{3}}{4} \times s^2 \approx \frac{\sqrt{3}}{4} \times (1 \text{ m})^2 \approx 0.433 \text{ m}^2$$

$$\begin{aligned}\text{Area of the rectangular floor} &= 10 \text{ m} \times 5 \text{ m} \\ &= 50 \text{ m}^2\end{aligned}$$

Determine the arrangement: Assume a pattern where one hexagon is surrounded by 6 triangles. The area covered by one hexagon and the 6 surrounding triangles:

Total area covered by 1 hexagon and 6 triangles

$$= 2.598 \text{ m}^2 + 6 \times 0.433 \text{ m}^2 \approx 2.598 \text{ m}^2 + 2.598 \text{ m}^2 = 5.196 \text{ m}^2$$

Calculate the total number of hexagons and triangles needed:

$$\text{Number of sets} = \frac{50 \text{ m}^2}{5.196 \text{ m}^2} \approx 9.62 \text{ sets}$$

Rounding up, you can fit 10 sets of the pattern. Therefore, we need:

- Hexagons: 10
- Triangles: $10 \times 6 = 60$

Example 18: Falak plans to tile a square patio with an area of 100 square metres. He decides to use both square tiles and triangular tiles, each with an area of 0.25 square metres. If 60% of the tiles will be square and 40% will be triangular, how many tiles of each shape are needed?

Solution:

$$\begin{aligned}\text{Total number of tiles} &= \frac{\text{Patio Area}}{\text{Tile Area}} = \frac{100}{0.25} \\ &= 400 \text{ tiles}\end{aligned}$$

$$\text{Number of square tiles} = 400 \times 0.6 = 240$$

$$\text{Number of triangular tiles} = 400 \times 0.4 = 160$$

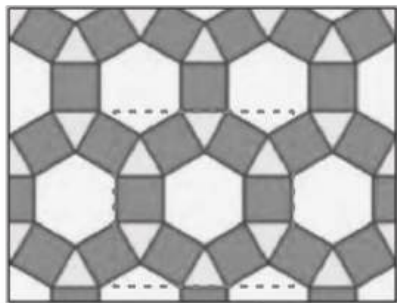
EXERCISE 9.4

1.
 - (i) What is the sum of the interior angles of a decagon (10-sided polygon)?
 - (ii) Calculate the measure of each interior angle of a regular hexagon.
 - (iii) What is each exterior angle of a regular pentagon?
 - (iv) If the sum of the interior angles of a polygon is 1260° , how many sides does the polygon have?
2. In a parallelogram $ABCD$, $m\overline{AB} = 10$ cm, $m\overline{AD} = 6$ cm and $m\angle BAD = 45^\circ$. Calculate the area of $ABCD$.
3. In a parallelogram $ABCD$ if $m\angle DAB = 70^\circ$, find the measures of all other angles in the parallelogram.
4. A shape is created by cutting a square in half diagonally and then attaching a right-angled triangle to the hypotenuse of each half. Explain why this shape can tessellate and calculate the interior angle of the new shape.
5. A tessellation is created by repeatedly reflecting a basic shape. The basic shape is a right-angled triangle with sides of length 3, 4, and 5 units. Find: The minimum number of reflections needed to create a tessellation that covers a square with an area of 3600 square units.
6. A tessellation is created using regular hexagons. Each hexagon has a side length of 5 cm. Find the total area of the tessellation if it consists of 25 hexagons and total perimeter of the outer edge of the tessellation, assuming it's a perfect hexagon.
7. A rectangular floor is 12 m by 15 m. How many square tiles, each 1 m by 1 m, are needed to cover the floor?
8. A rectangular wall is 10 m tall and 120 m wide. How many gallons of paint are needed to cover the wall, if one gallon covers 35 m^2 ?
9. A rectangular wall has a length of 10 m and a width of 4 meters. If 1 litre of paint covers 7 m^2 , how many liters of paint are needed to cover the wall?
10. A window has a trapezoidal shape with parallel sides of 3 m and 1.5 m and a height of 2 m. Find the area of the window.

REVIEW EXERCISE 9

1. Four options are given against each statement. Encircle the correct one.
- (i) If two polygons are similar, then:
- (a) their corresponding angles are equal.
 - (b) their areas are equal.
 - (c) their volumes are equal.
 - (d) their corresponding sides are equal.
- (ii) The ratio of the areas of two similar polygons is:
- (a) equal to the ratio of their perimeters.
 - (b) equal to the square of the ratio of their corresponding sides.
 - (c) equal to the cube of the ratio of their corresponding sides.
 - (d) equal to the sum of their corresponding sides.
- (iii) If the volume of two similar solids is 125 cm^3 and 27 cm^3 , the ratio of their corresponding heights is -----.
- (a) 3:5 (b) 5:3 (c) 25:9 (d) 9:25
- (iv) The exterior angle of regular pentagon is:
- (a) 40° (b) 45° (c) 60° (d) 72°
- (v) A parallelogram has an area of 64 cm^2 and a similar parallelogram has an area of 144 cm^2 . If a side of the smaller parallelogram is 8 cm, the corresponding side of the larger parallelogram is:
- (a) 10 cm (b) 12 cm (c) 18 cm (d) 16 cm
- (vi) The total number of diagonals in a polygon with 9 sides is:
- (a) 18 (b) 21 (c) 25 (d) 27
- (vii) Two spheres are similar, and their radii are in the ratio 4:5. If the surface area of the larger sphere is $500\pi \text{ cm}^2$, what is the surface area of the smaller sphere?
- (a) $256\pi \text{ cm}^2$ (b) $320\pi \text{ cm}^2$ (c) $400\pi \text{ cm}^2$ (d) $405\pi \text{ cm}^2$
- (viii) A regular polygon has an exterior angle of 30° . How many diagonals does the Polygon have?
- (a) 54 (b) 90 (c) 72 (d) 108
- (ix) In a regular hexagon, the ratio of the length of a diagonal to the side length is:
- (a) $\sqrt{3} : 1$ (b) 2 : 1 (c) 3 : 2 (d) 2 : 3

- (x) A regular polygon has an interior angle of 165° . How many sides does it have?
- (a) 15 (b) 16 (c) 20 (d) 24
2. If the sum of the interior angles of a polygon is 1080° , how many sides does the polygon has?
3. Two similar bottles are such that one is twice as high as the other. What is the ratio of their surface areas and their capacities?
4. Each dimension of a model car is $\frac{1}{10}$ of the corresponding car dimension. Find the ratio of:
- (a) the areas of their windscreens (b) the capacities of their boots
(c) the widths of the cars (d) the number of wheels they have.
5. Three similar jugs have heights 8 cm, 12 cm and 16 cm. If the smallest jug holds $\frac{1}{2}$ litre, find the capacities of the other two.
6. Three similar drinking glasses have heights 7.5 cm, 9 cm and 10.5 cm. If the tallest glass holds 343 millilitres, find the capacities of the other two.
7. A toy manufacturer produces model cars which are similar in every way to the actual cars. If the ratio of the door area of the model to the door area of the car is 1 cm to 2500 cm, find:
- (a) the ratio of their lengths
(b) the ratio of the capacities of their petrol tanks
(c) the width of the model, if the actual car is 150 cm wide
(d) the area of the rear window of the actual car if the area of the rear window of the model is 3 cm^2 .
8. The ratio of the areas of two similar labels on two similar jars of coffee is 144 : 169. Find the ratio of
- (a) the heights of the two jars (b) their capacities.
9. A tessellation of tiles on a floor has been made using a repeating pattern of a regular hexagon, six squares and six equilateral triangles. Find the total area of a single pattern with side length $\frac{1}{2}$ metre of each polygon.



Unit 10

Graphs of Functions

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Recall sketch graphs of linear functions (e.g. $y = ax + b$)
- Plot and interpret the graphs of quadratic, cubic, reciprocal and exponential functions.
 - Graph $y = ax^n$ where n is +ve integer, -ve integer, rational number for $x > 0$ and a is any real number.
 - Graph $y = ka^x$, where x is real $a > 1$.
- Discover exponential growth/decay of a practical phenomenon through its graph.
- Determine the gradients of curves by drawing tangents.
- Apply concepts of sketching and interpreting graphs to real-life problems (such as in tax payment, income and salary problems and cost and profit analysis)

INTRODUCTION

Graphs are powerful tools for visualizing and analyzing relationships between variables, making them essential in understanding various mathematical functions and their applications. In this unit, we will explore the graphs of linear, quadratic, cubic, reciprocal and exponential functions. We will also examine how to determine the gradient of curves by drawing tangents. Finally, we will connect these concepts to real-life scenarios, learning how to sketch and interpret graphs to solve practical problems.

10.1 Functions and their Graphs

Functions are essential tools for representing real-world phenomena using mathematical concepts. A function can be expressed in various forms, including an equation, a graph, a numerical table or a verbal description. For example, the area of a circle depends on its radius.

In such cases, one variable y depends on another variable x . This relationship is expressed as:

$$y = f(x)$$

Here, f denotes the function, x is the independent variable (input) and y is the dependent variable (output) determined by the value of x .

10.1.1 Graph of Linear Functions

A linear function is a mathematical expression that represents a straight-line relationship between two variables. Its general form is $f(x) = mx + c$, where “ m ” is the slope or gradient of the line, indicating how steep it is and “ c ” is the y -intercept (the point where the line crosses the y -axis). It can also be written as $y = mx + c$.

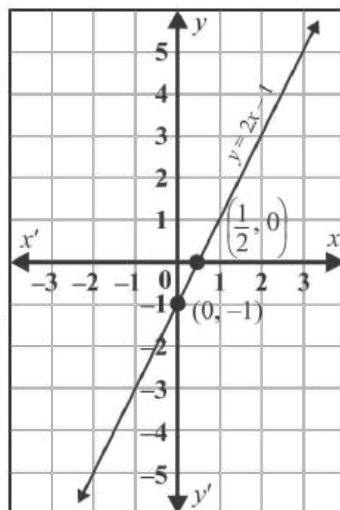
Example 1: Sketch the graph of $y = 2x - 1$.

Solution: To sketch the graph of linear function, we can find its x and y intercepts.

Put $x = 0$, we get $y = -1$. So $(0, -1)$ is the y -intercept.

Put $y = 0$, we get $x = \frac{1}{2}$. So $(\frac{1}{2}, 0)$ is the x -intercept.

The graph is a straight line that rises to the right because slope is positive.



10.1.2 Graph of Quadratic Functions

A quadratic function is a type of polynomial function that involves x^2 term. Its general form is:

$$y = ax^2 + bx + c$$

Where a, b, c are constants and $a \neq 0$.

Example 2: Plot the graphs of $y = x^2$ and $y = -x^2$ on the same diagram.

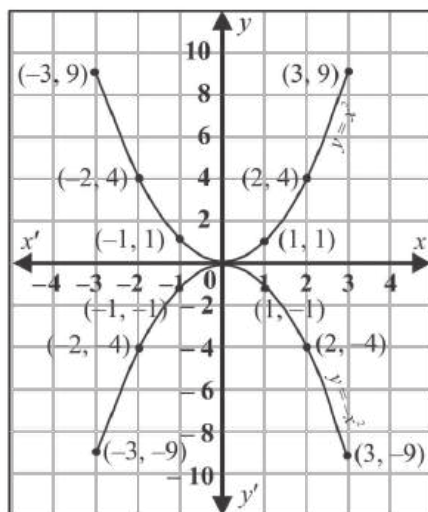
Solution: The following table shows several values of x and the given functions are evaluated at those values:

x	$y = x^2$	$y = -x^2$
-3	$(-3)^2 = 9$	-9
-2	$(-2)^2 = 4$	-4
-1	$(-1)^2 = 1$	-1
0	$(0)^2 = 0$	0
1	$(1)^2 = 1$	-1
2	$(2)^2 = 4$	-4
3	$(3)^2 = 9$	-9

Keep in mind!

The graph of a quadratic function is always a **parabola**.

- If $a > 0$, then the parabola opens upward like “ $\cup$ ”.
- If $a < 0$, then the parabola opens downward like “ $\cap$ ”.

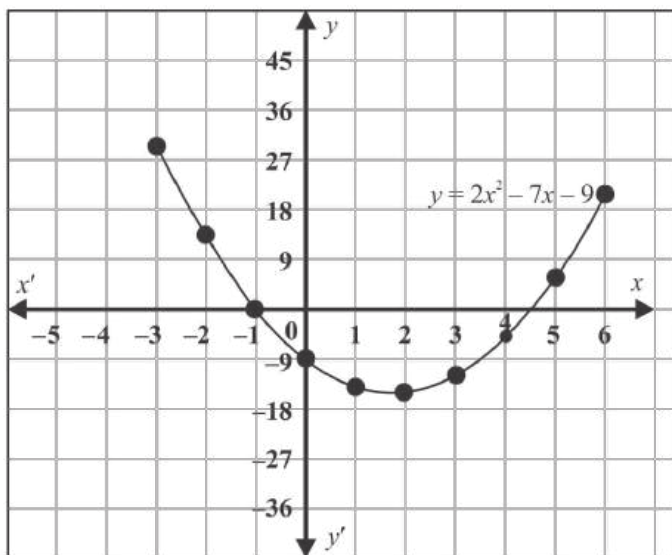


- (i) Graph of $y = x^2$ represents parabola, passing through origin and opens upward.
- (ii) Graph of $y = -x^2$ represents parabola, passing through origin and opens downward.

Example 3: Sketch the graph of $y = 2x^2 - 7x - 9$ for $-3 \leq x \leq 6$.

Solution: The values of x and y are given in the table and sketched in figure below:

x	y
-3	30
-2	13
-1	0
0	-9
1	-14
2	-15
3	-12
4	-5
5	6
6	21



Graph of $y = 2x^2 - 7x - 9$ represents parabola and opens upward. It intersects the y -axis at $(0, -9)$ and x -axis at $(-1, 0)$ and $(4.5, 0)$.

10.1.3 Graph of Cubic Functions

A cubic function is a type of polynomial function of degree 3. Its standard form is:

$$y = ax^3 + bx^2 + cx + d$$

Where a, b, c, d are constants and $a \neq 0$.

Remember!

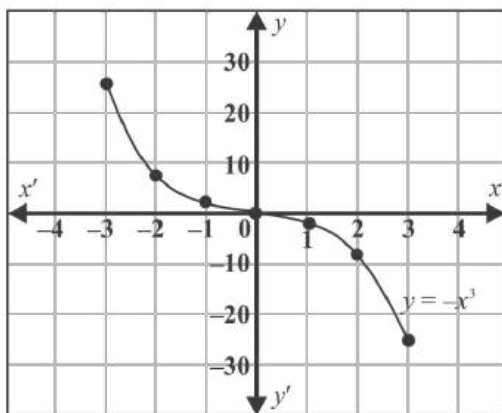
- The graph of a cubic function is a curve that can have at most two turning points.
- It has a general "S-shaped" appearance and depending on the coefficients, the shape may vary.
- Such functions are much more complicated and show more varied behaviour than linear and quadratic ones.

Example 4: Plot the graph of the following cubic function for $-3 \leq x \leq 3$:

$$y = -x^3$$

Solution: The following table shows several values of x and the given function is evaluated at those values:

x	$y = -x^3$
-3	27
-2	8
-1	1
0	0
1	-1
2	-8
3	-27



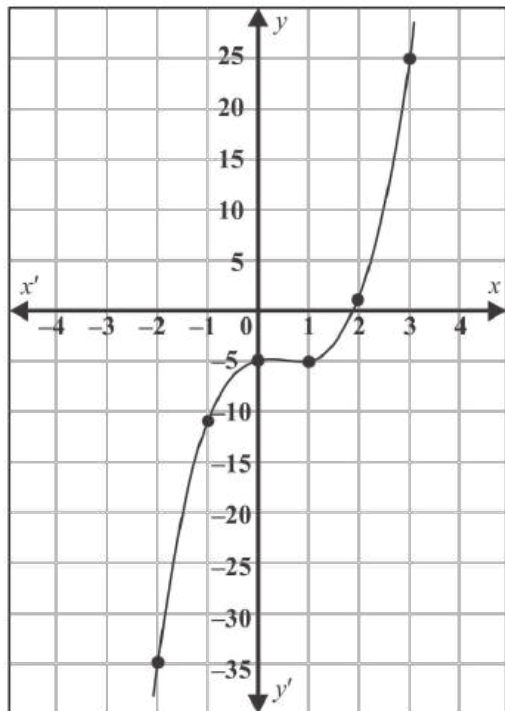
The curve passes through the origin.

Example 5: Plot the graph of $y = 2x^3 - 3x^2 + x - 5$ for $-2 \leq x \leq 3$.

Solution:

The following table shows several values of x and the given function is evaluated at those values:

x	y
-2	-35
-1	-11
0	-5
1	-5
2	1
3	25



The graph tells us that when $x = 0$, the function's value is -5 .

10.1.4 Graph of Reciprocal Functions

A reciprocal function is a function of the form:

$$y = \frac{a}{x}$$

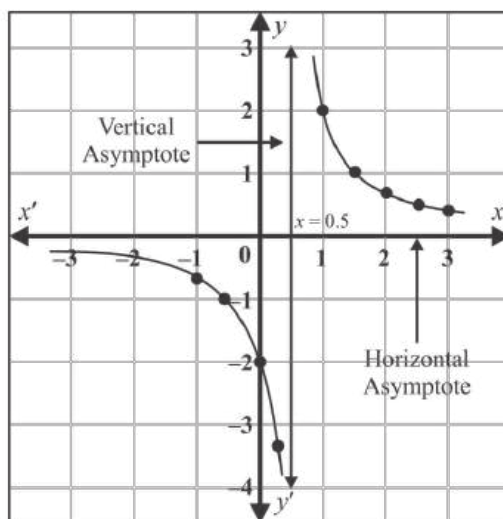
Where a is any real number and $x \neq 0$.

Example 6: Sketch the graph of the following reciprocal function:

$$y = \frac{1}{x-0.5}, x \neq 0.5$$

Solution: The following table shows several values of x and the given function is evaluated at those values:

x	y
-1	-0.67
-0.5	-1
-0.2	-1.43
0	-2
0.2	-3.3
0.5	undefined
1	2
1.2	1.43
1.5	1
2	0.67
2.2	0.59
2.5	0.5
3	0.4



Remember!

An asymptote is a line that a graph approaches but never touches.

10.1.5 Graph of Exponential Functions ($y = ka^x$ where x is real number, $a > 1$)

An exponential function is a mathematical function of the form:

$$y = ka^x$$

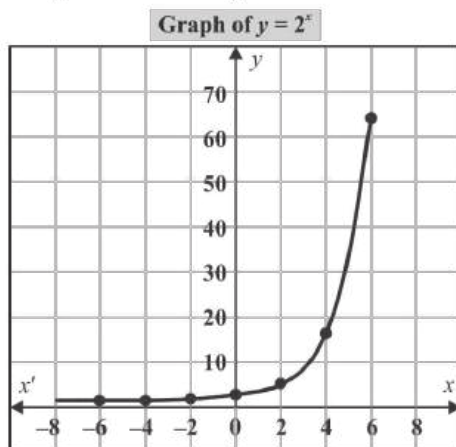
Where a, k are constants, x is variable and $a > 1$.

Example 7: Plot the graph of the exponential function $y = 2^x$ for $-6 \leq x \leq 6$.

Solution: The function $y = 2^x$ has base 2 and variable exponent x . Values of (x, y) are given in the table below:

x	-6	-4	-2	0	2	4	6
$y = 2^x$	0.02	0.06	0.25	1	4	16	64

Graph of the above points is given in the figure below:

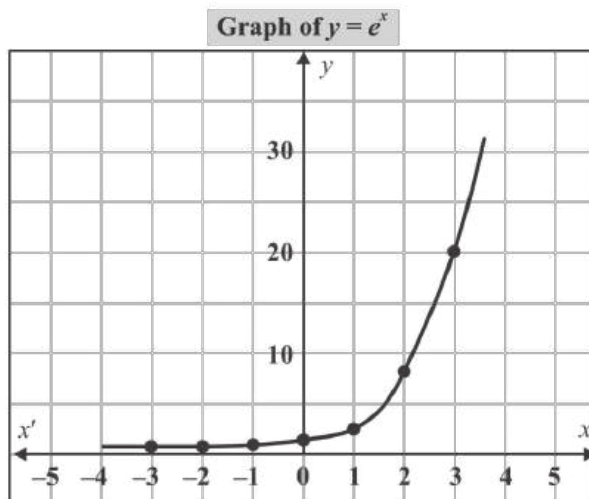


The graph of $y = 2^x$ represents the growth curve.

Example 8: Plot the graph of the exponential function, $y = e^x$.

Solution: The function $y = e^x$ has base e and variable power x . We know $e = 2.7182818$, correct to two decimal places $e = 2.72$. Table of x and y values is given below:

x	$y = e^x$
-3	0.05
-2	0.14
-1	0.37
0	1
1	2.72
2	7.40
3	20.09



10.1.6 Graphs of $y = ax^n$ (where n is +ve integer, -ve integer or rational number for $x > 0$ and a is any real number)

The graph of the function $y = ax^n$, where n is a positive integer, negative integer or rational number for $x > 0$ and a is any real number, exhibits distinct behaviours depending on the value of n . Following are the examples of these cases:

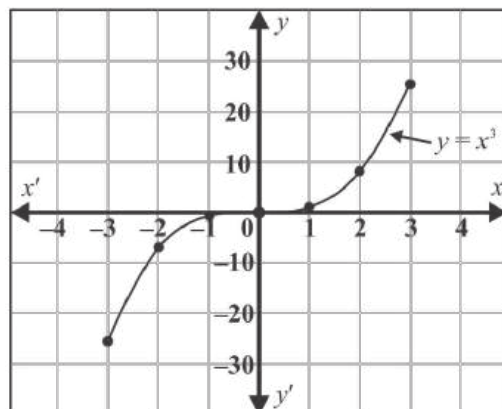
(i) When n is positive integer ($n = 3$)

Example 9: Plot the graph of $y = x^3$ for $-3 \leq x \leq 3$.

Solution: The table shows several values of x and the given function is evaluated at those values:

x	$y = x^3$
-3	-27
-2	-8
-1	-1
0	0
1	1
2	8
3	27

The curve passes through the origin.



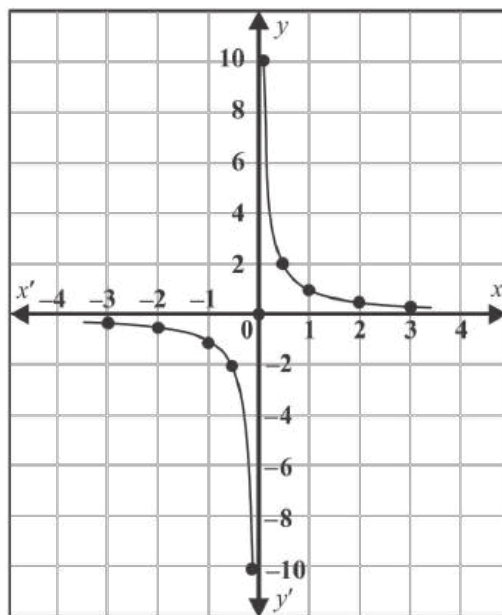
(ii) When n is negative integer ($n = -1$)

Example 10: Plot the graph of $y = x^{-1}$

Solution: $y = x^{-1} = \frac{1}{x}$

The following table shows several values of x and the given function is evaluated at those values:

x	$y = \frac{1}{x}$
-3	-0.3
-2	-0.5
-1	-1
-0.5	-2
-0.1	-10
0.1	10
0.5	2
1	1
2	0.5
3	0.3



The above graph consists of two branches, one in the first quadrant and the other in the third quadrant. Both branches approach but never touch the x -axis or the y -axis.

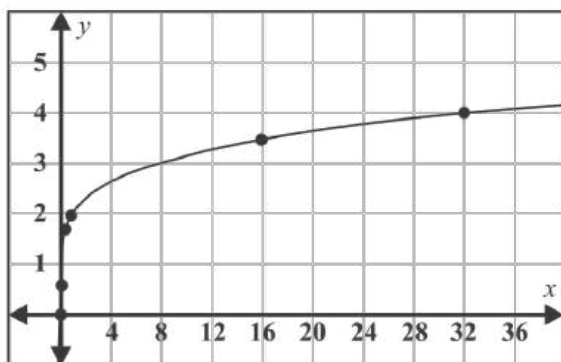
(iii) When n is rational number $\left(n = \frac{1}{5}\right)$

Example 11: Plot the graph of $y = 2x^{\frac{1}{5}}$.

Solution: $y = 2x^{\frac{1}{5}}$

The following table shows several values of x and the given function is evaluated at those values.

x	y
0	0
0.01	0.80
0.5	1.74
1	2
16	3.48
32	4



EXERCISE 10.1

- Sketch the graph of the following linear functions:
 - $y = 3x - 5$
 - $y = -2x + 8$
 - $y = 0.5x - 1$
- Plot the graph of the following quadratic and cubic functions:
 - $y = x^3 + 2x^2 - 5x - 6; -3.5 \leq x \leq 2.5$
 - $y = x^2 + x - 2$
 - $y = x^3 + 3x^2 + 2x; -2.5 \leq x \leq 0.5$
 - $y = 5x^2 - 2x - 3$
- Plot the graph of the following functions:
 - $y = 4^x$
 - $y = 5^{-x}$
 - $y = \frac{1}{x-3}, x \neq 3$
 - $y = \frac{2}{x} + 3, x \neq 0$
 - $y = x^{\frac{1}{2}}$
 - $y = 3x^{\frac{1}{3}}$
 - $y = 2x^{-2}$

10.2 Exponential Growth/Decay of a Practical Phenomenon through its Graph

Exponential growth and decay are widely observed in real-world phenomenon and their graphical representations offer critical insights into these processes. In exponential growth, such as population expansion, compound interest in finance or the spread of infectious diseases, the graph starts slowly but accelerates rapidly as time progresses. The curve increases steeply, showcasing how growth becomes more pronounced with time due to constant proportional changes. Conversely, in exponential decay, observed in cooling of objects or depreciation of assets, the graph starts high and decreases sharply before levelling off, indicating a gradual reduction over time. These graphs are essential for interpreting trends, making predictions and informing decision-making in diverse fields.

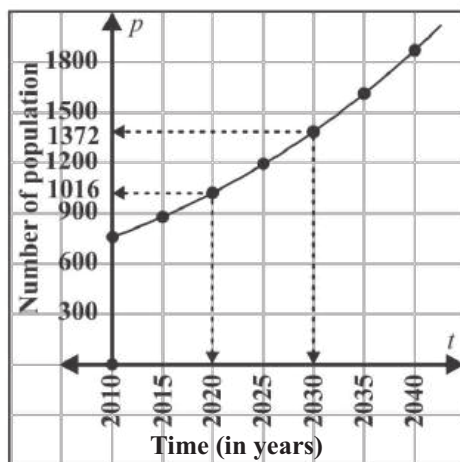
Example 12: The population of a village was 753 in 2010. If the population grows according to the equation $p = 753e^{0.03t}$, where p is the number of persons in the population at time t ,

- Graph the population equation for $t = 0$ (in 2010) to $t = 30$ (in 2040).
- From the graph, estimate the population (i) in 2020 and (ii) in 2030.

Solution: (a) The general shape of the exponential is known; however, since the graph is being used for estimations, an accurate graph over the required interval, $t=0$ to $t=30$, is required.

Calculate a table of values for different time periods and sketched in below figure:

t	p
0	753
5	874.9
10	1016.4
15	1180.9
20	1372.1
25	1594.1
30	1852.1



(b) From graph,

(i) In 2020 ($t = 10$) the population is 1016 persons.

(ii) In 2030 ($t = 20$) the population is 1372 persons.

10.2.1 Gradients of Curves by Drawing Tangents

The gradient or slope of a graph at any point is equal to the gradient of the tangent to the curve at that point. Remember that a tangent is a line that just touches a curve only at one point (and doesn't cross it).

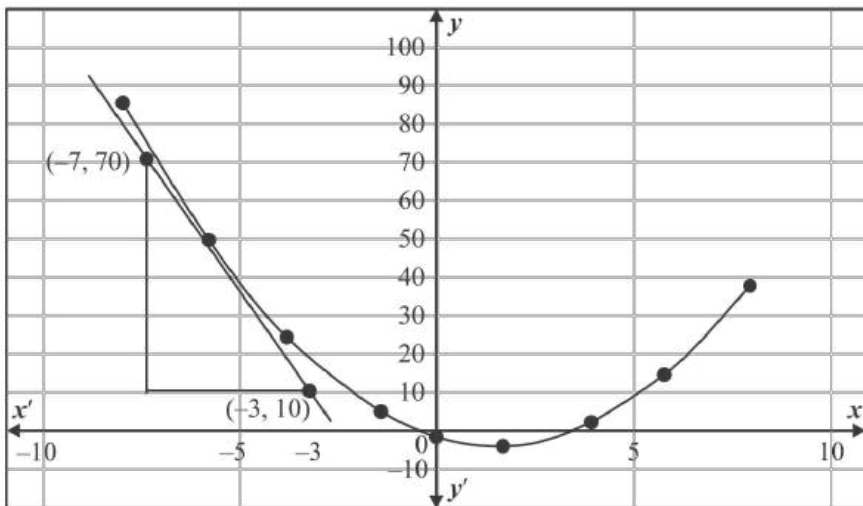
The gradient between two points is defined as:

$$\text{Gradient} = \frac{\text{Change in } y}{\text{Change in } x} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example 13: Sketch the graph of $y = x^2 - 3x - 2$ for values of x from -8 to 8 , draw a tangent line at $x = -6$ and determine the gradient.

Solution: Calculate the y -values for given values of x . The results are given in the table and sketched in below figure:

x	-8	-6	-4	-2	0	2	4	6	8
y	86	52	26	8	-2	-4	2	16	38



Consider two points $(-3, 10)$ and $(-7, 70)$ on the tangent line.

So, gradient = $\frac{70-10}{-7+3} = -15$. Since the gradient is negative, this indicates that the height of the graph decreases as the value of x increases.

10.2.2 Applications of Graph in Real-Life

Applying concepts of sketching and interpreting graphs to real-life problems enables individuals to visualize and analyse complex relationships, make informed decisions and optimize solutions. In tax payment scenarios, graphing concepts help identify optimal income levels, tax brackets, and liability. In income and salary problems, graphing facilitates analysis of compensation packages and income growth. By sketching salary against experience, patterns or anomalies in compensation structures become apparent. In cost and profit analysis, graphing enables businesses to visualize cost-profit relationships, determine break-even points, and optimize production levels.

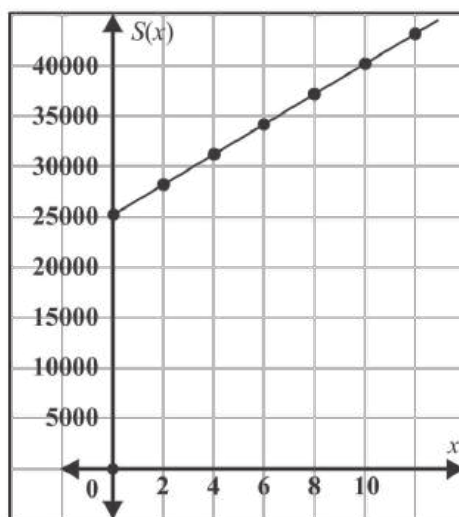
Example 14: Majid's salary $S(x)$ in rupees is based on the following formula:

$$S(x) = 25000 + 1500x,$$

where x is the number of years he worked. Sketch and interpret the graph of salary function for $0 \leq x \leq 10$.

Solution: Table values and graph are given below:

x	$S(x)$
0	25000
2	28000
4	31000
6	34000
8	37000
10	40000



Majid's salary increases linearly with years of service and rises by Rs. 1500 for every year.

Example 15: A company manufactures footballs. The cost of manufacturing x footballs is $C(x) = 90,000 + 600x$. The revenue from selling x footballs is $R(x) = 1,800x$. Find the break-even point and determine the profit or loss when 200 footballs are sold. Draw the graphs of both the functions and identify the break-even point.

Solution: Given that

$$\text{Cost function: } C(x) = 90,000 + 600x$$

$$\text{Revenue function: } R(x) = 1,800x$$

The break-even point occurs when $R(x) = C(x)$

$$1800x = 90000 + 600x$$

$$1200x = 90000$$

$$\Rightarrow x = \frac{90000}{1200}$$

$$x = 75$$

So, at the break-even point, 75 footballs are produced or sold.

Next, we find the profit for 150 footballs

When $x = 150$, revenue:

$$\begin{aligned} R(150) &= 1,800(150) \\ &= \text{Rs. } 270,000 \end{aligned}$$

$$\begin{aligned} \text{and } C(150) &= 90,000 + 600(150) \\ &= \text{Rs. } 180,000 \end{aligned}$$

Now profit: $P(x) = R(x) - C(x)$

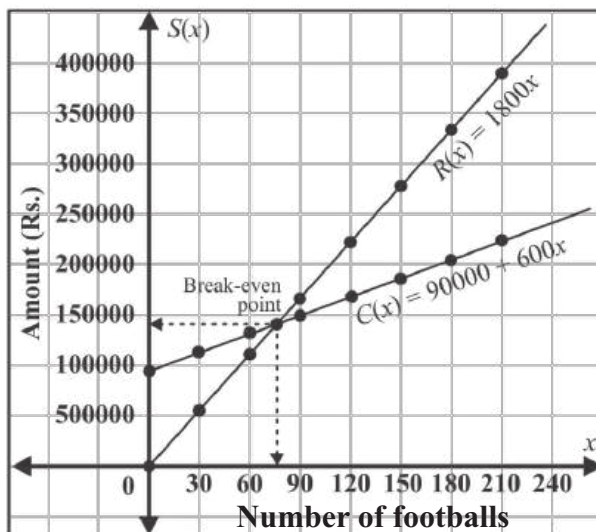
Substitute $x = 150$

$$\begin{aligned} P(150) &= R(150) - C(150) \\ &= \text{Rs. } 270,000 - \text{Rs. } 180,000 \\ &= \text{Rs. } 90,000 \end{aligned}$$

Thus, a company earns a profit of Rs. 90,000 when selling 150 footballs.

Table values and graph are given below:

x	$C(x)$	$R(x)$
0	90000	0
30	108000	54000
60	126000	108000
90	144000	162000
120	162000	216000
150	180000	270000
180	198000	324000
210	216000	378000



EXERCISE 10.2

1. Plot the graph of $y = 2x^2 - 4x + 3$ for x from -1 to 3 . Draw tangent at $(2, 3)$ and find the gradient.
2. Plot the graph of $y = 3x^2 + x + 1$ and draw tangent at $(1, 5)$. Also find gradient of the tangent line at this point.
3. The strength of students in a school was 1000 in 2016. If the strength decay according to the equation $S = 1000 e^{-t}$, where S is the number of students at time t .
 - (a) Graph the given equation for $t = 0$ (in 2016) to $t = 9$ (in 2025).
 - (b) From the graph, estimate the student's strength in 2019 and in 2023.
4. The demand and supply functions for a product are given by the equations $P_d = 400 - 5Q$, $P_s = 3Q + 24$:

Plot the graph of each function over the interval $Q = 0$ to $Q = 300$.

5. Shahid's salary $S(x)$ in rupees is based on the following formula:

$$S(x) = 45000 + 4500x,$$

where x is the number of years he has been with the company. Sketch and interpret the graph of salary function for $0 \leq x \leq 5$.

6. A company manufactures school bags. The cost function of producing x bags is $C(x) = 1200 + 20x$ and the revenue from selling x bags is $R(x) = 50x$.
 - (a) Find the break-even point.
 - (b) Determine the profit or loss when 250 bags are sold.
 - (c) Plot the graphs of both the functions and identify the break-even point.
7. A newspaper agency fixed cost of Rs. 70 per edition and marginal printing and distribution costs of Rs. 40 per copy. Profit function is $p(x) = 10x - 70$, where x is the number of newspapers. Plot the graph and find profit for 500 newspapers.
8. Ali manufactures expensive shirts for sale to a school. Its cost (in rupees) for x shirts is $C(x) = 1500 + 10x + 0.2x^2$, $0 \leq x \leq 150$. Plot the graph and find the cost of 200 shirts.

REVIEW EXERCISE 10

1. Four options are given against each statement. Encircle the correct option.

(i) $x = 5$ represents:

- (a) x -axis (b) y -axis
(c) line $\parallel$ to x -axis (d) line $\parallel$ to y -axis

(ii) Slope of the line $y = 5x + 3$ is:

- (a) 3 (b) -3 (c) 5 (d) -5

(iii) The y - intercepts of $y = -2x - 1$ is:

- (a) -2 (b) 2
(c) -1 (d) 1

(iv) The graph of $y = x^3$, cuts the x -axis at:

- (a) $x = 0$ (b) $x = 1$ (c) $x = -1$ (d) $x = 2$

(v) The graph of 3^x represents:

- (a) growth (b) decay (c) both (a) and (b) (d) a line

(vi) The graph of $y = -x^2 + 5$ opens:

- (a) upward (b) downward (c) left side (d) right side

(vii) The graph of $y = x^2 - 9$ opens:

- (a) upward (b) downward (c) left side (d) right side

(viii) $y = 5^x$ is _____ function.

- (a) linear (b) quadratic (c) cubic (d) exponential

(ix) Reciprocal function is:

- (a) $y = 7^x$ (b) $y = \frac{2}{x}$ (c) $y = 2x^2$ (d) $y = 5x^3$

(x) $y = -3x^3 + 7$ is _____ function.

- (a) exponential (b) cubic (c) linear (d) reciprocal

2. Plot the graph of the following functions:

- (i) $y = 3^{-x}$ for x from -2 to 4 (ii) $y = \frac{2}{x}$, $x \neq 0$

3. Sales for a new magazine are expected to grow according to the equation:
 $S = 200000 (1 - e^{-0.05t})$, where t is given in weeks.
- (a) Plot graph of sales for the first 50 weeks.
- (b) Calculate the number of magazines sold, when $t = 5$ and $t = 35$.
4. Plot the graph of following for x from -5 to 5 :
- (i) $y = x^2 - 3$ (ii) $y = 15 - x^2$
5. Plot the graph of $y = \frac{1}{2} (x + 4)(x - 1)(x - 3)$ for x from -5 to 4 .
6. The supply and demand functions for a particular market are given by the equations:
 $P_s = Q^2 + 5$ and $P_d = Q^2 - 10Q$, where P represents price and Q represents quantity,
Sketch the graph of each function over the interval $Q = -20$ to $Q = 20$.
7. A television manufacturer company make 40 inches LEDs. The cost of manufacturing x LEDs is $C(x) = 60,000 + 250x$ and the revenue from selling x LEDs is $R(x) = 1200x$. Find the break-even point and find the profit or loss when 100 LEDs are sold. Identify the break-even point graphically.

Unit 11

Loci and Construction

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Construct a triangle having given two sides and the included angle.
 - Construct a triangle having given one side and two of the angles.
 - Construct a triangle having given two of its sides and the angle opposite to one of them.
 - Draw angle bisectors, perpendicular bisectors, medians, altitudes of a given triangle and verify their concurrency.
 - Draw loci and intersection of loci for set of points in two dimensions which are
 - at a given distance from a given point.
 - at a given distance from a given line
 - equidistant from two given points
 - equidistant from two given intersecting lines
- Solve real life problems using the loci and interesting loci.

INTRODUCTION

A locus plural loci is a set of points that follow a given rule. Loci are also useful for understanding and predicting patterns. For instance, consider two people walking around a room, each maintaining a fixed distance from the other. The possible locations are where each person form a specific path. By studying these loci, we can predict where each person might be relative to the other at any time. In contexts like tracking satellites orbiting Earth, we use the concept of loci to predict where they will be at given times. This helps in areas like telecommunications and GPS technology.

Loci in two dimensions are triangle, circle, parallel lines, perpendicular bisector and angle bisector.

11.1 Construction of Triangles

A triangle is a closed figure having three sides and three angles. We construct triangle in the following cases:

- (a) When measure of all three sides are given.
- (b) When measure of two sides and their included angle are given.
- (c) When measure of one side and measure of two angles are given.
- (d) When measure of two sides and an angle opposite to one of them is given.

Remember!

There are three types of triangles w.r.t. sides:

Scalene triangle: All sides are of different length.

Isosceles triangles: Two sides are of equal length.

Equilateral triangle: All sides of equal length.

There are three types of triangles w.r.t. angles:

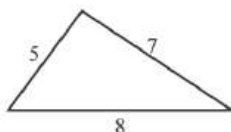
Acute angled triangle: All angles are of measure less than 90° .

Obtuse angled triangle: One angle is of measure greater than 90° .

Right angled triangle: One angle is of measure equal to 90° .

Triangle Inequality Theorem

The sum of the measure of any two sides of a triangle is always greater than the measure of the third side. For example, we can see in the figure adding any two lengths then this will be greater than the third side i.e., $5 + 7 > 8$, $5 + 8 > 7$ and $7 + 8 > 5$



Key fact!

- An equilateral triangle is acute angled triangle.
- A right angled triangle cannot be equilateral.

(a) Construction of a triangle when measure of three sides is given

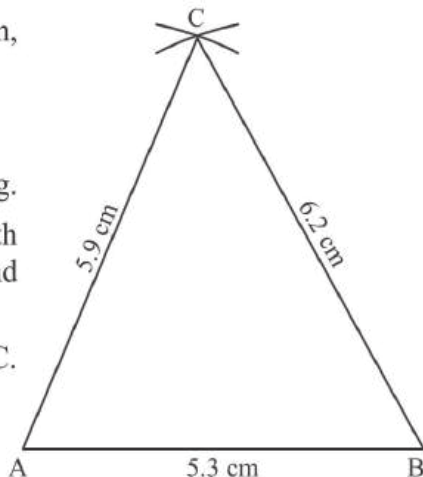
Example 1: Construct a triangle of sides 5.3 cm, 5.9 cm and 6.2 cm.

Solution: Steps of construction:

- Draw a line segment AB of length 5.3 cm long.
- Using a pair of compasses, draw two arcs with centres at points A and B of radii 5.9 cm and 6.2 cm respectively.
- These two arcs intersect each other at point C.
- Join A and B with C.

Hence, $\triangle ABC$ is the required triangle.

NOTE: The angles 30° , 45° , 60° , 75° , 90° , 105° , 120° , 135° and 150° are constructed with the help a pair of compasses. Other angles are drawn using protractor.



Do you know?

When three sides are given, we can draw any length first.

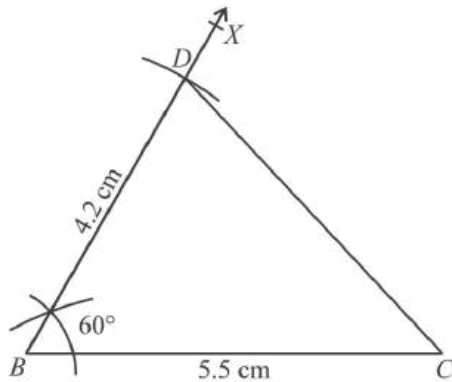
(b) Construction of a triangle when the measure of two sides and their included angle are given

Example 2: Construct a triangle BCD in which measures of two sides are 5.5 cm and 4.2 cm and measure of their included angle is 60° .

Solution: Steps of construction

- Draw a line segment BC of length 5.5 cm.
- Draw an angle 60° at point B using a pair of compasses and draw a ray BX through this angle.
- Draw an arc of radius 4.2 cm with centre at point B intersecting BX at point D.
- Join C and D.

Hence, $\triangle BCD$ is the required triangle.

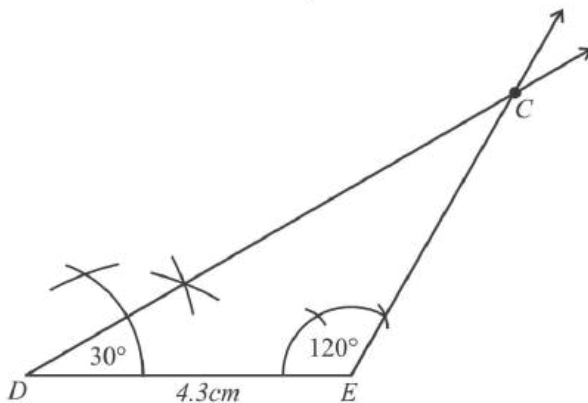


(c) Construction of a triangle when measure of one side and two angles are given

Example 3: Draw a triangle CDE when $m\overline{DE} = 4.3$ cm, $m\angle D = 30^\circ$ and $m\angle E = 120^\circ$.

Solution: Steps of construction:

- (i) Draw $m\overline{DE} = 4.3$ cm.
- (ii) Draw angles 30° and 120° at points D and E respectively using a pair of compasses and draw two rays through these angles from D and E .
- (iii) These two rays intersect each other at point C .



Hence, $\triangle CDE$ is the required triangle.

(d) Construction of a triangle when measure of two sides and angle opposite to one of them is given

Consider the given two cases:

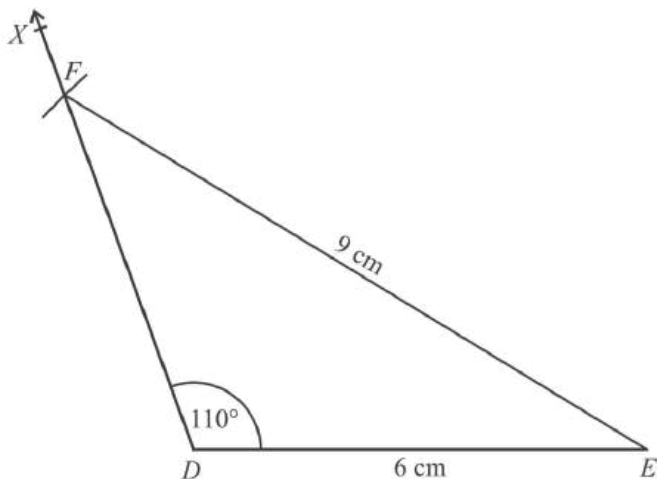
- (i) If the measure of one angle is greater than or equal to 90° .
- (ii) If the measure of angle is less than 90° .

Example 4: Construct a triangle DEF when $m\overline{DE} = 6$ cm, $m\angle D = 110^\circ$ and $m\overline{EF} = 9$ cm.

Solution:

Steps of construction:

- (i) Draw $m\overline{DE} = 6$ cm.
- (ii) Construct $m\angle D = 110^\circ$ using protractor and draw $\overrightarrow{DX}$ through this angle.
- (iii) Draw an arc of radius 9 cm with centre at point E intersecting $\overrightarrow{DX}$ at point F .
- (iv) Join E and F .



Hence, $\triangle DEF$ is the required triangle

If the given angle opposite to the given side is obtuse, only one triangle is possible.

Example 5:

Construct triangles DEF and DEF' when $\overline{mDE} = 6 \text{ cm}$, $m\angle D = 30^\circ$ and $\overline{mEF} = 3.6 \text{ cm}$

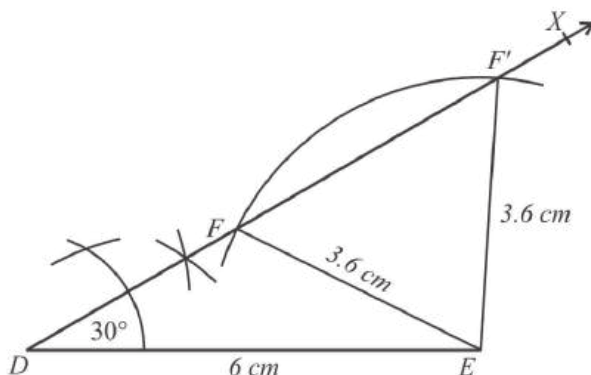
Solution:

Steps of construction:

- (i) Draw $\overline{mDE} = 6 \text{ cm}$.
- (ii) Construct an angle 30° at point D using a pair of compasses and draw $\overrightarrow{DX}$ through this angle.
- (iii) Draw an arc of radius 3.6 cm with centre at point E .
- (iv) This arc intersects $\overrightarrow{DX}$ at two points F and F' .
- (v) Join F and F' with E .

We get two triangles DEF and DEF' .

This is known as **ambiguous case**.



Do you know?

The **Ambiguous Case (SSA)** occurs when we are given two sides and the angle opposite one of these is less than 90° .

Example 6: In the above example if we take:

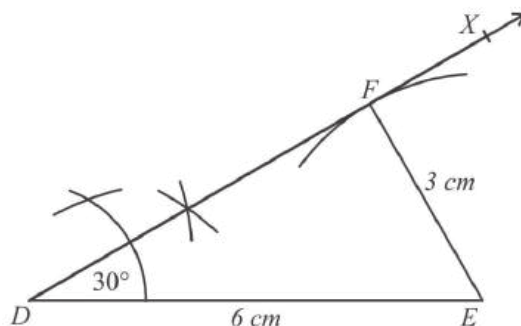
- (a) $\overline{mEF} = 3 \text{ cm}$
- (b) $\overline{mEF} = 2.5 \text{ cm}$

Solution: Steps of construction:

Follow the same steps (i) and (ii) as in Example 5.

Case (a)

- (iii) Draw an arc of radius 3 cm with centre at point E which touches $\overrightarrow{DX}$ at point F .
- (iv) Join E with F . Here, $\overline{EF}$ will be perpendicular to $\overrightarrow{DX}$.



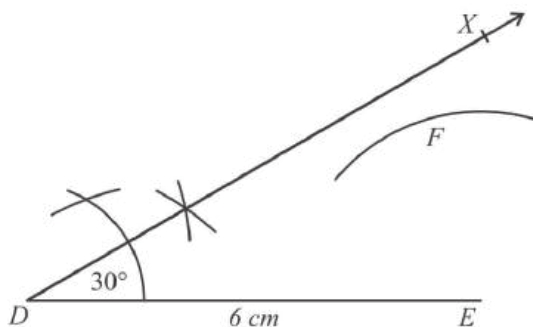
Hence, $\triangle DEF$ is the required triangle, which is a right angled triangle.

Case (b)

(iii) If we take $\overline{mEF} = 2.5\text{cm}$ less than 3cm and draw an arc of radius 2.5cm with centre at E .

(iv) This arc does not intersect $\overrightarrow{DX}$.

So, in this case, no triangle can be formed.

**Remember:**

We considered three cases when acute angle is given.

- If $\overline{mEF} > 3\text{ cm}$, two triangles are possible.
- If $\overline{mEF} = 3\text{ cm}$, only one triangle is possible.
- If $\overline{mEF} < 3\text{ cm}$, no triangle is possible.

11.2 Perpendicular Bisectors and Medians of a Triangle

Perpendicular Bisector: A perpendicular bisector is a line that intersects a line segment at right angle and dividing it into two equal parts. In other words, it intersects the line segment at its midpoint and forms right angles (90°) with it.

Median: A median of a triangle is a line segment that joins a vertex to the midpoint of the side that is opposite to that vertex.

Point of concurrency: A point of concurrency is the single point where three or more lines, rays or line segments intersect or meet in a geometric figure. This concept is commonly used in triangles, where several important types of points of concurrency exist.

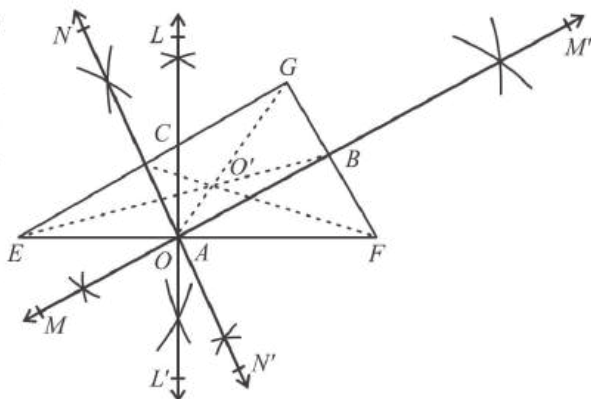
Example 7: Draw perpendicular bisector of the triangle EFG with $\overline{mEF} = 5\text{cm}$, $\overline{mFG} = 2.5\text{cm}$ and $\overline{mEG} = 4.3\text{ cm}$. Find the medians.

Solution: First we draw perpendicular bisectors and then medians.

Steps of construction:

- i. Draw $\triangle GEF$ as explained in the previous examples.
- ii. Draw two arcs above and below $\overline{EF}$ with more than half of $\overline{mEF}$ with centre at E .
- iii. Draw two arcs above and below $\overline{EF}$ with radius more than half of $\overline{mEF}$ with centre at F .

- iv. Draw a line through the points of intersection of the arcs in steps (ii) and (iii), we get the perpendicular bisector $\overleftrightarrow{LL'}$ of the side $\overline{EF}$ at A .
- v. Draw two more perpendicular bisectors $\overleftrightarrow{MM'}$ and $\overleftrightarrow{NN'}$ of the sides $\overline{FG}$ and $\overline{EG}$ at B and C respectively.



- vi. Join the point G with opposite midpoint A so $\overline{GA}$ is the median.
 - vii. Join the point F with opposite midpoint C , we get median $\overline{FC}$ and join point E with opposite midpoint B , we get median $\overline{EB}$.
- Hence, we see that the perpendicular bisector $\overleftrightarrow{LL'}$, $\overleftrightarrow{MM'}$ and $\overleftrightarrow{NN'}$ are concurrent at point O or A and the medians $\overline{GA}$, $\overline{EB}$ and $\overline{FC}$ are concurrent at point O' .

Circumcentre: The point of concurrency of perpendicular bisector of the sides of a triangle is called circumcentre.

Centroid: The point of concurrency of the medians of a triangle is called centroid of the triangle.

11.3 Angle Bisector of a Triangle

An angle bisector of a triangle is a line or ray that divides an angle into two equal parts, creating two smaller angles that are congruent (each having half the measure of the original angle).

Example 8: Draw angle bisector of a triangle FGH if:

$$m\overline{FG} = 5.2 \text{ cm}, m\overline{GH} = 4.1 \text{ cm and } m\angle FGH = 120^\circ$$

Solution: We first construct triangle FGH , then draw its angle bisector.

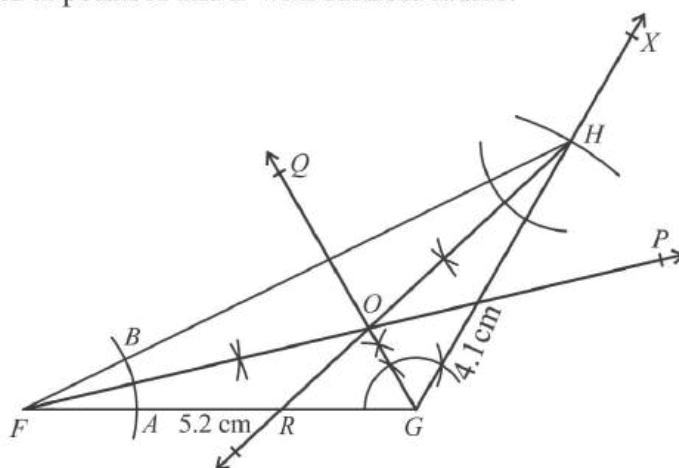
Steps of construction:

- (i) Construct $\triangle FGH$ with given lengths and angle.
- (ii) Draw an arc of suitable radius with centre at point F intersecting sides FG and FH at points A and B respectively.

(iii) Draw two arcs with centres at points A and B with suitable radius.

(iv) Draw a ray from F passing through the point of intersection of the arcs in step (iii). Which is the required angle bisector $\overrightarrow{FP}$ of the angle F .

(v) Draw two more angle bisectors $\overrightarrow{GQ}$ and $\overrightarrow{HR}$ of the angles G and H respectively.



We see that all the angle bisectors $\overrightarrow{FP}$, $\overrightarrow{GQ}$ and $\overrightarrow{HR}$ intersect at one point O . i.e, the angle bisectors of the triangle are concurrent.

Incentre: The point of concurrency of the angle bisectors of a triangle is called incentre of the triangle.

11.4 Altitudes of Triangle

Altitude is a ray drawn perpendicular from a vertex to the opposite side of the triangle. There are three altitudes of the triangle which meet at a single point i.e. the altitudes of a triangle are concurrent.

Orthocentre

The point of concurrency of the altitudes of the triangle is called orthocentre of the triangle.

Example 9:

Construct a triangle GHI in which $\overline{mGH} = 5.7$ cm, $m\angle G = 68^\circ$ and $m\angle H = 50^\circ$. Prove that altitudes of the $\triangle GHI$ are concurrent.

Solution:

First, we construct $\triangle GHI$ using the given measurements and then draw altitudes of the triangle.

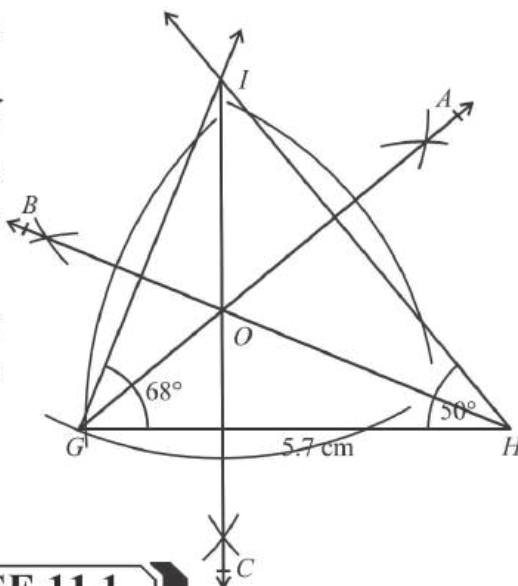
Steps of construction.

(i) Construct $\triangle GHI$ using the given measurements.

(ii) Draw perpendicular $\overrightarrow{GA}$ from G to the opposite side HI .

(iii) Draw two more perpendiculars $\overrightarrow{HB}$ and $\overrightarrow{IC}$. The first is from point H to the opposite side GI and the other is from point I to the opposite side GH .

So, $\overrightarrow{GA}$, $\overrightarrow{HB}$ and $\overrightarrow{IC}$ are the altitudes of $\triangle GHI$ and they intersect at one point O . i.e., the altitudes of $\triangle GHI$ are concurrent.



EXERCISE 11.1

- Construct $\triangle ABC$ with the given measurements and verify that the perpendicular bisectors of the triangle are concurrent.
 - $m\overline{AB} = 5$ cm, $m\overline{BC} = 6$ cm and $m\overline{AC} = 7$ cm
 - $m\overline{AB} = 7.1$ cm, $m\angle B = 135^\circ$ and $m\overline{BC} = 6.5$ cm
- Construct $\triangle LMN$ of the following measurements and verify that the medians of the triangle are concurrent.
 - $m\overline{LM} = 4.9$ cm, $m\angle L = 51^\circ$ and $m\angle M = 38^\circ$
 - $m\overline{MN} = 4.8$ cm, $m\angle N = 30^\circ$ and $m\overline{LM} = 8.1$ cm
- Verify that the angle bisectors of $\triangle ABC$ are concurrent with the following measurement:
 - $m\overline{AB} = 4.5$ cm, $m\angle A = 45^\circ$ and $m\overline{AC} = 5.3$ cm
 - $m\overline{AB} = 6$ cm, $m\angle A = 150^\circ$ and $m\angle B = 60^\circ$
- Given the measurements of $\triangle DEF$: $m\overline{DE} = 4.8$ cm, $m\overline{EF} = 4$ cm and $m\angle E = 45^\circ$, draw altitudes of $\triangle DEF$ and find orthocentre.
- Construct the following triangles and find whether there exists any ambiguous case.
 - $\triangle BCD$; $m\overline{BC} = 5$ cm, $m\angle B = 62^\circ$ and $m\overline{CD} = 4.7$ cm
 - $\triangle KLM$; $m\overline{LM} = 6$ cm, $m\angle M = 42^\circ$ and $m\overline{LN} = 5$ cm

11.5 Loci and Construction

A locus (plural loci) is a set of points that follow a given rule. In geometry, loci are often used to define the positions of points relative to one another or to other geometric figures. Some common types of loci along with detailed explanations will be discussed.

Do you know? In Latin, the word *locus* is defined by the English term, location.

Remember!

Equidistant: Let A be a fixed point and B be a set of points. If A is at equal distance from all points of B , then A is said to be equidistant from B .

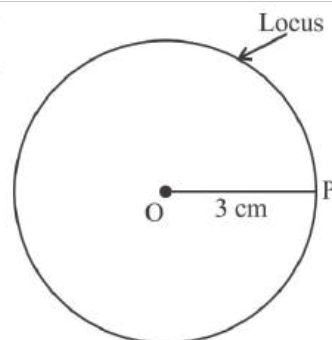
11.5.1 Loci in Two Dimensions

We study the loci (circle, parallel lines, perpendicular bisector and angle bisector) in two dimensions and apply them to real life situations.

Circle

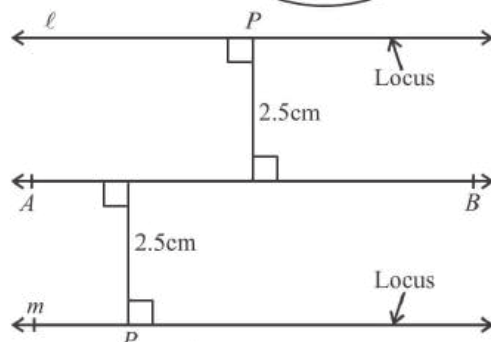
The locus of a point whose distance is constant from a fixed point is called a circle.

For example, the locus of a point P whose distance is 3 cm from a fixed-point O is a circle of radius 3 cm and centre at point O .

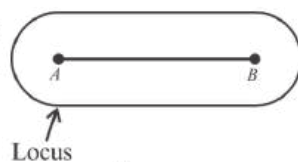


Parallel Lines

The locus of a point whose distance from a fixed line is constant are parallel lines, ℓ and m e. g. the locus of a point P whose distance is 2.5 cm from a fixed line AB are parallel lines at a distance of 2.5 cm from AB .



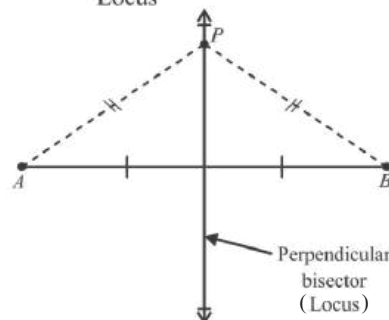
For example, a locus of points equidistant from a line segment creates a **sausage shape**. We can think of this type of locus as a track surrounding a line segment.



Perpendicular Bisector

The locus of a point whose distance from two fixed points is constant is called a perpendicular bisector.

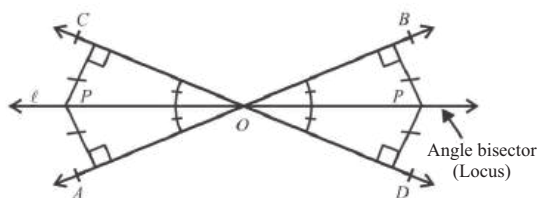
For example, the locus of a point P whose distance from fixed points A and B is constant is the perpendicular bisector of the line segment AB .



Angle Bisector

The locus of a point whose distance is constant from two intersecting lines is called an angle bisector.

For example, the locus of a point P whose distance is constant from two lines AB and CD intersecting at O is the angle bisector (ℓ) of $\angle AOC$ and $\angle BOD$.



Remember!

- Locus of points equidistant from a fixed point is a circle and equidistant from two fixed points is a perpendicular bisector.
- Locus of points equidistant from a fixed line are two parallel lines and equidistant from two fixed intersecting lines is angle bisector.

11.5.2 Intersection of Loci

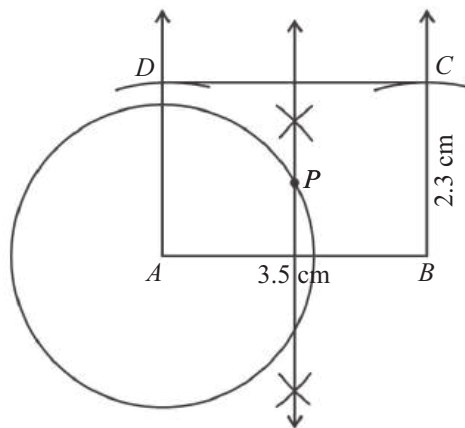
If two or more loci intersect at a point P , then P satisfies all given conditions of the loci. This will be explained in the following examples:

Example 10: Construct a rectangle $ABCD$ with $m\overline{AB} = 3.5$ cm and $m\overline{BC} = 2.3$ cm. Draw the locus of all points which are:

- at a distance of 2 cm from point A .
 - equidistant from A and B .
- Label the point P inside the rectangle which is 2 cm from point A and equidistant from A and B .

Solution: Construct rectangle $ABCD$ with the given lengths.

- Draw a circle of radius 2 cm with centre at A .
 - Draw perpendicular bisector of $\overline{AB}$.
- The two loci intersect at P inside the rectangle which is 2 cm from point A and equidistant from A and B .



Example 11: Construct an isosceles triangle DEF with vertical angle 80° at E and $m\overline{EF} = m\overline{DE} = 3.1$ cm. Draw the locus of all points which are:

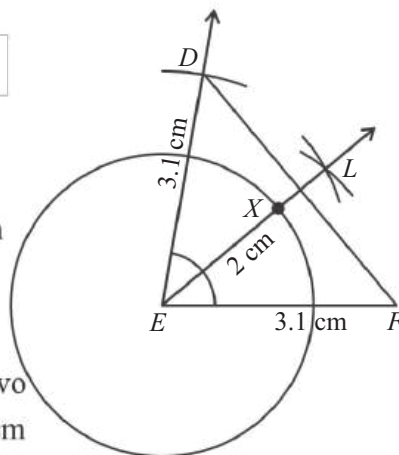
- at a distance of 2 cm from point E ,

- (ii) equidistant from $\overline{DE}$ and $\overline{EF}$.

Label the point X inside the triangle which is 2 cm from point E and equidistant from $\overline{ED}$ and $\overline{EF}$.

Solution: Construct triangle DEF with given measurements.

- Draw a circle of radius 2 cm with centre at E .
- Draw angle bisector $\overrightarrow{EL}$ of angle DEF . The two loci intersect at X inside the triangle which is 2 cm from point E and equidistant from $\overline{ED}$ and $\overline{EF}$.



Example 12: A field is in the form of a triangle LMN with $m\overline{LM} = 69$ m, $m\angle L = 60^\circ$ and $m\angle M = 45^\circ$.

- Construct $\triangle LMN$ with given measurements. [Scale: 10m = 1cm]
- Draw the locus of all points which are equidistant from L and M , equidistant from $\overline{LM}$ and $\overline{LN}$ and at a distance of 13 m from $\overline{LM}$ inside the triangular field.
- Two trees are to be planted at points P and Q inside the field.

- Mark the position of point P which is equidistant from L and M and equidistant from $\overline{LM}$ and $\overline{LN}$.
- Mark the position of point Q which is equidistant from $\overline{LM}$ and $\overline{LN}$ and 13 m from $\overline{LM}$.
- Find $m\angle PQ$.

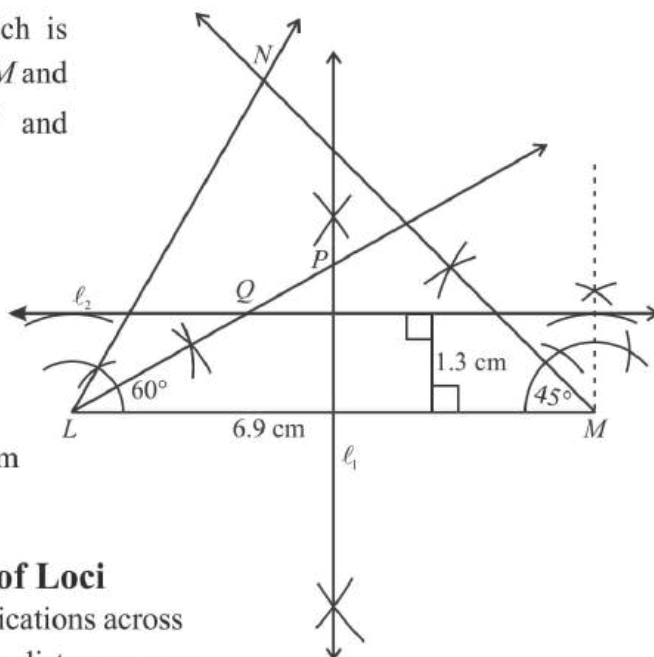
Solution:

- Construct triangle LMN with given measurements using a scale of 10 m to represent 1 cm.
- Draw perpendicular bisector ℓ_1 of $\overline{LM}$. Draw angle bisector of angle MLN . Draw a parallel line ℓ_2 inside the triangle LMN , 1.3 cm from $\overline{LM}$.

- (iii) (a) Label the point P which is equidistant from L and M and equidistant from $\overline{LM}$ and $\overline{LN}$.

- (b) Label the point Q which is equidistant from $\overline{LM}$ and $\overline{LN}$ and 1.3 cm from $\overline{LM}$.

- (c) $m\overline{PQ} = 1.3 \times 10 = 13$ m



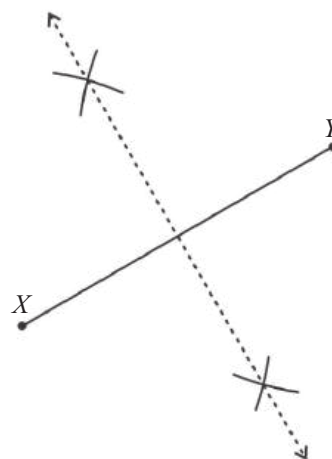
11. 6 Real Life Application of Loci

The concept of loci has many applications across fields where spatial relationships, distances, or specific constraints are important. Here, are detailed examples illustrating the use of loci in different contexts.

- (i) A park has two water sources at two different points. A fire hydrant needs to be placed so it is equally accessible to both sources.

Let X and Y represent the two water sources in the park. Draw the perpendicular bisector of $\overline{XY}$ which represents the locus of all points equidistant from X and Y .

- (ii) A robotic arm in a factory is programmed to work within a specific area without crossing into areas where it could interfere with other equipment. The loci of the robot's possible positions would be a defined space, such as a circle or rectangular region, ensuring it operates safely within its designated zone.



EXERCISE 11.2

- Two points A and B are 8.2 cm apart. Construct the locus of points 5 cm from point A .
- Construct a locus of point 2.2 cm from line segment CD of measure 5.7 cm.
- Construct an angle $ABC = 105^\circ$. Construct a locus of a point P which moves such that it is equidistant from $\overline{BA}$ and $\overline{BC}$.
- Two points E and F are 5.4 cm apart. Construct a locus of a point P which moves such that it is equidistant from E and F .
- The island has two main cities A and B 8 km apart. Kashif lives on the island exactly 6.8 km from city A and exactly 7.3 km from city B . Mark with a cross the points on the island where Kashif could live.
- Construct a triangle CDE with $m\overline{CD} = 7.6$ cm, $m\angle D = 45^\circ$ and $m\overline{DE} = 5.9$ cm. Draw the locus of all points which are:
 - equidistant from C and D
 - equidistant from $\overline{CD}$ and $\overline{CE}$
 - mark the point X where the two loci intersect.
- Construct a triangle LMN with $m\overline{LM} = 7$ cm, $m\angle L = 70^\circ$ and $m\angle M = 45^\circ$. Find a point within the triangle LMN which is equidistant from L and M and 3 cm from L .
- Construct a right angled triangle RST with $m\overline{RS} = 6.8$ cm, $m\angle S = 90^\circ$ and $m\overline{ST} = 7.5$ cm. Find a point within the triangle RST which is equidistant from $\overline{RS}$ and $\overline{RT}$ and 4.5 cm from R .
- Construct a rectangle $UVWX$ with $m\overline{UV} = 7.2$ cm and $m\overline{VW} = 5.6$ cm. Draw the locus of points at a distance of 2 cm from $\overline{UV}$ and 3.5 cm from W .
- Imagine two cell towers located at points A and B on a coordinate plane. The GPS-enabled device, positioned somewhere on the plane, receives signals from both towers. To ensure accurate navigation, the device is placed equidistant from both towers to estimate its position. Draw this locus of navigation.
- Epidemiologists use loci to determine infection zones, especially for contagious diseases, to predict the spread and take containment measures. In the case of a disease outbreak, authorities might determine a quarantine zone within 10 km

- of the infection source. Draw the locus of all points 10 km from the source defining the quarantine area to monitor and control the disease's spread.
12. There is a treasure buried somewhere on the island. The treasure is 24 kilometres from A and equidistant from B and C . Using a scale of 1cm to represent 10 km, find where the treasure could be buried.
13. There is an apple tree at a distance of 90 metres from banana tree in the garden of Sara's house. Sara wants to plant a mango tree M which is 64 metres from apple tree and between 54 and 82 metres from the banana tree. Using a scale of 1cm to represent 10m, Find the points where the mango tree should be planted.

REVIEW EXERCISE 11

1. Four options are given against each statement. Encircle the correct option.
- (i) A triangle can be constructed if the sum of the measure of any two sides is _____ the measure of the third side.
- (a) less than (b) greater than
(c) equal to (d) greater than and equal to
- (ii) An equilateral triangle _____.
- (a) can be isosceles (b) can be right angled
(c) can be obtuse angled (d) has each angle equal to 50° .
- (iii) If the sum of the measures of two angles is less than 90° , then the triangle is _____.
- (a) equilateral (b) acute angled
(c) obtuse angled (d) right angled
- (iv) The line segment joining the midpoint of a side to its opposite vertex in a triangle is called _____.
- (a) median (b) perpendicular bisector
(c) angle bisector (d) circle
- (v) The angle bisectors of a triangle intersect at _____.
- (a) one point (b) two points
(c) three points (d) four points
- (vi) Locus of all points equidistant from a fixed point is _____.
- (a) circle (b) perpendicular bisector
(c) angle bisector (d) parallel lines

- (vii) Locus of points equidistant from two fixed points is -----
 (a) circle (b) perpendicular bisector
 (c) angle bisector (d) parallel lines
- (viii) Locus of points equidistant from a fixed line is/are -----
 (a) circle (b) perpendicular bisector
 (c) angle bisector (d) parallel lines
- (ix) Locus of points equidistant from two intersecting lines is _____.
 (a) circle (b) perpendicular bisector
 (c) angle bisector (d) parallel lines
- (x) The set of all points which is farther than 2 km from a fixed point B is a region outside a circle of radius _____ and centre at B .
 (a) 1 km (b) 1.9 km
 (c) 2 km (d) 2.1 km
2. Construct a right angled triangle with measures of sides 6 cm, 8 cm and 10 cm.
3. Construct a triangle ABC with $m\overline{AB} = 5.3$ cm, $m\angle A = 30^\circ$ and $m\angle B = 120^\circ$. Draw the locus of all points which are equidistant from A and B .
4. Construct a triangle with $m\overline{DE} = 7.3$ cm, $m\angle D = 42^\circ$ and $m\overline{EF} = 5.4$ cm.
5. Construct a triangle XYZ with $m\overline{YX} = 8$ cm, $m\overline{YZ} = 7$ cm and $m\overline{XZ} = 6.5$ cm. Draw the locus of all points which are equidistant from $\overline{XY}$ and $\overline{XZ}$.
6. Construct a triangle FGH such that $m\overline{FG} = m\overline{GH} = 6.4$ cm, $m\angle G = 122^\circ$. Draw the locus of all points which are:
 (a) equidistant from F and G ,
 (b) equidistant from $\overline{FG}$ and $\overline{GH}$.
 (c) Mark the point where the two loci intersect.
7. Two houses Q and R are 73 metres apart. Using a scale of 1 cm to represent 10 m, construct the locus of a point P which moves such that it is:
 (i) at a distance of 32 metres from Q
 (ii) at a distance of 48 metres from the line joining Q and R .
8. The field is in the form of a rectangle $ABCD$ with $m\overline{AB} = 70$ m and $m\overline{BC} = 60$ m. Construct the rectangle $ABCD$ using a scale of 1cm to represent 10 m. Show the region inside the field which is less than 30 m from C and farther than 25 m from $\overline{AB}$.

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Construct a grouped frequency table, histogram (with unequal class interval) and frequency polygon.
- Calculate the mean, modal class and median of a grouped frequency distribution.
- Solve real life situations involving mean, weighted mean, median and mode for given data (such as allocation of funds in different projects, forecasting future demographics, marketing, forecasting government budgets).

INTRODUCTION

Before knowing about information let us think how can we answer the question like how many students were there in each class of a particular school.

How many patients visited in a hospital within a particular week.

To answer these questions, we need to have quantitative information and it can be obtained by counting.

It should be kept in mind that simple numbers 85, 96, 70, 80, 73, 70, 65, 83, 89, 75 are not data but if we say that the above data indicates the marks of students of different classes, the above figures are considered as data and have precise meaning.

Hence, to know about something is known as “Information” and to represent that information in a manageable way so that useful conclusions can be drawn is called information handling. So, the

collection of meaningful information in the form of facts and numerical figures is known as data.

The numerical figures are obtained from any field of study e. g., the mass of the students of your class, the number of pair of shoes sold by a shopkeeper in a month etc. Data can be obtained from existing sources i.e., office records, published papers or the same can be obtained directly from the field according to needs.

History

In statistics, information handling is also known as data handling. “Data Handling” plays vital role to represent the information in a manageable way.

The word “Data Handling” was first used by Sir Ronald Fisher.



Sir Ronald Aylmer Fisher
(17 February 1890 – 29 July 1962)

Information Handling

Information handling is the process of collecting, organizing, summarizing, analyzing and interpreting numerical data.

Data is further classified into two categories.

- (i) **Discrete data:** It can take only some specific values. whole numbers are used to write discrete data. e.g., number of books sold by a shopkeeper, number of patients visited a hospital in a week etc. This data is only obtained by counting.
- (ii) **Continuous data:** It can take every possible value in a given interval. Decimal numbers are used to write continuous data. The data is only obtained by measuring e.g., the mass of students in class i.e., 28.5 kg, 26.5 kg, 27.5 kg etc.

12.1 Ungrouped and Grouped Data

Data which is not arranged in any systematic order (groups or classes) is called ungrouped data. For example, the number of toys sold by a shopkeeper in a month is given below:

10, 5, 8, 12, 15, 20, 25, 30, 23, 15, 23, 21, 18, 15, 17, 23, 22, 15, 20, 21, 24, 18, 16, 21, 23, 21, 17, 19, 21, 23. This data is called ungrouped data.

If we arrange the above given data in groups or classes, then it is called grouped data.

Do you know?

Ungrouped data is also known as raw data.

Classes	Tally marks	No. of toys sold
5 – 9		2
10 – 14		2
15 – 19		10
20 – 24		14
25 – 29		1
30 – 34		1

Teachers' note!

By using more examples, clear the concept of grouped data and ungrouped data to the students.

In above grouped data, 5, 10, 15, 20, 25 and 30 are lower class limits and 9, 14, 19, 24, 29 and 34 are upper class limits.

12.1.1 Frequency Distribution

A distribution or table that represents classes or groups along with their respective class frequencies is called frequency distribution. In other words, the various items of data

are classified into certain groups or classes and the number of items lying in each group or class is put against that group or class. The data organised and summarized in this way is known as frequency distribution.

Think!

If the size of class limits is 6. The greatest value is 80 and the smallest value is 25. Can you find the number of class limits for the data?

Formation of Frequency distribution

In this method, the raw data or the ungrouped data is presented into a grouped data. Choice is yours to select the number of classes.

Generally, the size of class limits is determined on the basis of the greatest value, smallest value and the desired number of groups or classes.

Following are the major steps to construct frequency distribution:

- (i) Find the range of the data. Range is the difference between the greatest value and the smallest value i.e., $\text{Range} = X_{\max} - X_{\min}$
- (ii) Find the size of the class by dividing the range by the number of classes or groups you wish to make.

Keep in mind!

The number of times a value occurs in a data is called the frequency of that value. It is denoted by “ f ”.

For example, the greatest value is 136, the smallest value is 30 and if we have to make 10 classes or groups, then the size of class limits is found by the given formula.

$$\begin{aligned}\text{Size of class} &= \frac{\text{Range}}{\text{Number of classes}} = \frac{\text{Greatest Value} - \text{Smallest Value}}{\text{Number of classes}} \\ &= \frac{136 - 30}{10} = \frac{106}{10} = 10.6 \approx 11\end{aligned}$$

So, size of class limits = 11

- (iii) Prepare four columns.

- | | |
|------------------|----------------------|
| (a) Class limits | (b) Tally marks |
| (c) Frequencies | (d) Class Boundaries |

- (iv) Make classes having size of 11. Start from the smallest value.

For example, 30 – 40, 41 – 51, 52 – 62 and so on.

- (v) Look for the class in which each element of ungrouped data falls. Draw a small tally mark (|) against that class and also tick the element concerned with a sign (✓). In this way you can remember that you have counted for the element. Continue this way with the next element that upto the last element of the data set. If 5 or more tallies appear in any class, mark every 5th tally diagonally as |||.

(vi) Class boundaries usually are found by the following method:

- Chose the upper class limit of the 1st class and lower class limit of the 2nd class.
- Find the difference between these two limits.
- The difference is divided by 2 and subtract it from the lower class limit and add it to the upper class limit.

Do you know?

Class boundaries may also be obtained from the midpoints (x) as $\left[x \pm \frac{h}{2} \right]$, where h is the difference between any two consecutive values of x .

Example 1: Following are the number of telephone calls made in a week to 30 teachers of a high school.

5 8 11 25 13 16 20 17 15 16 30 21 14 18 19
6 22 26 15 19 35 29 31 23 25 20 10 9 7 26

Construct a frequency distribution with number of classes 7.

Solution: (i) Find range

Greatest value (maximum value) = 35, Smallest value (minimum value) = 5

$$\text{Range} = X_{\max} - X_{\min} = 35 - 5 = 30$$

(ii) Size of class limits = $\frac{\text{Range}}{\text{Number of classes}} = \frac{30}{7} = 4.28 \approx 5$

(iii) Make class limits having size 5. For example, 5 – 9, 10 – 14, 15 – 19 and so on. (see 1st column of table: 1).

(iv) Tally marks are used to count the values, fall in the given class limits. (See 2nd column of table: 1).

(v) Now, count the number of tally marks and write the number as frequency in the third column (see 3rd column of table: 1).

(vi) **Class boundaries**

The difference between lower class limit of the second class and upper class limit of the first class is 1. i.e., $10 - 9 = 1$. Now, divide the difference of the limits by 2 i.e. $\frac{1}{2} = 0.5$.

Activity

Collect data of height of 50 students in your class, and convert the data into grouped data.

Lower class boundaries are obtained by “subtracting 0.5” from the lower class limits.

Upper class boundaries are obtained by “adding 0.5” to the upper class limits.

Lower class boundaries

$$5 - 0.5$$

4.5 and so on.

(see 4th column of the table: 1)

Upper class boundaries

$$9 + 0.5$$

9.5 and so on.

Table 1

Class limits	Tally marks	Frequency (f)	Class Boundaries (C.B)
5 – 9		5	4.5 – 9.5
10 – 14		4	9.5 – 14.5
15 – 19		8	14.5 – 19.5
20 – 24		5	19.5 – 24.5
25 – 29		5	24.5 – 29.5
30 – 34		2	29.5 – 34.5
35 – 39		1	34.5 – 39.5

12.1.2 Graph of Frequency Distribution

The following are the types of graphs which can be used to represent a frequency distribution on a graph.

- (a) Histogram (b) Frequency polygon

(a) Histogram (with equal class limits)

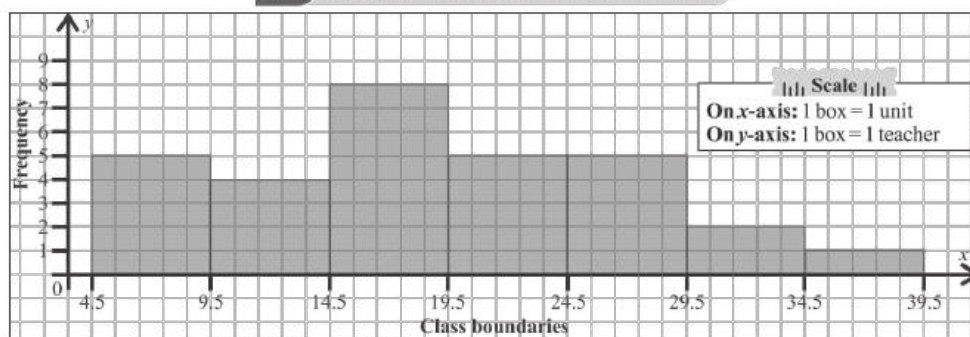
This is a graph of adjacent rectangles constructed on xy plane. A histogram is similar to bar graph but it is constructed for a frequency distribution. In a histogram, the values of the data (classes) are represented along the horizontal axis and the frequencies are shown by bars perpendicular to the horizontal axis. Bars of equal width are used to represent individual classes of frequency table. The procedure for making histogram is explained below:

Do you know?

Continuous data is mostly represented by using histogram and frequency polygon.

- Draw lines as x – axis and as y – axis on a graph paper perpendicular to each other.
- Class boundaries are marked on x – axis and a rectangle is made against each group with its width proportional to the size of the class limits and height proportional to the class frequencies.
- Setting a scale, draw frequencies on y – axis. The resulting figure is called histogram. Histogram of table: 1 is given below:

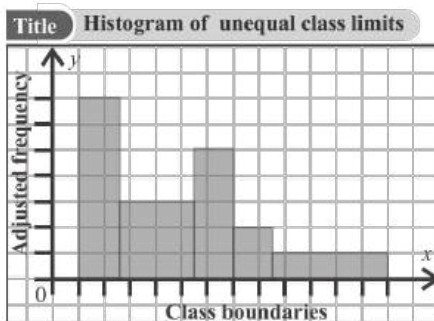
Title Histogram of telephone calls made in a week



12.1.3 Histogram (with unequal class limits)

The procedure for making histogram is explained below:

- Draw lines as x – axis and y – axis on a graph paper perpendicular to each other.
- Class boundaries are marked on x – axis and a rectangle is made against each group with its width proportional to the size of class limits and height proportional to the class frequencies.
- This can be achieved by adjusting the heights of rectangle. The height of each rectangle is obtained by dividing each class frequency on its class limit size.



Example 2: The frequency distribution of ages (in years) of 76 members of a locality is available. Draw a histogram for this data.

Class limits	2 – 4	4 – 9	9 – 12	12 – 17	17 – 20	20 – 27	27 – 30
Frequency (f)	7	10	18	20	10	7	4

Solution: Look at the table, indicates that the width of the class limits is not equal as first class has width 2, second has 5, the third has 3, the fourth has 5, the fifth has 3, sixth class has 7, seventh class has width 3. So, there is need to adjust the heights of the

rectangles i.e., for the first class we have 2 as width of class and 7 as a frequency, so the height of the first class is $\frac{7}{2} = 3.5$, similarly

for the other $\frac{10}{5} = 2$, $\frac{18}{3} = 6$,

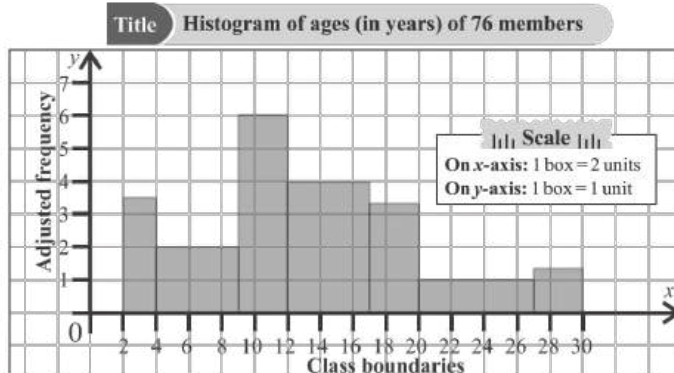
$\frac{20}{5} = 4$, $\frac{10}{3} = 3.3$, $\frac{7}{7} = 1$,

$\frac{4}{3} = 1.3$.

These proportional heights are also called adjusted frequencies.

Class limits	Frequency (f)	Width of Class	Height of rectangle (Adjusted frequency)
2 – 4	7	$4 - 2 = 2$	$\frac{7}{2} = 3.5$
4 – 9	10	$9 - 4 = 5$	$\frac{10}{5} = 2$
9 – 12	18	$12 - 9 = 3$	$\frac{18}{3} = 6$
12 – 17	20	$17 - 12 = 5$	$\frac{20}{5} = 4$
17 – 20	10	$20 - 17 = 3$	$\frac{10}{3} = 3.3$
20 – 27	7	$27 - 20 = 7$	$\frac{7}{7} = 1$
27 – 30	4	$30 - 27 = 3$	$\frac{4}{3} = 1.3$

Taking class boundaries along x – axis and corresponding adjusted frequencies along y – axis, rectangles are drawn and the histogram is given below.



12.1.4 Frequency Polygon

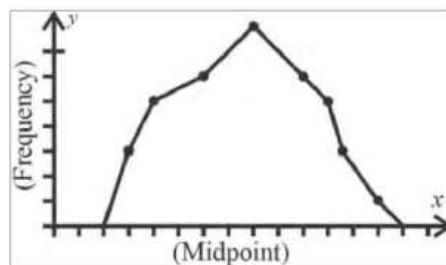
A frequency polygon is a closed geometrical figure used to display a frequency distribution graphically. A line graph of a frequency distribution is known as frequency polygon in which frequencies are plotted against their midpoints.

Midpoint is the average value of the lower and upper class limits. Midpoint is also known as class mark. Midpoint is calculated by the given formula:

$$\text{Midpoint} = \frac{\text{Lower class limit} + \text{Upper class limit}}{2}$$

The following steps are followed to draw a frequency polygon for a frequency distribution:

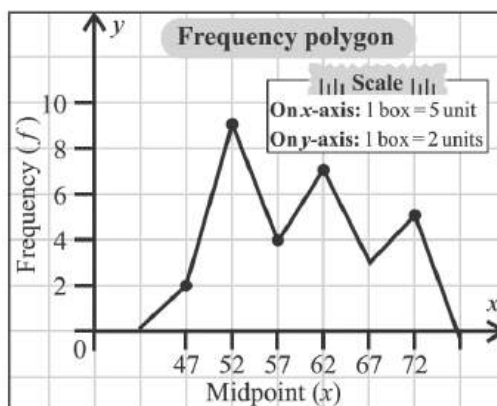
- Draw lines as x – axis and y – axis perpendicular to each other.
- Take midpoints on x – axis and class frequencies on y – axis.
- Put a dot mark against each midpoint corresponding to its class frequency. Join all the dotted marks by straight lines to get the required frequency polygon.
- The lines at both ends are joined together with the next midpoints to touch the bases of x – axis.



Example 3: The following are the marks obtained by 30 students out of 100 in the subject of Mathematics at their final examination. Construct frequency polygon for the following frequency table.

Marks	45 – 49	50 – 54	55 – 59	60 – 64	65 – 69	70 – 74
Frequency	2	9	4	7	3	5

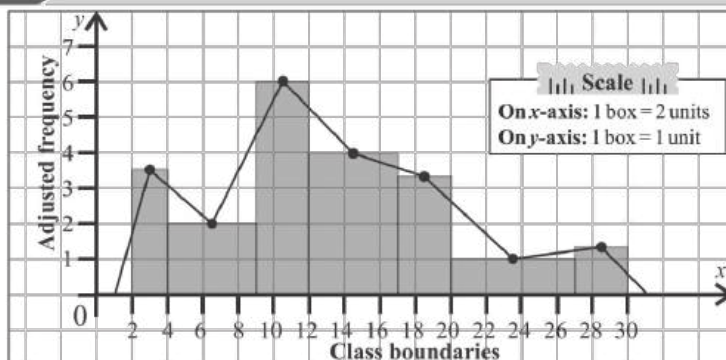
Solution:	Marks	f	Midpoints
	45 – 49	2	$\frac{45 + 49}{2} = 47$
	50 – 54	9	$\frac{50 + 54}{2} = 52$
	55 – 59	4	$\frac{55 + 59}{2} = 57$
	60 – 64	7	$\frac{60 + 64}{2} = 62$
	65 – 69	3	$\frac{65 + 69}{2} = 67$
	70 – 74	5	$\frac{70 + 74}{2} = 72$



Remember!

Frequency polygon on histogram: In histogram, we mark the midpoints on the top of rectangles and join all the points. To touch the base of x -axis, we extend the line at both ends to the next midpoints. The resulting (Example 2) graph is a frequency polygon.

Title Frequency Polygon on Histogram of ages (in years) of 76 members



EXERCISE 12.1

- The following distribution represents the scores achieved by a group of chemistry students in the chemistry laboratory.

Scores	24 – 28	29 – 33	34 – 38	39 – 43	44 – 48	49 – 53	Total
No. of students	3	6	12	23	15	6	65

Answer the following questions.

- What is the upper limit of the last class?
- What is the lower limit of the class 39 – 43?

- (iii) What is the midpoint of the class (34 – 38)?
 - (iv) What are the class frequencies of the classes 29 – 33 and 44 – 48?
 - (v) What is the size of the class limits in the above frequency distribution?
 - (vi) In which class or group does minimum number of students fall?
 - (vii) What is the lower limit of the class having 15 as its class frequency?
 - (viii) What is the number of students having scores between 24 and 43?
2. For a school staff, the following expenditures (rupees in hundred) are required for the repair of chairs.

145, 152, 153, 156, 158, 160, 146, 152, 155, 159,
 161, 163, 165, 147, 148, 151, 154, 156, 158, 160,
 144, 167, 151, 150, 152, 149, 145, 153, 152, 155

Prepare a frequency distribution by tally bar method by taking number of classes 8, also write down what are the frequencies of the last three classes?

3. Given below are the weights in kg of 30 students of a high school.

30, 33, 24, 21, 15, 39, 37, 44, 42, 33,
 33, 28, 29, 32, 31, 28, 26, 32, 34, 35,
 38, 36, 41, 30, 35, 41, 23, 26, 18, 34

Taking 6 as the number of classes, prepare a frequency table and construct a frequency polygon.

4. A group of Grade - 10 students obtained the following marks out of 100 marks in English test.

58, 59, 58, 33, 40, 58, 45, 46, 43, 45, 45,
 50, 52, 49, 50, 57, 52, 55, 49, 50, 62, 49,
 48, 44, 42, 47, 46, 47, 46, 53, 40, 44

Classify the data into a frequency distribution by (direct method) taking 5 as the number of classes. Also find the class limit with least class frequency and construct histogram for the data.

5. From the table given below. Draw a frequency polygon on histogram for the given frequency distribution.

Weight (kg)	10 – 14	15 – 19	20 – 24	25 – 29	30 – 34	35 – 39
Frequency (f)	06	17	23	30	22	13

6. The following data shows the number of heads in an experiment of 50 sets of

tossing a coin 5 times. Make a discrete frequency distribution from the information.

3, 3, 4, 0, 5, 4, 3, 3, 1, 2, 4, 5, 0, 3, 2, 4, 4, 0, 0, 0, 5, 5, 3, 2, 1
4, 3, 2, 5, 3, 2, 1, 3, 5, 4, 3, 2, 1, 3, 2, 1, 3, 1, 3, 1, 4, 3, 2, 2, 4

7. The marks obtained by the students of Grade - 10 in mathematics test were grouped into the following frequency distribution.

Marks	35 – 37	38 – 44	45 – 54	55 – 61	62 – 67	68 – 72
Frequency	2	12	16	13	9	3

Draw a histogram for the above distribution.

8. Make a frequency polygon on histogram for the following grouped data:

Class limits	5 – 8	8 – 12	12 – 20	20 – 25	25 – 27	27 – 32
Frequency (f)	2	12	25	32	14	5

12.2 Measures of Location (Central Tendency)

The measure that gives the centre of the data is called measure of central tendency. Therefore, measure of central tendency is used to find out the middle or central value of a data set.

We have seen that when the raw data has been condensed into a frequency distribution, the information was easily understood. The information given in the data can be further condensed to a single representative value for the entire distribution. It is more or less the central value around which the data appear to be crowded. For example, usually, we make statements such as:

- Hassan studies 6 hours daily.
- The monthly expenditure of Ayesha's house is Rs.50,000.
- The speed of Maham's car is 72 km per hour.
- In a country, yearly income is 70,000 rupees per head.
- The price of onion in the market is Rs.150 per kg etc.

If we look at the first statement, we come to know that Hassan does not study exactly 6 hours daily. Sometimes, he studies more than 6 hours and sometimes less. But still why do we say that he studies 6 hours daily? As he studies near about 6 hours daily so in his study time, 6 hours becomes an important figure because of its approximated statement, which we call Average. Such an average value is known a measure of central tendency because it is a representative value of the daily study time. Similarly, other statements can also be treated as representative values. As each statement locates the centre of a distribution so it is also known as a measure of central tendency.

The following measure of central tendency will be discussed in this section:

- | | |
|----------------------------|--------------------|
| (i) Arithmetic Mean (A.M.) | (ii) Median |
| (iii) Mode | (iv) Weighted mean |

12.2.1 Arithmetic Mean (A.M.)

It is defined as a value of variable which is obtained by dividing the sum of all the values (observations) by their number of observations. Thus, the arithmetic mean of a set of values $x_1, x_2, x_3, \dots, x_n$ is denoted by $\bar{X}$ (read as X -bar) and is calculated as:

$$\bar{X} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum x}{n} \text{ (Direct method)}$$

where, the sign Σ stands for the sum and n is the number of observations.

Example 4: The marks of a student in five examinations were 64, 75, 81, 87, 90. Find the arithmetic mean of the marks.

Solution:

$$\begin{aligned} \text{A.M.} = \bar{X} &= \frac{\sum x}{n} \\ &= \frac{64 + 75 + 81 + 87 + 90}{5} \\ \text{or } \bar{X} &= \frac{397}{5} = 79.4 \text{ marks} \end{aligned}$$

Try Yourself !

The mean of 10, 30, 40, x , 67 and 81 is 50. Find the value of the x .

Example 5: A government allocates funds of Rs.200,000 to five sectors of a school i.e.,

- School Library: Rs. 35,000
- Sports facilities: Rs. 25,000
- Parking area: Rs. 40,000
- Room renovation: Rs. 45,000
- Furniture: Rs. 55,000

Try Yourself !

The mean of 15 values was 50. It was found on rechecking that the value 25 was wrongly copied as 52. Find the correct mean.

Find the average of fund allocation in each sector of a school.

Solution: To find out the average of each sector, we will find the mean of the given data.

$$\begin{aligned} \bar{X} &= \frac{35,000 + 25,000 + 40,000 + 45,000 + 55,000}{5} \\ \bar{X} &= \frac{200,000}{5} \\ \bar{X} &= \text{Rs. } 40,000 \end{aligned}$$

On average, each sector takes Rs.40,000 in funding.

Method of finding Arithmetic Mean for Grouped Data

Let $x_1, x_2, x_3, \dots, x_n$ be the midpoints of the class limits with corresponding frequencies say $f_1, f_2, f_3, \dots, f_n$. Then the arithmetic mean is obtained by dividing sum of the products of f and x by the sum of all the frequencies.

$$\bar{X} = \frac{f_1x_1 + f_2x_2 + \dots + f_nx_n}{f_1 + f_2 + \dots + f_n} = \frac{\Sigma fx}{\Sigma f}$$

Example 6: Given below are the marks out of 100 obtained by 100 students in a examination. Find the average marks of the students.

Marks	30 – 35	35 – 40	40 – 45	45 – 50	50 – 55	55 – 60
No. of students	14	16	18	23	18	11

Solution:

Marks	Midpoint (x)	Frequency (f)	fx
30 – 35	32.5	14	455.0
35 – 40	37.5	16	600.0
40 – 45	42.5	18	765.0
45 – 50	47.5	23	1092.5
50 – 55	52.5	18	945.0
55 – 60	57.5	11	632.5
Total	—	$\Sigma f = 100$	$\Sigma fx = 4490$

$$\bar{X} = \frac{\Sigma fx}{\Sigma f} = \frac{4490}{100}$$

or $\bar{X} = 44.9$ marks

Hence, the average marks is 44.9 of the students.

Short Formula for Computing Arithmetic Mean

The computation of arithmetic mean using direct method for ungrouped data as well as for grouped data is no doubt easy for small values. If x and f become very large, it becomes difficult to deal with the problems so to minimize our time and calculations we take deviations from an assumed or provisional mean. Let A be considered as assumed or provisional mean (may be any value from the values of x or any number) and D denotes the deviations of x from A i.e., $D = x - A$. For $x = D + A$, the formula of

arithmetic mean becomes;

$$\bar{X} = A + \frac{\Sigma D}{n} \quad (\text{for ungrouped data}) \quad \dots(i)$$

$$\bar{X} = A + \frac{\Sigma fD}{\Sigma f} \quad (\text{for grouped data}) \quad \dots(ii)$$

Example 7: Find the arithmetic mean using short formula for the runs made by a batsman.

Runs: 40, 45, 50, 52, 50, 60, 56, 70.

Solution: Taking deviations from $A = 52$ (assumed mean)

Try Yourself !

If $\bar{X} = 120$; $A = 85$ and $n = 25$, then can you find the value of ΣD ?

x	40	45	50	52	50	60	56	70
$D = x - A$	-12	-7	-2	0	-2	8	4	18

Now: $\Sigma D = -23 + 30 = 7$

$$\therefore \bar{X} = A + \frac{\Sigma D}{n}$$

$$\text{So, } \bar{X} = 52 + \frac{7}{8}$$

$$= 52 + 0.875 = 52.88 \text{ or } 53 \text{ runs.}$$

Example 8: Deviations from 12.5 of ten different values are 6, -2, 3.5, 9, 8.7, -5.5, 14, 11.3, -6.8, -4.2, find the arithmetic mean.

Solution: Deviations from 12.5 are:

6, -2, 3.5, 9, 8.7, -5.5, 14, 11.3, -6.8, -4.2

Now, $\Sigma D = 34$. Also, $A = 12.5$, using the formula we have.

$$\begin{aligned} \bar{X} &= A + \frac{\Sigma D}{n} \\ &= 12.5 + \frac{34}{10} \end{aligned}$$

$$\text{or } \bar{X} = 12.5 + 3.4 = 15.9$$

Example 9: The heights (in inches) of 200 students are recorded in the following frequency distribution. Find the mean height of the student by short formula.

Height (x) (in inches)	51	52	53	54	55	56	57	58	59	60
Frequency (f)	2	5	8	24	55	45	38	16	6	1

Solution:

Heights (x) (in inches)	Frequency (f)	$A = 55$ $D = x - A$	fD
51	2	-4	-8
52	5	-3	-15
53	8	-2	-16
54	24	-1	-24
$A \leftarrow 55$	55	0	0
56	45	1	45
57	38	2	76
58	16	3	48
59	6	4	24
60	1	5	5
Total	$\Sigma f = 200$	$\Sigma fD = 135$	

Now, using the formula (ii), we get

$$\bar{X} = A + \frac{\Sigma fD}{\Sigma f}$$

$$\bar{X} = 55 + \frac{135}{200}$$

or $\bar{X} = 55 + 0.675$

$\therefore \bar{X} = 55.68$ inches approx.

Hence, the mean height of the students is 55.68 inches.

Example 10: Ten students each from Grade-V section A and B of a well reputed school were taken randomly. Their weights were measured in kg. and recorded as given below:

Weights (kg) Section A	30	28	32	29.5	35	34	31	33	40	37.5
Weights (kg) Section B	35	31.5	34.5	35	32.8	38	29.5	36	36.5	34

- Compute the mean weight for section A and B.
- Conclude which section is better on Average?

Solution: (i) We find arithmetic mean for both the sections by direct method. (Any method can be applied).

As number of observations $n = 10$

$$\text{and } \bar{X}_{(A)} = \frac{\Sigma X_{(A)}}{n}$$

$$\therefore \bar{X}_{(A)} = \frac{330}{10} = 33 \text{ kg}$$

$$\text{and } \bar{X}_{(B)} = \frac{\Sigma X_{(B)}}{n}$$

$$\therefore \bar{X}_{(B)} = \frac{342.8}{10} = 34.28 \text{ kg}$$

$X_{(A)}$	$X_{(B)}$
30	35
28	31.5
32	34.5
29.5	35
35	32.8
34	38
31	29.5
33	36
40	36.5
37.5	34
$\Sigma X_{(A)} = 330$	$\Sigma X_{(B)} = 342.8$

(ii) We have seen from the results that

$\bar{X}_{(B)}$ is greater than $\bar{X}_{(A)}$. Therefore, we conclude that section B is better on the average.

12.2.2 Median

Median is the middle most value in an arranged (ascending or descending order) data set. Median is the value which divides the data into two equal parts i.e., 50% data is before the median and 50% data after it. Median is denoted by $\tilde{X}$.

Median for ungrouped data

The median of n observations $x_1, x_2, \dots, x_n$ is obtained as:

$$\text{Median } (\tilde{X}) = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ observation} \quad \left(\begin{array}{l} \text{when } n \text{ is} \\ \text{odd number} \end{array} \right)$$

$$\text{Median } (\tilde{X}) = \frac{1}{2} \left(\left(\frac{n}{2} \right)^{\text{th}} \text{ observation} + \left(\frac{n+2}{2} \right)^{\text{th}} \text{ observation} \right) \quad \left(\begin{array}{l} \text{when } n \text{ is} \\ \text{even number} \end{array} \right)$$

Example 11: The following are the scores made by a batsman. Find the median of the data. 8, 12, 18, 13, 16, 5, 20.

Solution: Writing the scores in an ascending order, we have

5, 8, 12, 13, 16, 18, 20

Since, number of observations is odd i.e., $n = 7$

$$\begin{aligned} \text{Median } (\tilde{X}) &= \left(\frac{n+1}{2} \right)^{\text{th}} \text{ observation} \\ &= \left(\frac{7+1}{2} \right)^{\text{th}} \text{ observation} = 4^{\text{th}} \text{ observation} = 13 \end{aligned}$$

Hence, 13 is the median of the given data.

Example 12: Following are the marks out of 100 obtained by 10 students in English. 23, 15, 35, 48, 41, 5, 8, 9, 11, 51. Find the median of the data.

Solution: Arranging the data in an ascending order.

$$\underline{5, 8, 9, 11, 15, 23, 35, 41, 48, 51}$$

Since, number of observation is even, i.e., $n = 10$

$$\therefore \text{Median } (\tilde{X}) = \frac{1}{2} \left(\left(\frac{n}{2} \right)^{\text{th}} \text{ observation} + \left(\frac{n+2}{2} \right)^{\text{th}} \text{ observation} \right)$$

$$\text{As, } \frac{n}{2} = \frac{10}{2} = 5 \text{ and } \frac{n+2}{2} = \frac{12}{2} = 6$$

$$\therefore \text{Median} = \frac{1}{2} [5^{\text{th}} \text{ observation} + 6^{\text{th}} \text{ observation}]$$

$$\text{or Median} = \frac{1}{2} [15 + 23] = \frac{38}{2} = 19$$

Hence, 19 is the median of the data.

Median for Grouped Data

The median for grouped data is obtained by the following formula:

$$\text{Median } (\tilde{X}) = \ell + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

Where, ℓ = Lower class boundary of median class,

h = The size of class limits of median class,

f = Frequency of the median class,

n = Total frequency i.e., Σf ,

and c = Cumulative frequency preceding the median class.

Remember the following points:

- (i) The groups of classes must be in a continuous form i.e., we need class boundaries.
- (ii) Make the column of cumulative frequencies ($c.f.$) from the column of frequencies.
- (iii) Locate median class i.e., $\left(\frac{n}{2} \right)^{\text{th}}$ see value in $c.f.$ column wherever it lies.
- (iv) Underline the median class, then take the values of f and h of the median class thus obtained

Example 13: The heights of 100 athletes, measured to the nearest (inches) are given in the following table. Find the median.

Heights (in inches)	62.5–63.5	63.5–64.5	64.5–65.5	65.5–66.5	66.5–67.5	67.5–68.5	68.5–69.5	69.5–70.5	70.5–71.5
No. of Students	4	6	10	20	30	13	12	3	2

Solution: In the above data, class boundaries have already been given:

Heights (inches)	Frequency (f)	$c.f.$	
62.5 – 63.5	4	4	
63.5 – 64.5	6	$4 + 6 = 10$	
64.5 – 65.5	10	$10 + 10 = 20$	
65.5 – 66.5	20	$20 + 20 = 40 \rightarrow c$	
66.5 – 67.5	30	$30 + 40 = 70 \rightarrow$	Median class
67.5 – 68.5	13	$13 + 70 = 83$	
68.5 – 69.5	12	$12 + 83 = 95$	
69.5 – 70.5	3	$3 + 95 = 98$	
70.5 – 71.5	2	$2 + 98 = 100 \rightarrow n$	
Total	$\Sigma f = 100$	---	

Here, $n = 100$

$$\text{so, } \frac{n}{2} = \frac{100}{2} = 50$$

50^{th} item lies in the class boundaries 66.5 – 67.5.

$$\ell = 66.5, \quad h = 1, \quad f = 30, \quad c = 40$$

$$\begin{aligned}
 \therefore \text{Median} &= \ell + \frac{h}{f} \left(\frac{n}{2} - c \right) \\
 &= 66.5 + \frac{1}{30} (50 - 40) \quad (\text{Putting the values}) \\
 &= 66.5 + \frac{10}{30} \\
 &= 66.5 + 0.33
 \end{aligned}$$

$$\therefore \text{Median} = 66.83 \text{ inches}$$

Example 14: Following are the weights (in kg) of 50 men. Find the median weight.

Weights (kg)	110 – 114	115 – 119	120 – 124	125 – 129	130 – 134
No. of men	5	12	23	6	4

Solution: As class boundaries are not given so, first of all we make class boundaries by the usual procedure.

Weight (kg)	Frequency(f)	Class Boundaries	c.f.	
110 – 114	5	109.5 – 114.5	5	
115 – 119	12	114.5 – 119.5	17 → c	
120 – 124	23	119.5 – 124.5	40 →	Median class
125 – 129	6	124.5 – 129.5	46	
130 – 134	4	129.5 – 134.5	50 → n	
Total	Σf = 50	---	---	

Here $n = 50$ so, $\frac{n}{2} = \frac{50}{2} = 25$. 25th item lies in 119.5 – 124.5.

$$\ell = 119.5, h = 5, f = 23, c = 17$$

$$\begin{aligned} \text{Median} &= \ell + \frac{h}{f} \left(\frac{n}{2} - c \right) \\ &= 119.5 + \frac{5}{23} (25 - 17) \quad (\text{Putting the values}) \\ &= 119.5 + \frac{40}{23} = 119.5 + 1.74 \end{aligned}$$

$$\therefore \text{Median} = 121.24 \text{ kg}$$

12.2.3 Mode

In a data the values (observation) which appears or occurs most often is called mode of the data. It is the most common value. Mode is denoted by $\hat{X}$.

Mode for Ungrouped Data

Example 15: The marks in mathematics of Jamal in eight monthly tests were 75, 76, 80, 80, 82, 82, 82, 85. Find the mode of the marks.

Solution: As 82 is repeated more than any other number so, clearly mode is 82.

Example 16: Ten students were asked about the number of questions they have solved out of 20 questions last week. Records were 13, 14, 15, 11, 16, 10, 19, 20, 18, 17. Find the mode of the data.

Solution: It is obvious that the given data contains no mode. It is ill-defined. Sometimes data contains several modes. If the data is: 10, 15, 15, 15, 20, 20, 20, 25, 32, then data contains two modes i.e., 15 and 20.

Example 17: A survey was conducted from the 15 students of a school and asked the students about their favourite colour.

The responses are: purple, yellow, purple, yellow, yellow, red, blue, green, yellow, yellow, red, blue, yellow, purple, green. Find mode of the data.

Solution: Mode is the most frequent colour.

Mode = yellow

So, the colour “yellow” is the mode of the given data.

Mode for Grouped Data

Mode can be calculated by the following formula:

$$\text{Mode} = \ell + \frac{(f_m - f_1)}{(f_m - f_1)(f_m - f_2)} \times h$$

Where, ℓ = Lower class boundary of the modal class.

f = Frequency of the modal class.

f_1 = Frequency preceding the modal class.

f_2 = Frequency following the modal class and

h = Size of the modal class.

Remember!

A data can have more than one mode. A data may or may not have a mode.

Note:

Mode cannot be easily calculated from the data presented in a frequency distribution. As it has no individual values, so we do not know which value appears most frequently. We only assume the class with the highest frequency as a modal class.

Example 18: Following are the heights in (inches) of 40 students in Grade - 8.

Heights (inches)	48 – 50	50 – 52	52 – 54	54 – 56	56 – 58	58 – 60
No. of students	5	7	10	9	6	3

Find mode of the above data.

Solution:

Heights (inches)	Frequency (f)
48 – 50	5
50 – 52	$7 \rightarrow f_1$
52 – 54	$10 \rightarrow f_m$
54 – 56	$9 \rightarrow f_2$
56 – 58	6
58 – 60	3
Total	$\Sigma f = 40$

Activity

Collect data of weights of 50 students. Make a frequency distribution and find mean, median and mode of the data.

In the above data, class boundaries have already been given. Using the formula for grouped data we find mode as:

$$\ell = 52, h = 2, f_m = 10, f_1 = 7, f_2 = 9$$

$$\text{Mode} = \ell + \frac{(f_m - f_1) \times h}{(f_m - f_1) + (f_m - f_2)}$$

$$\text{or Mode} = 52 + \frac{(10 - 7) \times 2}{(10 - 7) + (10 - 9)}$$

$$\text{or Mode} = 52 + \frac{3 \times 2}{3 + 1} = 52 + \frac{6}{4}$$

$$\text{or Mode} = 52 + 1.5 = 53.5 \text{ (inches)}$$

Skill practice!

Find the mean, median and mode of the first twenty whole numbers.

12.2.4 Weighted Mean

Arithmetic Mean is used when all the observations are given equal importance / weight but there are certain situations in which the different observations get different weights.

In this situation, weighted mean denoted by $\bar{X}_w$ is preferred. The weighted mean of $X_1, X_2, X_3, \dots, X_n$ with corresponding weights $W_1, W_2, W_3, \dots, W_n$ is calculated as:

$$\bar{X}_w = \frac{W_1X_1 + W_2X_2 + W_3X_3 + \dots + W_nX_n}{W_1 + W_2 + W_3 + \dots + W_n} = \frac{\sum_{i=1}^n W_iX_i}{\sum_{i=1}^n W_i} = \frac{\sum WX}{\sum W}$$

Example 19: The following data describes the marks of a student in different subjects and weights assigned to these subjects are also given:

Mark(X)	74	78	74	90
Weights(W)	4	3	5	6

Find its weighted mean.

Solution: Weighted mean $(\bar{X}_w) = \frac{\sum WX}{\sum W}$

$$\begin{aligned}\bar{X}_w &= \frac{4(74) + 3(78) + 5(74) + 6(90)}{4 + 3 + 5 + 6} \\ &= \frac{296 + 234 + 370 + 540}{18} = \frac{1440}{18} \\ \bar{X}_w &= 80\end{aligned}$$

Example 20: A medicine company started marketing of a sample of medicine in seven different areas of a city. The company distributed the packets of medicine in each area of the city and the weight of each area based on the demand of the medicine. Find the mean and weighted mean of the given data.

Areas of a city	Number of packets (X)	Weights (W)
A	15	5
B	25	4
C	18	3
D	23	4
E	15	2
F	10	1
G	8	2

Solution: Mean = $\frac{\sum X}{n}$

$$= \frac{15 + 25 + 18 + 23 + 15 + 10 + 8}{7}$$

$$= \frac{114}{7} = 16.29 \approx 16 \text{ packets}$$

So, the average number of packets of the medicine distributed by the company per area is 16.

Weighted mean = $\frac{\sum WX}{\sum W}$

$$= \frac{15(5) + 25(4) + 18(3) + 23(4) + 15(2) + 10(1) + 8(2)}{5 + 4 + 3 + 4 + 2 + 1 + 2}$$

$$= \frac{377}{21} = 17.95 \approx 18$$

12.2.5 Real Life Situations Involving Mean, Weighted Mean, Median and Mode

Sales and Marketing

Example 21: A toy factory sold toys in a month. Consider the following data:

Class limits	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
f	15	28	45	29	20

- Calculate mean, median and mode of the number of toys sold by the factory.
- Also tell the modal class of the distribution.

Solution: (i) For mean

Class limits	f	X	fX	$c.f.$	
10 – 20	15	15	225	15	
20 – 30	28	25	700	$28 + 15 = 43$	
30 – 40	45	35	1575	$45 + 43 = 88$	Modal / Median class
40 – 50	29	45	1305	$29 + 88 = 117$	
50 – 60	20	55	1100	$20 + 117 = 137$	
Total	$\Sigma f = 137$		4905		

$$\text{Mean } (\bar{X}) = \frac{\Sigma fx}{\Sigma f} = \frac{4905}{137} = 35.8 \approx 36$$

Average sale of the toys is 36.

For median: Here, $n = 137$, so, $\frac{137}{2} = 68.5$; 68.5 lies in 30 – 40. $\ell = 30$, $h = 10$, $f = 45$, $n = 137$, $c = 43$

$$\begin{aligned} \text{Median } (\tilde{X}) &= \ell + \frac{h}{f} \left(\frac{n}{2} - c \right) \\ &= 30 + \frac{10}{45} \left(\frac{137}{2} - 43 \right) \\ &= 30 + \frac{10}{45} (68.5 - 43) \\ &= 30 + \frac{10}{45} (25.5) \\ &= 30 + 5.67 \end{aligned}$$

$$\text{Median} = 35.67 \approx 36$$

Thus, median of the sold toys by the factory is 36.

For mode: $\ell = 30$, $h = 10$, $f_m = 45$, $f_1 = 28$, $f_2 = 29$

$$\begin{aligned} \text{Mode } (\hat{X}) &= \ell + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h \\ &= 30 + \frac{(45 - 28)}{(45 - 28) + (45 - 29)} \times 10 \end{aligned}$$

$$\begin{aligned}
 &= 30 + \frac{17}{17+16} \times 10 \\
 &= 30 + \frac{17}{33} \times 10 \\
 &= 30 + 5.15
 \end{aligned}$$

$$\text{Mode } (\hat{X}) = 35.15 \approx 35$$

Thus, mode of the sold toys by the factory is 35.

- (ii) The modal class of sold toys by the factory is (30 – 40).

EXERCISE 12.2

- Find the arithmetic mean in each of the following:
 - 4, 6, 10, 12, 15, 20, 25, 28, 30.
 - 12, 18, 19, 0, -19, -18, -12
 - 6.5, 11, 12.3, 9, 8.1, 16, 18, 20.5, 25
 - 8, 10, 12, 14, 16, 20, 22
- Following are the heights in (inches) of 12 students. Find the median height.
55, 53, 54, 58, 60, 61, 62, 56, 57, 52, 51, 63.
- Following are the earnings (in Rs.) of ten workers:
88, 70, 72, 125, 115, 95, 81, 90, 95, 90. Calculate
 - Arithmetic Mean
 - Median
 - Mode
- The Marks obtained by the students in the subject of English are given below.

Marks obtained	15 – 19	20 – 24	25 – 29	30 – 34	35 – 39
Frequency	9	18	35	17	5

- Find: (i) Arithmetic mean of their marks by direct and short formula.
(ii) Median of their marks.

5. Given below is a frequency distribution.

Class Interval	5 – 9	10 – 14	15 – 19	20 – 24	25 – 29
Frequency	1	8	18	11	2

Find the mode of the frequency distribution.

6. Ten boys work on a petrol pump station. They get weekly wages as follows:
Wages (in Rs.) 4250, 4350, 4400, 4250, 4350, 4410, 4500, 4300, 4500, 4390.
Find the arithmetic mean by short formula, median and mode of their wages.

7. The arithmetic mean of 45 numbers is 80. Find their sum.
8. Five numbers are 1, 4, 0, 7, 9. Find their mean, median and mode.
9. A set of data contains the values as 148, 145, 160, 157, 156, 160.

Show that Mode > Median > Mean.

10. The monthly attendance of 10 students for their lunch in the hostel is recorded as: 21, 15, 16, 18, 14, 17, 15, 12, 13, 11.

Find the median and mode of the attendance. Also find the mean if $D = A - 20$.

11. On a prize distribution day, 50 students brought pocket money as under:

Rupees	5 – 10	10 – 15	15 – 20	20 – 25	25 – 30
Frequency (f)	12	9	18	7	4

- (i) Find the median and mode of the above data.
- (ii) Find the arithmetic mean of the data given above using coding method.
12. The arithmetic mean of the ages of 20 boys is 13 years, 4 months and 5 days. Find the sum of their ages. If one of the boys is of age exactly 15 years. What is the average age of the remaining boys?
13. Calculate the arithmetic mean from the following information:
 - (i) If $D = X - 140$, $\Sigma D = 500$ and $n = 10$
 - (ii) If $U = \frac{x-130}{6}$, $\Sigma U = -150$ and $n = 15$
 - (iii) If $D = x - 25$, $\Sigma fD = 300$ and $\Sigma f = 20$
 - (vi) If $U = \frac{x-120}{5}$, $\Sigma fU = 60$ and $\Sigma f = 100$
14. The three children Haris, Maham and Minal made the following scores in a game conducted by a group of teachers in the school.

Haris scores	50	55	70	85	90
Maham scores	75	60	60	45	53
Minal scores	80	77	66	42	48

It is decided that the candidate who gets the highest average score will be awarded rupees 1000. Who will get the awarded amount?

15. Given below is a frequency distribution derived by making a substitution as $D = X - 20$. Calculate the arithmetic mean.

<i>D</i>	-6	-4	-2	0	2	4	6
<i>f</i>	1	3	6	20	26	12	2

16. Being partners Hafsa and Fatima took part in a quiz programme. They made the following number of points 45, 51, 58, 61, 74, 48, 46 and 50. Compute the average number of points using deviation $D = x - 58$.
17. A person purchased the following food items:

Food item	Quantity (in Kg)	Cost per Kg (in Rs.)
Rice	10	96
Flour	12	48
Ghee	4	190
Sugar	3	49
Mutton	2	650

What is the weighted mean of cost of food items per kg?

18. For the following data, find the weighted mean.

Item	Quantity	Cost of item (in thousands)
Washing Machine	5	35
Heater	3	5
Stove	2	13
Dispenser	6	18

19. A company is planning its next year marketing budget across five years: yearly budgets (in million) are: 5, 7, 8, 6, 7. Find the average budget for the next year.
20. Ahmad obtained the following marks in a certain examination. Find the weighted mean if weights 5, 4, 2, 3, 2, 4 respectively are allotted to the subjects.

Urdu	English	Science	Math	Islamiyat	Computer
78	65	80	90	85	72

REVIEW EXERCISE 12

1. Four options are given against each statement. Encircle the correct option.
- (i) Which data takes only some specific values?
 - (a) continuous data
 - (b) discrete data
 - (c) grouped data
 - (d) ungrouped data
 - (ii) The number of times a value occurs in a data is called:
 - (a) frequency
 - (b) relative frequency
 - (b) class limit
 - (d) class boundaries.
 - (iii) Midpoint is also known as:
 - (a) mean
 - (b) median
 - (c) class limit
 - (d) class mark
 - (iv) Frequency polygon is also drawn /constructed by using:
 - (a) histogram
 - (b) bar graph
 - (c) class boundaries
 - (d) class limit
 - (v) The difference between the greatest value and the smallest value is called:
 - (a) class limits
 - (b) midpoint
 - (c) relative frequency
 - (d) range
 - (vi) Measure of central tendency is used to find out the _____ of a data set.
 - (a) class boundaries
 - (b) cumulative frequency
 - (c) middle or centre value
 - (d) frequency
 - (vii) If the mean of 5, 7, 8, 9 and x is 7.5, what will be the value of x ?
 - (a) 10
 - (b) 8
 - (c) 8.5
 - (d) 5.8
 - (viii) Find the mode of the given data: 2, 5, 8, 9, 0, 1, 3, 7 and 10
 - (a) 5
 - (b) 7
 - (c) 0
 - (d) no mode
 - (ix) In a data the values (observations) which appears or occurs most often is called:
 - (a) mean
 - (b) mode
 - (c) median
 - (d) weighted mean
 - (x) Find the median of the given data: 110, 125, 122, 130, 124, 127 and 120
 - (a) 124
 - (b) 120
 - (c) 125
 - (d) 127
2. Define the following:
- (i) frequency distribution
 - (ii) histogram (unequal class limits)
 - (iii) mean
 - (iv) median

3. Following are the weights of 40 students recorded to the nearest (lbs).
 138, 164, 150, 132, 144, 125, 149, 157, 146, 158, 140, 147, 136, 148, 152, 144,
 168, 126, 138, 176, 163, 119, 154, 165, 146, 173, 142, 147, 135, 153, 140, 135
 161, 145, 135, 142, 150, 156, 145, 128.
- (a) Make a frequency table by taking number of classes as 6
 (b) Draw histogram (c) Draw a frequency polygon of the given data.
4. From the table given below. Draw a frequency polygon on histogram for the given frequency distribution.

Weight (kg)	50 – 56	57 – 59	60 – 64	65 – 72	73 – 75	76 – 80
Frequency (f)	25	32	40	30	15	8

5. Given below are marks obtained by 45 students in the monthly test of Biology:

Marks	20 – 24	25 – 29	30 – 34	35 – 39	40 – 44	45 – 49
No. of students	05	08	12	15	03	02

With reference to the above table find the following:

- (i) upper class boundary of the 5th class.
 (ii) lower class boundaries of all the classes.
 (iii) midpoint of all the classes.
 (iv) the class interval with the least frequency.
6. Given below is frequency distribution.

Draw frequency polygon and histogram for the distribution.

Class limits	5 – 9	10 – 14	15 – 19	20 – 24	25 – 29	30 – 34
Frequency	1	8	18	11	2	5

7. For the following data, find the weighted mean.

Item	Quantity	Cost of item (Rs.)
Chair	20	500
Table	20	400
Black board	10	750
Tube light	25	230
Cupboard	09	950

8. A principal of a school allocates funds of Rs.50, 000 to five different sectors:
- (i) chairs: Rs. 15000
 - (ii) tables: Rs. 12,000
 - (iii) black boards: Rs.6,000
 - (iv) room renovation: Rs. 10,000
 - (v) gardening: Rs. 7,000

Find the average of funds allocation in each sector of the school.

9. The marks of a student Saad in six tests were 84, 91, 72, 68, 87, 78. Find the arithmetic mean of his marks.
10. Adjoining distribution showed maximum load (in kg) supported by certain ropes. Find the mean load using short method.

Max-Load kg	93 – 97	98 – 102	103 – 107	108 – 112	113 – 117	118 – 122
No. of ropes	2	5	8	12	6	2

11. Usman rolled a fair dice eight times. Each time their sum was recorded as 8, 5, 6, 6, 9, 4, 3, 11. Find the median and mode of the sum.
12. Two partners Mr. Aslam and Mrs. Kalsoom run a company. In the following data the weekly wages (in Rs.) of employees who work in the company are given:

Wages (Rs.)	600 – 700	700 – 800	800 – 900	900 – 1000	1000 – 1100
Employees	3	5	7	21	11

Unit 13

Probability

Students' Learning Outcomes

At the end of the unit, the students will be able to:

- Calculate the probability of a single event and the probability of an event not occurring.
- Solve real life problems involving probability.
- Calculate relative frequency as an estimate of probability.
- Calculate expected frequencies.
- Solve real life problems involving relative and expected frequencies.

INTRODUCTION

In our daily life, we normally say that manufacturing companies give warranty on their products, there is chance that some product might not meet warranty time period. A person judges the chances of winning cricket match of a team based on previous performances etc. All the above statements have lack prediction

with certainty. In such situations, what makes it easier for us to represent the chance of an event occurring numerically i.e., probability.

Hence, Probability is the chance of occurrence of a particular event.

Probability is calculated by using the given formula:

$$\text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

It is written as: $P(A) = \frac{n(A)}{n(S)}$

$P(A)$ = Probability of an event A

$n(A)$ = Number of favourable outcomes

$n(S)$ = Total number of possible outcomes

Basic Concepts of Probability

Experiment: The process which generates results e.g., tossing a coin, rolling a dice, etc. is called an experiment.

History!

The word “probability” is derived from the Latin word “Probabilitas”. It means “probtity”. Girolamo Cardano is known as the father of probability. He was an Italian doctor and mathematician.



Outcomes: The results of an experiment are called outcomes e.g., the possible outcomes of tossing a coin are head or tail, the possible outcomes of rolling a dice are 1, 2, 3, 4, 5, or 6.

Favourable Outcome: An outcome which represents how many times we expect the things to be happened e.g., while tossing a coin, there is 1 favourable outcome of getting head or tail. While rolling a dice, there are 3 favourable outcomes of getting multiples of 2 i.e. {2, 4, 6}

Sample Space: The set of all possible outcomes of an experiment is called sample space. It is denoted by ' S ' e.g., while tossing a coin, the sample space will be $S = \{H, T\}$.

While rolling a dice, the sample space will be $S = \{1, 2, 3, 4, 5, 6\}$.

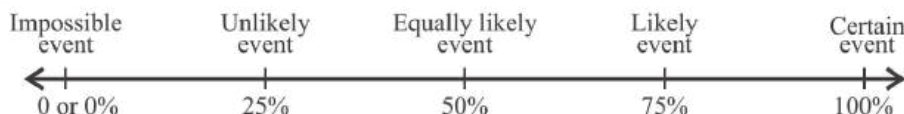
Event: The set of results of an experiment is called an event e.g., while rolling a dice getting even number is an event i.e., $A = \{2, 4, 6\}$; $n(A) = 3$.

Remember!

Each element of the sample space is called sample point.

Recall! Types of Events:

- **Certain event:** An event which is sure to occur. The probability of sure event is 1.
- **Impossible event:** An event cannot occur in any trial. The probability of this event is 0.
- **Likely event:** An event which will probably occur. It has greater chance to occur.
- **Unlikely event:** An event which will not probably occur. It has less chance to occur.
- **Equally likely events:** The events which have equal chance of occurrence. The probability of these events is 0.5.



13.1 Probability of Single Event

Example 1: Abdul Raheem rolls a fair dice, what is the probability of getting the number divisible by 3?

Solution: When a dice is rolled, the sample space will be:

$$S = \{1, 2, 3, 4, 5, 6\}; n(S) = 6$$

Let " A " be the event of getting the number divisible by 3.

$$A = \{3, 6\}; n(A) = 2$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

The probability of getting the number divisible by 3 is $\frac{1}{3}$.

Keep in mind

The range of probability for an event is:
 $0 \leq P(A) \leq 1$

Teachers' note:

Clear the concept of all the types of events by using different colours of balls or pencils etc.

Example 2: If Zeeshan rolled two fair dice, find the probability of getting:

- Even numbers on both dice.
- Multiples of 3 on both dice.
- Even number on the first dice and the number 3 on the second dice.
- At least the number 3 on the first dice and number 4 on the second dice.

Solution: When a pair of fair dice is rolled, the sample space will be:

1 st \ 2 nd	1	2	3	4	5	6
1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
4	4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
6	6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

Try Yourself!

Can you find out the sample space when 3 dice are rolled.

- Even numbers on both dice.

Let “A” be the event of getting even numbers on both dice.

$$A = \{(2, 2), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (6, 2), (6, 4), (6, 6)\}$$

$$n(A) = 9; n(S) = 36$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{9}{36} = \frac{1}{4}$$

Thus, the probability of getting even numbers on both dice is $\frac{1}{4}$.

- Multiple of 3 on both dice.

Let “B” be the event of getting multiples of 3 on both dice.

$$B = \{(3, 3), (3, 6), (6, 3), (6, 6)\}$$

$$n(B) = 4; n(S) = 36$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

Thus, the probability of getting multiples of 3 on both dice is $\frac{1}{9}$.

- (iii) Even number on the first dice and the number 3 on the second dice.

Let “C” be the event of getting even numbers on the first dice and the number 3 on the second dice.

$$C = \{(2,3), (4,3), (6,3)\}$$

$$n(C) = 3; n(S) = 36$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

Thus, the probability of getting an even number on the first dice and the number 3 on the second dice is $\frac{1}{12}$.

- (iv) At least the number 3 on the first dice and number 4 on the second dice.

Let “D” be the event of getting at least the number 3 on the first dice and number 4 on the second dice.

$$D = \{(3, 4), (4, 4), (5, 4), (6, 4)\}$$

$$n(D) = 4; n(S) = 36$$

$$P(D) = \frac{n(D)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

Thus, the probability of getting at least the number 3 on the first dice and number 4 on the 2nd dice is $\frac{1}{9}$.

13.2 Probability of an Event Not Occurring

Sometimes, we are interested in the probability that the head will not occur while tossing a coin.

Let “A” be the event of getting head while tossing a coin, then the event “A’ ” be the event of not getting head while tossing a coin.

The probability of not getting head while tossing a coin is known as the complement of that event. It is written as $P(A')$ or $P(A^c)$.

The complement of an event “A” is calculated by the given formula:

$$P(A') = 1 - P(A)$$

For example, while tossing a coin, the probability of getting a head is:

$$P(A) = \frac{1}{2}$$

Teachers’ note:

Give more examples to explain complement of events e.g., if the desired outcome is head on a flipping coin, the complement is tail. The compliment rule states that the sum of the probability of an event and its complement must be equal to 1.

and the probability of not getting a head is:

$$\begin{aligned} P(A') &= 1 - P(A) \\ &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

Thus, the complement of the event of getting a head is $\frac{1}{2}$.

Example 3: Zubair rolls a dice, what is the probability of not getting the number 6?

Solution: Let “A” be the event of getting the number 6.

The sample space while rolling a dice is: $S = \{1, 2, 3, 4, 5, 6\}$

$$n(S) = 6$$

$$A = \{6\}; n(A) = 1$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{6}$$

To find out probability of not getting the number 6, we have

$$\begin{aligned} P(A') &= 1 - P(A) \\ &= 1 - \frac{1}{6} = \frac{6-1}{6} = \frac{5}{6} \end{aligned}$$

Thus, the probability of not getting the number 6 is $\frac{5}{6}$.

Example 4: If two fair dice are rolled. What is the probability of getting:

- (i) not a double six (ii) not the sum of both dice is 8

Solution: Sample space of two fair dice is given by:

$$\begin{aligned} S = \{ & (1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), \\ & (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), \\ & (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), \\ & (6,5), (6,6) \} \end{aligned}$$

$$n(S) = 36$$

- (i) not a double six.

Let “A” be the event that a double six occurs.

$$A = \{(6, 6)\}; n(A) = 1$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{36}$$

Remember!

The sum of the probability of an event “A” and the probability of an event not occurring “A” is always “1”

$$P(A) + P(A') = 1$$

Let “ A' ” be the event that not a double six occurs

As we know that

$$\begin{aligned} P(A') &= 1 - P(A) \\ &= 1 - \frac{1}{36} = \frac{36-1}{36} = \frac{35}{36} \end{aligned}$$

Thus, the probability of not getting the double six is $\frac{35}{36}$.

- (ii) not the sum of both dice is 8.

Let “ B ” be the event that the sum of both dice is 8.

$$B = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$$

$$n(B) = 5$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{5}{36}$$

Let “ B' ” be the event not sum of both dice is 8.

$$\begin{aligned} P(B') &= 1 - P(B) \\ &= 1 - \frac{5}{36} = \frac{36-5}{36} = \frac{31}{36} \end{aligned}$$

Thus, the probability of not the sum of both dice be 8 is $\frac{31}{36}$.

13.3 Real Life Problems Involving Probability

Example 5: Let A , B and C are three missiles and they are fired at a target. If the probabilities of hitting the target are $P(A) = \frac{1}{4}$, $P(B) = \frac{3}{7}$, $P(C) = \frac{5}{9}$, respectively.

Find the probabilities of

- (i) missile A does not hit the target. (ii) missile B does not hit the target.
(iii) missile C does not hit the target.

Solution: (i) missile A does not hit the target.

$$\text{Since, } P(A) = \frac{1}{4}$$

Let ‘ A' ’ be the event that missile A does not hit the target

$$\begin{aligned} P(A') &= 1 - P(A) \\ &= 1 - \frac{1}{4} = \frac{4-1}{4} = \frac{3}{4} \end{aligned}$$

Thus, the probability of missile ‘ A ’ does not hit the target is $\frac{3}{4}$.

- (ii) missile 'B' does not hit the target.

$$\text{Since, } P(B) = \frac{3}{7}$$

Let 'B' be the event missile B does not hit the target

$$\begin{aligned} P(B') &= 1 - P(B) \\ &= 1 - \frac{3}{7} \\ &= \frac{7-3}{7} = \frac{4}{7} \end{aligned}$$

Thus, the probability of missile 'B' does not hit the target is $\frac{4}{7}$.

- (iii) missile 'C' does not hit the target.

$$\text{Since, } P(C) = \frac{5}{9}$$

Let 'C' be the event missile C of not hitting the target

$$\begin{aligned} P(C') &= 1 - P(C) \\ &= 1 - \frac{5}{9} = \frac{9-5}{9} = \frac{4}{9} \end{aligned}$$

Thus, the probability of missile 'C' does not hit the target is $\frac{4}{9}$.

Example 6: A bag contains 5 blue balls and 8 green balls. Find the probability of selecting at random:

- (i) a blue ball (ii) a green ball. (iii) not a green ball.

Solution:

- (i) a blue ball

Let 'A' be the event that the ball is blue

$$\text{Blue balls} = n(A) = 5$$

$$\text{Total balls} = n(S) = 5 + 8 = 13$$

$$\begin{aligned} P(A) &= \frac{n(A)}{n(S)} \\ &= \frac{5}{13} \end{aligned}$$

Try Yourself!

Can you find out the complement of selecting a blue ball?

Thus, the probability of selecting a blue ball is $\frac{5}{13}$.

- (ii) a green ball

Let 'B' be the event that ball is green

$$\text{Green balls} = n(B) = 8$$

$$\text{Total balls} = n(S) = 5 + 8 = 13$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{8}{13}$$

Thus, the probability of selecting green ball is $\frac{8}{13}$.

- (iii) not a green ball

Let 'B' be the event that the ball is not green.

$$P(B') = 1 - P(B)$$

$$= 1 - \frac{8}{13}$$

$$= \frac{13-8}{13} = \frac{5}{13}$$

Thus, the probability of not selecting a green ball is $\frac{5}{13}$.

Think!

The probability that a person A will be alive 0.75. Can you find out the complement of that event?

Example 7: A card is drawn at random, from a pack of 52 playing cards. What is the probability of getting:

- (i) a card of heart

- (ii) neither spade nor heart

Solution: (i) a card of heart

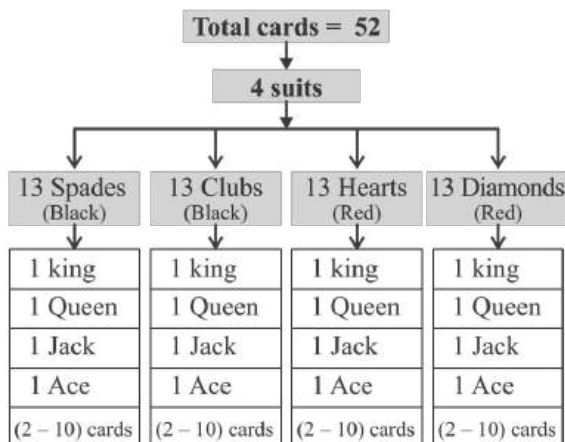
Total number of cards = 52 ; $n(S) = 52$

Let 'A' be the event of selecting a card of heart.

Number of heart cards = 13 ; $n(A) = 13$

$$\begin{aligned} P(A) &= \frac{n(A)}{n(S)} \\ &= \frac{13}{52} = \frac{1}{4} \end{aligned}$$

Thus, the probability of getting a card of heart is $\frac{1}{4}$.



- (ii) neither spade nor heart

Let ' B ' be the event of selecting a card of spade or heart

Number of spade and heart cards = 26 ; $n(B) = 26$

$$\begin{aligned}P(B) &= \frac{n(B)}{n(S)} \\&= \frac{26}{52} \\&= \frac{1}{2}\end{aligned}$$

Let ' B' ' be the event of selecting neither spade nor heart card.

$$\begin{aligned}P(B') &= 1 - P(B) \\&= 1 - \frac{1}{2} \\&= \frac{1}{2}\end{aligned}$$

Thus, the probability of getting neither spade nor heart cards is $\frac{1}{2}$.

EXERCISE 13.1

- Arshad rolls a dice, with sides labelled L, M, N, O, P, U. What is the probability that the dice lands on consonant?
- Shazia throws a pair of fair dice. What will be the probability of getting:
 - sum of dots is at least 4.
 - product of both dots is between 5 to 10.
 - the difference between both the dots is equal to 4.
 - number at least 5 on the first dice and the number at least 4 on the second dice.
- One alphabet is selected at random from the word "MATHEMATICS". Find the probability of getting:

(i) vowel	(ii) consonant	(iii) an E
(iv) an A	(v) not M	(vi) not T
- Aslam rolled a dice. What is the probability of getting the numbers 3 or 4? Also find the probability of not getting the numbers 3 or 4.

5. Abdul Hadi labelled cards from 1 to 30 and put them in a box. He selects a card at random. What is the probability that selected card containing:
- (i) the number 25
 - (ii) number between 17 to 22
 - (iii) number at least 20
 - (iv) number not 27 and 29
 - (v) number not between 12 to 15
6. The probability that Ayesha will pass the examination is 0.85. What will be the probability that Ayesha will not pass the examination?
7. Taabish tossed a fair coin and rolled a fair dice once. Find the probability of the following events:
- (i) tail on coin and at least 4 on dice.
 - (ii) head on coin and the number 2,3 on dice.
 - (iii) head and tail on coin and the number 6 on dice.
 - (iv) not tail on coin and the number 5 on dice.
 - (v) not head on coin and the number 5 and 2 on dice.
8. A card is selected at random from a well shuffled pack of 52 plying cards. What will be the probability of selecting:
- (i) a queen
 - (ii) neither a queen nor a jack
9. A card is chosen at random from a pack of 52 playing cards. Find the probability of getting:
- (i) a jack
 - (ii) no diamond

13.4 Relative Frequency as an Estimate of Probability

Relative frequency tells us how often a specific event occurs relative to the total number of frequency event or trials. It is calculated by using the following method:

$$\text{Relative frequency} = \frac{\text{Frequency of specific event}}{\text{Total frequency}} = \frac{x}{N}, \text{ where } N = \sum f$$

Example 8: Find the relative frequency of the given date.

<i>X</i>	2	3	4	5	6	7	8
<i>f</i>	3	5	6	9	10	8	2

Solution:	<i>X</i>	<i>f</i>	Relative frequency
	2	3	$\frac{3}{43} = 0.07$
	3	5	$\frac{5}{43} = 0.12$
	4	6	$\frac{6}{43} = 0.14$
	5	9	$\frac{9}{43} = 0.21$
	6	10	$\frac{10}{43} = 0.23$
	7	8	$\frac{8}{43} = 0.19$
	8	2	$\frac{2}{43} = 0.04$
	Total	$\Sigma f = 43$	

13.5 Real Life Application of Relative Frequency

Example 9: A survey was conducted on 80 students of Grade – IX and asked about their favourite colour. The responses are:

- | | |
|---------------------------------|---------------------------------|
| (i) Red colour = 23 students | (ii) Green colour = 15 students |
| (iii) Pink colour = 25 students | (iv) Blue colour = 10 students |
| (v) White colour = 7 students. | |

Find the relative frequency for each colour.

Solution: Total number of students = 80

- (i) Relative frequency for red colour = $\frac{23}{80} = 0.29$

It means that 29% students prefer red colour.

- (ii) Relative frequency for green colour = $\frac{15}{80} = 0.19$

It means that 19% students prefer green colour.

Keep in mind

The sum of all the relative frequencies is always equal to or approximately equal to 1.

Remember!

Relative frequency is an estimated probability of an event occurring when an experiment is repeated a fixed number of times.

(iii) Relative frequency for pink colour = $\frac{25}{80} = 0.31$

It means that 31% students prefer pink colour.

(iv) Relative frequency for blue colour = $\frac{10}{80} = 0.12$

It means that 12% students prefer blue colour.

(v) Relative frequency for white colour = $\frac{7}{80} = 0.09$

It means that 9% students prefer white colour.

Try Yourself!

Out of 200 students in a school, 80 play cricket, 50 play football, 25 play volleyball and 45 do not play any game. Can you find out the probability of the students who do not play any game and relative frequency of the students who play cricket?

Example 10: Abdul Rehman obtained different marks in different subjects out of 100 marks. The detail is as under:

Subject	Urdu	English	Islamiyat	Mathematics	Science	Computer Science
Marks Obtained	75	80	72	95	81	85

Find the relative frequency of above given data.

Solution:

Subject	Marks obtained	Relative frequency
Urdu	75	$\frac{75}{488} = 0.15$
English	80	$\frac{80}{488} = 0.16$
Islamiyat	72	$\frac{72}{488} = 0.15$
Mathematics	95	$\frac{95}{488} = 0.19$
Science	81	$\frac{81}{488} = 0.17$
Computer Science	85	$\frac{85}{488} = 0.17$
Total	$\Sigma f = 488$	

13.6 Expected Frequency

Expected frequency is a measure that estimate how often an event should be occurred depended on probability. Expected frequency is found by using the following method:

Expected frequency = Total number of trials $\times$ Probability of the event.

$$= N \times P(A)$$

Teachers' note

Clear the concept to the students that relative frequency as an estimate of probability by using different real life problems.

Example 11: Six fair dice are rolled 50 times. The probability of occurrence of different number of sixes are given below. Find the expected frequency of the following data:

x	0	1	2	3	4	5	6
$P(x)$	0.09	0.10	0.12	0.24	0.10	0.20	0.15

Find the expected frequency of occurrence of each six.

Solution:	No. of Sixes (x)	$P(x)$	Expected frequency $= N \times P(x) = 50 \times P(x)$
	0	0.09	$50 \times 0.09 = 4.5$
	1	0.10	$50 \times 0.10 = 5$
	2	0.12	$50 \times 0.12 = 6$
	3	0.24	$50 \times 0.24 = 12$
	4	0.10	$50 \times 0.10 = 5$
	5	0.20	$50 \times 0.20 = 10$
	6	0.15	$50 \times 0.15 = 7.5$

13.7 Real Life Application on Expected Frequency

Example 12: Find the average number of times getting 1 or 6, when a fair dice is rolled 300 times.

Solution: Let “S” be the sample space when a fair dice is rolled:

$$S = \{1, 2, 3, 4, 5, 6\} ; n(S) = 6$$

Let “B” be the event that 1 or 6 comes up.

$$B = \{1, 6\} ; n(B) = 2$$

Remember!

Sum of all expected frequencies is always equal to or approximately equal to a fixed number of trials.

So,
$$P(B) = \frac{n(B)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

Therefore, $E(B) = N \times P(B)$

$$= 300 \times \frac{1}{3} = 100$$

Thus, the average number of times 1 or 6 comes up is 100.

Example 13: If the probability of a defective bolt is 0.3. Find the number of non-defective bolts in a total to 800.

Solution:

The probability of defective bolt is = 0.3

Probability of non-defective bolt = $1 - 0.3 = 0.7$

Number of non-defective bolts = $0.7 \times 800 = 560$

Thus, the non-defective bolts will be 560.

EXERCISE 13.2

1. A researcher collected data on number of deaths from Horse-Ricks in Russian Army crops over to years. The table is as follows:

No. of death	0	1	2	3	4	5	6
Frequency	60	50	87	40	32	15	10

Find the relative frequency of the given data.

2. The frequency of defective products in 750 samples are shown in the following table. Find the relative frequency for the given table.

No. of defectives per sample	0	1	2	3	4	5	6	7	8
No. of sample	120	140	94	85	105	50	40	66	50

3. A quiz competition on general knowledge is conducted. The number of corrected answers out of 5 questions for 100 sets of questions is given below.

X	0	1	2	3	4	5
f	10	23	15	25	18	9

Find the relative frequencies for the given data.

4. A survey was conducted from the 108 students of a class and asked about their favourite food. The responses are as under:

Name of food item	Biryani	Fresh Juice	Chicken	Bar. B.Q	Sweets
No. of students	40	07	21	15	25

- how many percentages of students like biryani?
 - how many percentages of students like chicken?
 - which food is the least like by the students?
 - which food is the most prefer by the students?
5. In 500 trials of a thrown of two dice, what is expected frequency that the sum will be greater than 8?
6. What is the expectation of a person who is to get Rs. 120 if he obtains at least 2 heads in single toss of three coins?
7. Find the expected frequencies of the given data if the experiment is repeated 200 times.

x	0	1	2	3	4	5	6
$P(x)$	0.11	0.21	0.17	0.18	0.09	0.17	0.07

8. The probability of getting 5 sixes while tossing six dice is $\frac{2}{5}$, the dice is rolled 200 times. How many times would you expect it to show 5 sixes?

REVIEW EXERCISE 13

1. Four options are given against each statement. Encircle the correct option.
- Each element of the sample space is called:
 - event
 - experiment
 - sample point
 - outcomes
 - An outcome which represents how many times we expect the things to be happened is called:
 - outcomes
 - favourable outcome
 - sample space
 - sample point

- (iii) Which one tells us how often a specific event occurs relative to the total number of frequency event or trials?
- (a) expected frequency (b) sum of relative frequency
(c) relative frequency (d) frequency
- (iv) Estimated probability of an event occurring is also known as:
- (a) relative frequency (b) expected frequency
(c) class boundaries (d) sum of expected frequency
- (v) The sum of all expected frequencies is equal to the fixed number of:
- (a) trials (b) relative frequencies
(c) outcomes (d) events
- (vi) The chance of occurrence of a particular event is called:
- (a) sample space (b) estimated probability
(c) probability (d) expected frequency
- (vii) An event which will probably occur. It has greater chance to occur is called:
- (a) equally likely event (b) likely event
(c) unlikely event (d) certain event
- (viii) Find out the total number of possible sample space when 4 dice are rolled.:
- (a) 6^2 (b) 6^3 (c) 6^4 (d) 6^6
- (ix) While rolling a pair of dice, what will be the probability of double 2?
- (a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{5}{6}$ (d) $\frac{1}{36}$
- (x) A card is chosen from a pack of 52 playing cards, find the probability of getting no jack and king:
- (a) $\frac{2}{13}$ (b) $\frac{11}{13}$ (c) $\frac{2}{52}$ (d) $\frac{11}{52}$

2. Define the following:

- (i) relative frequency (ii) expected frequency

3. An urn contains 10 red balls, 5 green balls and 8 blue balls. Find the probability of selecting at random.

- (i) a green ball (ii) a red ball (iii) a blue ball
(iv) not a red ball (v) not a green ball

4. Three coins are tossed together. what is the probability of getting:
- (i) exactly three heads
 - (ii) at least two tails
 - (iii) not at least two heads
 - (iv) not exactly two heads
5. A card is drawn from a well shuffled pack of 52 playing cards. What will be the probability of getting:
- (i) king or jack of red colour
 - (ii) not “2” of club and spade
6. Six coins are tossed 600 times. The number of occurrence of tails are recorded and shown in the table given below:

No. of tails	0	1	2	3	4	5	6
Frequency	110	90	105	80	76	123	16

Find the relative frequency of given table.

7. From a lot containing 25 items, 8 items are defective. Find the relative frequency of non-defective items, also find the expected frequency of non-defective items.

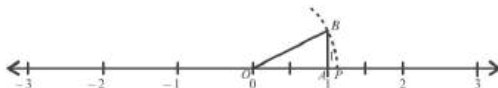
Answers

EXERCISE 1.1

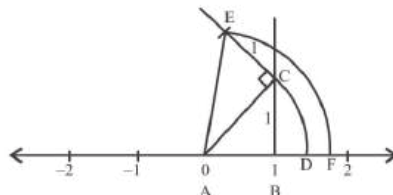
1. (i) Rational (ii) Rational (iii) Irrational (iv) Irrational (v) Irrational
(vi) Irrational (vii) Irrational (viii) Irrational (ix) Rational (x) Irrational

2.

(i)



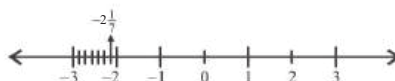
(ii)



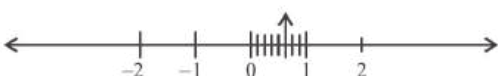
(iii)



(iv)



(v)



(vi)



3. (i) $\frac{4}{9}$ (ii) $\frac{37}{99}$ (iii) $\frac{21}{99}$

4. (i) Associative property over addition (ii) Commutative property over addition
(iii) Additive inverse (iv) Left distributive property of multiplication over addition
(v) Additive identity (vi) Multiplicative identity
(vii) Associative property under multiplication (viii) Commutative property under multiplication
5. (i) Additive property (ii) Reciprocal property (iii) Additive property
(iv) Multiplicative property (v) Multiplicative property (vi) Trichotomy property

EXERCISE 1.2

1. (i) $4 - \sqrt{3}$ (ii) $\frac{\sqrt{6} + \sqrt{15}}{3}$ (iii) $\frac{\sqrt{10} - \sqrt{5}}{5}$ (iv) $17 - 12\sqrt{2}$ (v) $5 - 2\sqrt{6}$
- (vi) $2\sqrt{3}(\sqrt{7} - \sqrt{5})$ 2. (i) $\frac{8}{27}$ (ii) 12 (iii) $\frac{10}{3}$ (iv) x^2yz^4 (v) $\frac{1}{6}$
- (vi) $\frac{9}{2}$ (vii) $\frac{27}{16}$ (viii) 243 (ix) 19 3. (i) 6 (ii) $2\sqrt{8}$ (iii) 34
- (iv) $12\sqrt{8}$ (v) 1154 (vi) 32 4. $P = -25, q = 18$ 5. (i) $\frac{3375}{512}$ (ii) $\frac{2}{3}$
- (iii) $\frac{6}{5}$ (iv) $a + b^2$

EXERCISE 1.3

1. 13, 14, 15 2. $\overline{AB} = 4\sqrt{3} - 2\sqrt{5}$ 3. $(11\sqrt{2} - 2)m^2$ 4. 45, 23
 5. 118.4 6. 20 years 7. 1.33% 8. Rs.6225 9. Rs.52500

REVIEW EXERCISE 1

1. (i) c (ii) d (iii) d (iv) d (v) a (vi) b (vii) b (viii) a (ix) d (x) d
 7. (i) $\frac{x^3 y^7}{z^4}$ (ii) 3^{2x} (iii) 27 8. 15, 17, 19 9. 34, 62 10. 540750

EXERCISE 2.1

1. (i) 2×10^6 (ii) 4.89×10^4 (iii) 4.2×10^{-3} (iv) 9×10^{-7} (v) 7.3×10^4
 (vi) 6.5×10^1 2. (i) 804 (ii) 300000 (iii) 0.015 (iv) 17700000
 (v) 0.0000055 (vi) 0.00004 3. 300,000,000 m/sec 4. 4.0075×10^7 m
 5. 6779 km 6. 12756 km

EXERCISE 2.2

1. (i) $\log_{10} 1000 = 3$ (ii) $\log_2 256 = 8$ (iii) $\log_3 \frac{1}{27} = -3$ (iv) $\log_{20} 400 = 2$
 (v) $\log_{16} \frac{1}{2} = -\frac{1}{4}$ (vi) $\log_{11} 121 = 2$ (vii) $\log_q p = r$ (viii) $\log_{32} \frac{1}{2} = -\frac{1}{5}$
 2. (i) $5^3 = 125$ (ii) $2^4 = 16$ (iii) $23^0 = 1$ (iv) $5^1 = 5$
 (v) $2^{-3} = \frac{1}{8}$ (vi) $9^{\frac{1}{2}} = 3$ (vii) $10^5 = 100000$ (viii) $4^{-2} = \frac{1}{16}$
 3. (i) $x = 4$ (ii) $x = 0$ (iii) $x = 8$ (iv) $x = \frac{1}{1000}$ (v) $x = 8$ (vi) $x = 10$

EXERCISE 2.3

1. (i) 3 (ii) 1 (iii) -2 (iv) 2 (v) -5 (vi) 5
 2. (i) 1.6335 (ii) 2.7627 (iii) 0.2971 (iv) -1.0575 (v) -1.3279 (vi) -3.4510
 3. (i) 3.5019 (ii) 1.5019 (iii) -1.4981 4. (i) $x = 1.015$ (ii) $x = 15.56$
 (iii) $x = 0.0003681$ (iv) $x = 0.02675$ (v) $x = 2270$ (vi) $x = 0.009585$

EXERCISE 2.4

1. (i) 1 (ii) 7 (iii) -2 (iv) 2 (v) 5 (vi) 1
 2. (i) $\log 45$ (ii) $\log 27$ (iii) $6 \log_a b$ (iv) $\log_3 x^2 y$ (v) $\log_5 \frac{x^4 z}{y}$ (vi) $\ln \frac{a^2 b^3}{c^4}$
 3. (i) $\log 11 - \log 5$ (ii) $\frac{3}{2} \log_3 2 + 3 \log_5 a$ (iii) $2 \ln a + \ln b - \ln c$ (iv) $\frac{1}{9} [\log x + \log y - \log z]$
 (v) $\frac{4}{3} \ln 2 + \ln x$ (vi) $5 [\log_2 (1-a) - \log_2 b]$ 4. (i) $x = 5$ (ii) $x = 4$ (iii) $x = -10$

- (iv) $x = 5$ (v) $x = 22$ (vi) $x = 5\frac{2}{3}$ 5. (i) 2.960 (ii) 23.62 (iii) 1.339
 (iv) 14.21 6. $M = 3$ 7. 14 years 8. 17.17°C

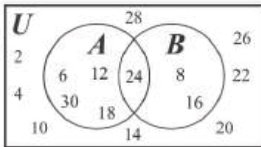
REVIEW EXERCISE 2

1. (i) c (ii) b (iii) b (iv) d (v) a (vi) c (vii) d (viii) c (ix) d (x) c
 2. (i) 5.67×10^{-4} (ii) 7.34×10^2 (iii) 3.3×10^2 3. (i) 2600 (ii) 0.0008794
 (iii) 0.000006 4. (i) $\log_3 2187 = 7$ (ii) $\log_a c = b$ (iii) $\log_{12} 144 = 2$
 5. (i) $4^x = 8$ (ii) $9^3 = 729$ (iii) $4^5 = 1024$
 6. (i) $x = 3$ (ii) $x = -\frac{1}{2}$ (iii) $x = -\frac{3}{5}$ 7. (i) $\log \frac{x^7}{y^6}$ (ii) $\log 2$ (iii) $\log_5 2$
 8. (i) $\log x + \log y + 6 \log z$ (ii) $\frac{1}{6}[5 \log_3 m + 3 \log_3 n]$ (iii) $\frac{3}{2}[\log 2 + \log x]$
 9. (i) 4.086 (ii) 1133 (iii) 24.01 10. 2035

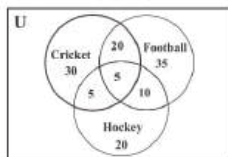
EXERCISE 3.1

1. (i) $\{x | x = n^2, n \in N \wedge 1 \leq n \leq 22\}$ (ii) $\{x | x = 2^n, n \in N \wedge 1 \leq n \leq 8\}$
 (iii) $\{x | x \in Z \wedge -1000 \leq x \leq 1000\}$ (iv) $\{x | x = 6n, n \in N \wedge 1 \leq n \leq 20\}$
 (v) $\{x | x = 100 + 2n, n \in W \wedge 0 \leq n \leq 150\}$ (vi) $\{x | x = 3^n, n \in W\}$
 (vii) $\{x | x \text{ is a divisor of } 100\}$ (viii) $\{x | x = 5n, n \in N \wedge 1 \leq n \leq 20\}$
 (ix) $\{x | x \in Z \wedge -100 < x < 1000\}$ 2. (i) $\{3, 6, 9, \dots, 36\}$ (ii) $\left\{-\frac{1}{2}\right\}$
 (iii) $\{2, 3, 5, 7, 11\}$ (iv) $\{1, 2, 4, 8, 16, 32, 64, 128\}$ (v) $\{2, 4, 8, 16, 32, 64, 128\}$
 (vi) $\{\}$ (vii) $\{1, 2, 3, 4, 5, \dots\}$ (viii) $\{\}$
 4. yes, $\{\}$ or $\emptyset$ 5. $\{a, b\}$ is a set containing two elements a and b while $\{\{a, b\}\}$ is a set containing one element $\{a, b\}$
 6. (i) 1 (ii) 4 (iii) 128 (iv) 256 (v) 4 (vi) 8
 7. (i) $\{\emptyset, \{9\}, \{11\}, \{9, 11\}\}$
 (ii) $\{\emptyset, \{+\}, \{-\}, \{\times\}, \{\div\}, \{+, -\}, \{+, \times\}, \{+, \div\}, \{-, \times\}, \{-, \div\}, \{\times, \div\}, \{+, -\times\}, \{+, -\div\}, \{+, \times\div\}, \{-, \times\div\}, \{+, -\times\div\}\}$
 (iii) $\{\emptyset, \{\emptyset\}\}$ (iv) $\{\emptyset, \{a\}, \{b, c\}\}, \{a, \{b, c\}\}$

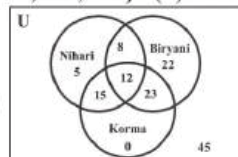
EXERCISE 3.2

1. (i) $A = \{6, 12, 18, 24, 30\}$, $B = \{8, 16, 24\}$ (ii) $A \cap B = \{24\}$ (iii) 
 2. (i) $G = \{1, 2, 4, 8, 16, 32, 64, 128\}$,
 $H = \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144\}$
 (ii) $G \cup H = \{1, 2, 4, 8, 9, 16, 25, 32, 36, 49, 64, 81, 100, 121, 128, 144\}$
 (iii) $G \cap H = \{1, 4, 16, 64\}$
 3. (i) $P \cap Q = \{2, 3, 5, 7\}$ (ii) $P \cup Q = \{1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 15, 17, 19\}$
 7. 9 8. 130 9. 9 10. 18 11. (a) $\{1, 2, \dots, 49, 90, 91, \dots, 100\}$ (b) 40

12. (a) 5 (b)

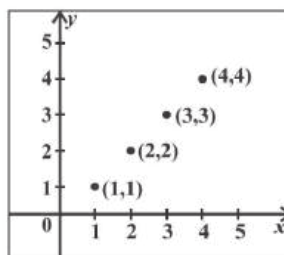


13. (a) 85 (b) 45 (c) 27 (d)

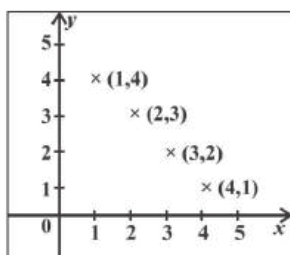


EXERCISE 3.3

1. (i) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$
 Domain of (i) = $\{1, 2, 3, 4\}$
 Range of (i) = $\{1, 2, 3, 4\}$

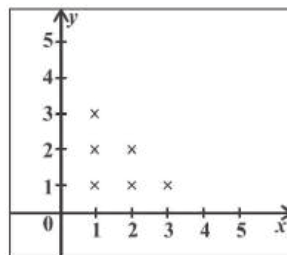


(ii)

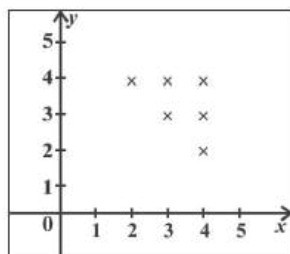


- $\{(1, 4), (2, 3), (3, 2), (4, 1)\}$
 Domain of (ii) = $\{1, 2, 3, 4\}$
 Range of (ii) = $\{1, 2, 3, 4\}$

- (iii) $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1)\}$
 Domain of (iii) = $\{1, 2, 3\}$
 Range of (iii) = $\{1, 2, 3\}$



(iv)



- $\{(2, 4), (3, 3), (3, 4), (4, 2), (4, 3), (4, 4)\}$
 Domain of (iv) = $\{2, 3, 4\}$
 Range of (iv) = $\{2, 3, 4\}$

2. Fig (1) does not represent a function. Fig (2) represents a function, which is a bijective function.
 Fig (3) represents a function, which is a bijective function.
 Fig (4) represents a function, which is an into function.

3. (i) 2 (ii) -7 (iii) 4 (iv) 2 (v) 17 (vi) $\frac{5}{4}$ 4. $a = 2, b = 1$ 5. $a = \frac{10}{3}, b = -\frac{5}{3}$
 6. $x = 6$ 7. $c = \frac{4}{3}, d = \frac{14}{3}$

REVIEW EXERCISE 3

1. (i) b (ii) c (iii) a (iv) d (v) d (vi) b (vii) b (viii) d (ix) a (x) b
 2. (i) $\{2, 4, 6, 8, 10, \dots\}$ (ii) $\{3, 5, 7, 9, 11, \dots\}$ (iii) $\{0, 11, 22, 33, 44, 55, 66, 77, 88, 99, 110\}$
 (iv) $\emptyset$ (v) $\emptyset$ (vi) $\emptyset$ (vii) $\{0\}$ (viii) Q 3. (i) $\{1, 3, 5, 7, 9\}$
 (ii) $\{6, 7, 8, 9, 10\}$ (iii) $\{1, 2, 3, 4, 5, 6, 8, 10\}$ (iv) $\{6, 8, 10\}$ (v) $\emptyset$
 (vi) $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ (vii) $\{1, 3, 5, 7, 9\}$ (viii) $\emptyset$

8. $\{10, 20, 30, 40, 50, \dots\}$ 10. (i) -2 (ii) -9 (iii) $-\frac{41}{3}$ (iv) 5 (v) 645 (vi) 95
 11. $a = \frac{7}{6}, b = \frac{16}{3}$ 12. 15 13. $m = \frac{15}{16}, n = 5$
 14. $A = \{1, 2, 3, \dots, 30\}, B = \{31, 32, 33, \dots, 55\}, C = \{76, 77, 78, \dots, 100\}$.
 $A \cup B \cup C = \{1, 2, 3, \dots, 30, 31, 32, \dots, 55, 76, 77, \dots, 100\}$
 15. (a) 150 (b) 30 (c) 30 (d) 90 16. (a) 260 (b) 160 (c) 40 (d) 50

EXERCISE 4.1

1. (i) $6(x+2)$ (ii) $5y(3y+4)$ (iii) $-3x(4x+1)$ (iv) $4ab(a+2b)$ (v) $x(y-3x+2)$
 (vi) $3ab(a-3b+5)$ 2. (i) $5(x+3)$ (ii) $(x+1)(x+3)$ (iii) $(x+2)(x+4)$ (iv) $(x+2)^2$
 3. (i) $(x+4)(x-3)$ (ii) $(x+5)(x+2)$ (iii) $(x-4)(x-2)$ (iv) $(x-8)(x+7)$
 (v) $(x-12)(x+2)$ (vi) $(y+6)(y-2)$ (vii) $(y+9)(y+4)$ (viii) $(x-2)(x+1)$
 4. (i) $(2x+1)(x+3)$ (ii) $(2x+5)(x+3)$ (iii) $(4x+1)(x+3)$ (iv) $(3x+2)(x+1)$
 (v) $(3y-2)(y-3)$ (vi) $(2y-1)(y-2)$ (vii) $(4z-3)(z-2)$ (viii) $(3x+2)(3-x)$

EXERCISE 4.2

1. (i) $(2x^2 - 6xy + 9y^2)(2x^2 + 6xy + 9y^2)$ (ii) $(a^2 - 4ab + 8b^2)(a^2 + 4ab + 8b^2)$
 (iii) $(x^2 - 2x + 4)(x^2 + 2x + 4)$ (iv) $(x^2 - 4x + 1)(x^2 + 4x + 1)$
 (v) $(x^2 - 6xy + 3y^2)(x^2 + 6xy + 3y^2)$ (vi) $(x^2 - 3xy + y^2)(x^2 + 3xy + y^2)$
 2. (i) $(x^2 + 5x + 5)^2$ (ii) $(x^2 - 5x + 3)(x^2 - 5x - 13)$
 (iii) $(2x^2 + 7x + 4)^2$ (iv) $(3x^2 + 5x + 6)(3x^2 + 5x + 2)$
 (v) $(x^2 + 4x + 6)(x^2 + 8x + 6)$ (vi) $(x^2 - 5x + 2)(x^2 + 5x + 2)$
 3. (i) $(2x+1)^3$ (ii) $(3a+4b)^3$ (iii) $(x+6y)^3$ (iv) $(2x-5y)^3$
 4. (i) $(5a-1)(25a^2+5a+1)$ (ii) $(4x+5)(16x^2-20x+25)$ (iii) $(x^2-3)(x^4+3x^2+9)$
 (iv) $(10a+1)(100a^2-10a+1)$ (v) $(7x+6)(49x^2-42x+36)$ (vi) $(3-8y)(9+24y+64y^2)$

EXERCISE 4.3

1. (i) $\text{HCF} = 7xy$ (ii) $\text{HCF} = 2x-3y$ (iii) $\text{HCF} = x^2+x+1$ (iv) $\text{HCF} = a(a+3)$
 (v) $\text{HCF} = t+1$ (vi) $\text{HCF} = x+8$ 2. (i) $\text{HCF} = 3x-2$ (ii) $\text{HCF} = x^2-4x+3$
 (iii) $\text{HCF} = 2(x^2+1)$ (iv) $\text{HCF} = x(x-2)$ 3. (i) $\text{LCM} = 12a^2b^2$ (ii) $\text{LCM} = x^2(x+1)$
 (iii) $\text{LCM} = a(a-2)^2$ (iv) $\text{LCM} = x(x^4-16)$ (v) $\text{LCM} = 4(4-x^2)(x+3)$ 4. $y^2-12y+35$
 5. $q(x) = 9x^3(x^3-a^3)$ 6. $12x^2(x-a)(x+a)^3$

EXERCISE 4.4

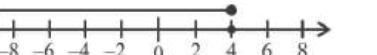
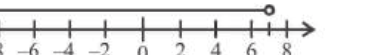
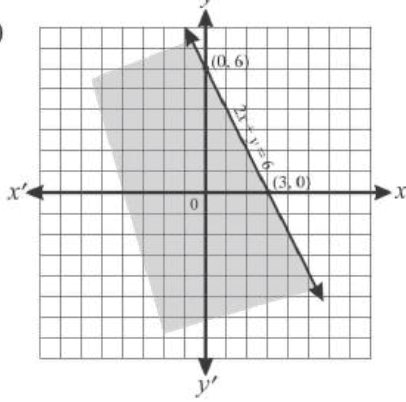
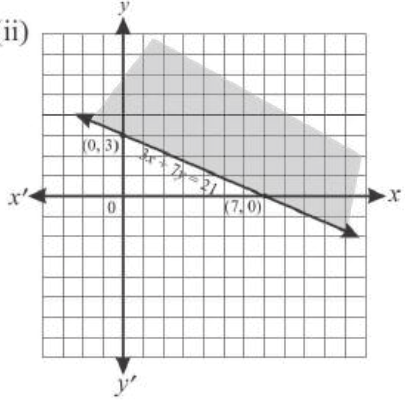
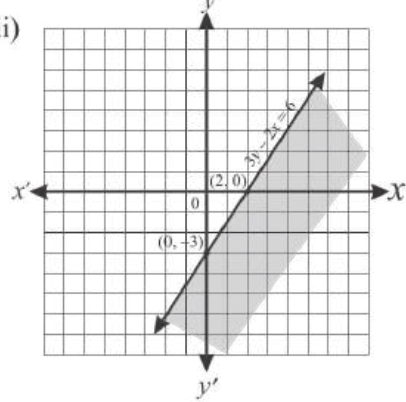
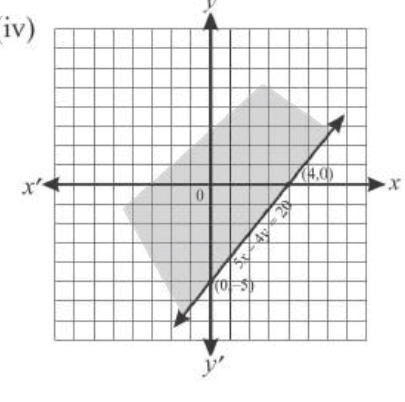
1. (i) $\pm(x-4)$ (ii) $\pm(3x \pm 2)$ (iii) $\pm(6a+7)$ (iv) $\pm(8y-2)$
 (v) $\pm\sqrt{2}(10t-3)$ (vi) $\pm\sqrt{10}(2x+3)$
 2. (i) $\pm(2x^2-7x-3)$ (ii) $\pm(11x^2-9x-12)$ (iii) $\pm(x^2-5xy+y^2)$ (iv) $\pm(2x^2-3x+7)$
 3. $x=2$ or $x=4$ 4. $x=5$ 5. $x=0, x=1$ or $x=2$ 6. $x=1$ or $x=3$

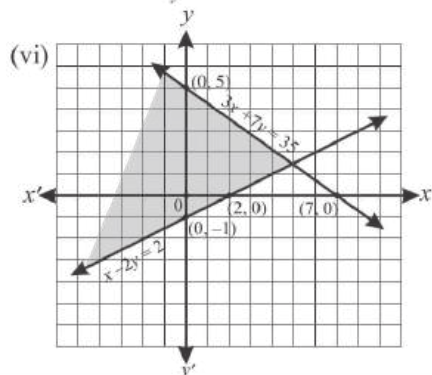
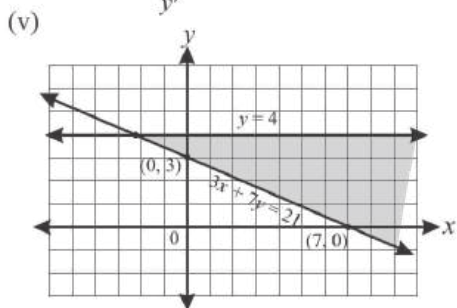
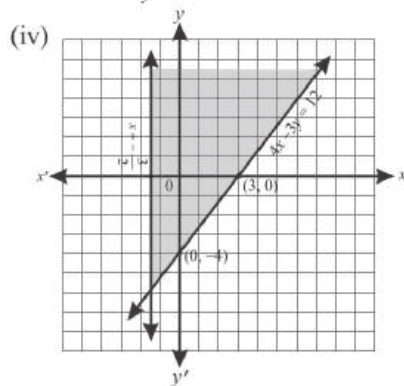
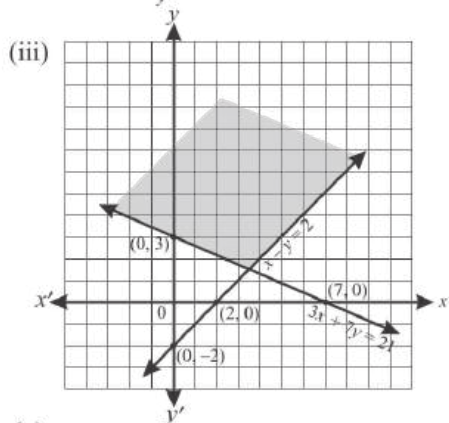
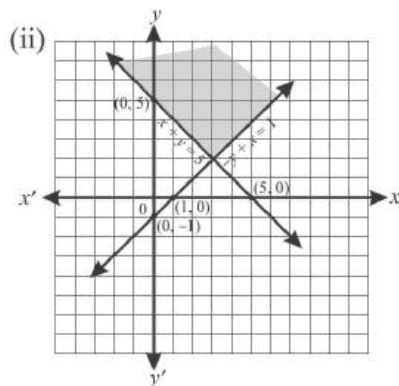
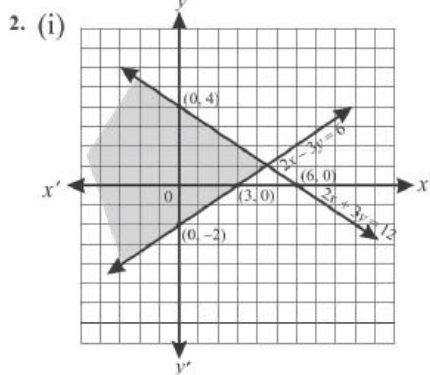
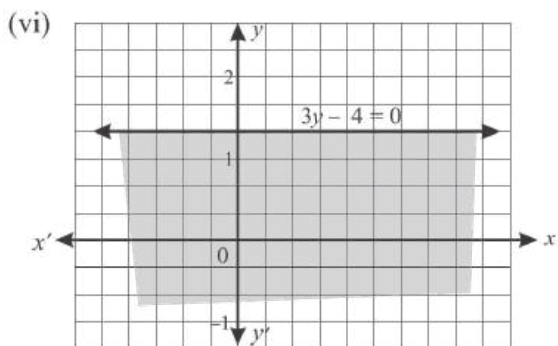
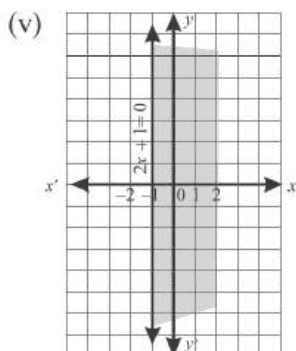
REVIEW EXERCISE 4

1. (i) a (ii) b (iii) b (iv) c (v) c (vi) a (vii) c (viii) a (ix) c (x) a

2. (i) $2x(2x^2+9x-6)$ (ii) $(x+4y)(x^2-4xy+16y^2)$ (iii) $(xy-2)(x^2y^2+2xy+4)$
 (iv) $-(x+3)(x+20)$ (v) $(2x+1)(x+3)$ (vi) $(x^2+4x+8)(x^2-4x+8)$ (vii) $(x^2+2x+3)(x^2-2x+3)$
 (viii) $x(x+9)(x^2+9x+38)$ (ix) $(x^2+6x-3)^2$
 3. (i) LCM = $8x^2(x+2)(x+3)$, HCF = $4x$ (ii) LCM = $x(x-1)(x-3)(x+4)$, HCF = $x-1$
 (iii) LCM = $(x-4)(x+4)^2$, HCF = $x+4$ (iv) LCM = $x(x+2)(x^2-9)$, HCF = $x-3$
 4. $\pm(4x^2+1)$ 5. 3 years or 5 years

EXERCISE 5.1

1. (i) $x = -3$  (ii) $x = -54$ 
- (iii) $x = -\frac{1}{3}$  (iv) $x = -1$ 
- (v) $x = -14$  (vi) $x = 6$ 
2. (i) $x \leq 4$  (ii) $x < 7$ 
- (iii) $x \geq 0$  (iv) $x \leq -10$ 
- (v) $x < \frac{2}{5}$  (vi) $x \geq 2$ 
3. (i)  (ii) 
- (iii)  (iv) 

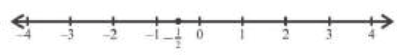
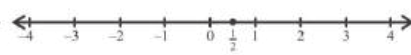


EXERCISE 5.2

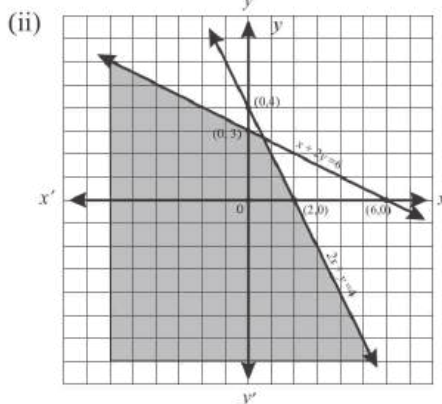
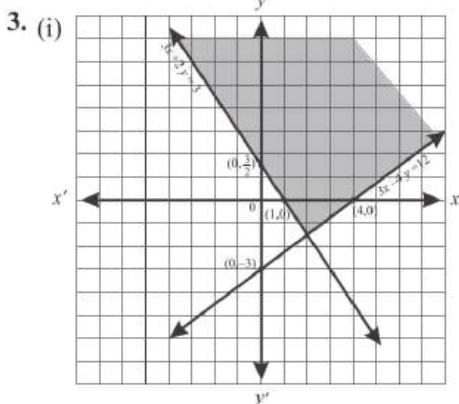
- Maximum at the corner point (16, 12)
- Maximum at the corner point (0, 5)
- Maximum at the corner point (0, 4)
- Minimum at the corner point (0, 3)
- Maximum at the corner point (2, 6)
- Maximum at the corner point (9, 0) and minimum at the corner point (0, 3)

REVIEW EXERCISE 5

1. (i) c (ii) c (iii) c (iv) d (v) b (vi) b (vii) b (viii) c (ix) b (x) b

2. (i) $x = -\frac{1}{2}$  (ii) $x = \frac{1}{2}$ 

(iii) $x < 3$  (iv) $x \leq -1$ 



4. Maximum at the corner point (0, 4).

5. Minimum at the corner point $\left(\frac{3}{2}, \frac{1}{2}\right)$.

EXERCISE 6.1

- (i) $1^{\text{st}}, 425^{\circ}$ (ii) $2^{\text{nd}}, -225^{\circ}$ (iii) $4^{\text{th}}, 320^{\circ}$ (iv) $3^{\text{rd}}, -150^{\circ}$ (v) $3^{\text{rd}}, 210^{\circ}$
- (i) $123^{\circ} 27' 21.6''$ (ii) $58^{\circ} 47' 20.76''$ (iii) $90^{\circ} 34' 4.08''$
- (i) 65.5375° (ii) 42.3125° (iii) 78.76°
- (i) $\frac{\pi}{5}$ rad (ii) $\frac{\pi}{8}$ (iii) $\frac{3\pi}{8}$ rad
- (i) 11.25° (ii) 396° (iii) 210°
- (i) (a) 6.28 cm (b) 18.84 cm^2 (ii) (a) 4 cm (b) 3.06 cm^2
- $75.4 \text{ cm}^2, 16.67\%$ 8. 6.25% 9. 12 cm, 5 cm

EXERCISE 6.2

- (a) (i) $\frac{4}{5}$ (ii) $\frac{3}{5}$ (iii) $\frac{4}{3}$ (iv) $\frac{5}{3}$ (v) $\frac{5}{4}$ (vi) $\frac{4}{3}$ (vii) $\frac{3}{4}$ (viii) $\frac{5}{3}$ (ix) $\frac{5}{4}$ (x) $\frac{4}{5}$
 (b) (i) $\frac{8}{17}$ (ii) $\frac{15}{17}$ (iii) $\frac{8}{15}$ (iv) $\frac{17}{15}$ (v) $\frac{17}{8}$ (vi) $\frac{8}{15}$ (vii) $\frac{15}{8}$ (viii) $\frac{17}{15}$ (ix) $\frac{17}{8}$ (x) $\frac{8}{17}$

(c) (i) $\frac{5}{13}$ (ii) $\frac{12}{13}$ (iii) $\frac{5}{12}$ (iv) $\frac{13}{5}$ (v) $\frac{13}{12}$ (vi) $\frac{5}{12}$ (vii) $\frac{12}{5}$ (viii) $\frac{13}{12}$ (ix) $\frac{13}{5}$ (x) $\frac{5}{13}$

2. (i) $\frac{c}{b}$ (ii) $\frac{a}{b}$ (iii) $\frac{c}{a}$ (iv) $\frac{a}{b}$ (v) $\frac{c}{b}$ (vi) $\frac{a}{c}$

4. (i) $\cos 60^\circ$ (ii) $\sin 60^\circ$ (iii) $\cot 60^\circ$ (iv) $\cot 30^\circ$ (v) $\cos 30^\circ$ (vi) $\sin 30^\circ$ (vii) $\cos 45^\circ$
 (viii) $\cot 45^\circ$ (ix) $\sin 45^\circ$ 5. (i) $\frac{a}{b}$ (ii) $\frac{a}{b}$ (iii) $\frac{c}{a}$ (iv) $\frac{b}{c}$ (v) $\frac{a}{c}$
 (vi) $\frac{a}{b}$ (vii) $\frac{c}{b}$ (viii) $\frac{a}{c}$ (ix) $\frac{b}{c}$ (x) $\frac{c}{a}$

EXERCISE 6.3

1. (i) $\cos \theta = \frac{\sqrt{5}}{3}$, $\tan \theta = \frac{2}{\sqrt{5}}$, $\operatorname{cosec} \theta = \frac{3}{2}$, $\sec \theta = \frac{3}{\sqrt{5}}$, $\cot \theta = \frac{\sqrt{5}}{2}$
 (ii) $\sin \theta = \frac{\sqrt{7}}{4}$, $\tan \theta = \frac{\sqrt{7}}{3}$, $\operatorname{cosec} \theta = \frac{4}{\sqrt{7}}$, $\sec \theta = \frac{4}{3}$, $\cot \theta = \frac{3}{\sqrt{7}}$
 (iii) $\sin \theta = \frac{1}{\sqrt{5}}$, $\cos \theta = \frac{2}{\sqrt{5}}$, $\operatorname{cosec} \theta = \sqrt{5}$, $\sec \theta = \frac{\sqrt{5}}{2}$, $\cot \theta = 2$
 (iv) $\sin \theta = \frac{2\sqrt{2}}{3}$, $\cos \theta = \frac{1}{3}$, $\tan \theta = 2\sqrt{2}$, $\operatorname{cosec} \theta = \frac{3}{2\sqrt{2}}$, $\cot \theta = \frac{1}{2\sqrt{2}}$
 (v) $\sin \theta = \sqrt{\frac{2}{5}}$, $\cos \theta = \sqrt{\frac{3}{5}}$, $\tan \theta = \sqrt{\frac{2}{3}}$, $\operatorname{cosec} \theta = \sqrt{\frac{5}{2}}$, $\sec \theta = \sqrt{\frac{5}{3}}$

EXERCISE 6.4

1. (i) $\frac{1}{2}$ (ii) $\frac{\sqrt{3}}{2}$ (iii) $\frac{\sqrt{3}}{3}$ (iv) $\sqrt{3}$ (v) 2 (vi) $\frac{1}{2}$ (vii) $\frac{\sqrt{3}}{3}$ (viii) $\frac{\sqrt{3}}{2}$
 (ix) $\frac{2\sqrt{3}}{3}$ (x) 2 (xi) $\frac{\sqrt{2}}{2}$ (xii) $\frac{\sqrt{2}}{2}$
 2. (i) $\frac{\sqrt{3}}{2}$ (ii) $\frac{\sqrt{3}}{2}$ (iii) $2\sqrt{2}$ (iv) 1 (v) 0 (vi) $\frac{1}{2}$ (vii) $\frac{\sqrt{3}}{2}$ (viii) 2
 3. (i) 0 (ii) $\frac{7}{\sqrt{2}}$ (iii) $\sqrt{2}$

EXERCISE 6.5

1. (i) $x = \frac{4}{\sqrt{3}} \text{ cm}$, $z = \frac{8}{\sqrt{3}} \text{ cm}$ (ii) $x = \sqrt{3} \text{ cm}$, $z = \sqrt{6} \text{ cm}$ (iii) $x = 1 \text{ cm}$, $y = \sqrt{3} \text{ cm}$ (iv) $x = 4 \text{ cm}$, $z = 4\sqrt{2} \text{ cm}$
 2. (i) $b = 4 \text{ cm}$, $m\angle A = 25.64^\circ$, $m\angle C = 64.36^\circ$ (ii) $b = 4\sqrt{2} \text{ cm}$, $m\angle A = m\angle C = 45^\circ$ 3. $60\sqrt{2} \text{ m}$
 4. (i) $a = 3 \text{ cm}$, $b = 6 \text{ cm}$, $m\angle A = 30^\circ$ (ii) $b = 8\sqrt{2} \text{ cm}$, $c = 8 \text{ cm}$, $m\angle A = 45^\circ$

- (iii) $b = 6\sqrt{5}$ cm, $m\angle A = 63.4^\circ$, $m\angle C = 26.6^\circ$ (iv) $b = 8$ cm, $a = 4\sqrt{3}$ cm, $m\angle C = 30^\circ$
 (v) $a = \frac{4}{\sqrt{3}}$ cm, $b = \frac{8}{\sqrt{3}}$ cm, $m\angle C = 60^\circ$ (vi) $c = 8$ cm, $m\angle A = 36.9^\circ$, $m\angle C = 53.1^\circ$
 5. 12 m, 1.18 rad 6. $5\sqrt{5}$ cm 7. 7.75 m 8. 8 m 9. (i) 16 cm (ii) 5 cm

EXERCISE 6.6

1. 69.28 m 2. 2.89 cm 3. 35.7° 4. 11.55 m 5. 86.6 m 6. 49.98° 7. 33.69°
 8. 87.4 m 9. 142.5 m, 109.2 m 10. 91.92 m

REVIEW EXERCISE 6

1. (i) d (ii) a (iii) a (iv) b (v) c (vi) b (vii) d (viii) a (ix) d (x) a
 2. (a) (i) $\frac{17\pi}{12}$ rad (ii) $\frac{101\pi}{240}$ rad (iii) $\frac{19\pi}{24}$ rad (b) (i) $127^\circ 30'$ (ii) 105° (iii) $123^\circ 45'$
 4. $\sin \theta = \frac{3}{\sqrt{11}}$, $\cos \theta = \sqrt{\frac{2}{11}}$, $\csc \theta = \frac{\sqrt{11}}{3}$, $\sec \theta = \sqrt{\frac{11}{2}}$, $\cot \theta = \frac{\sqrt{2}}{3}$
 5. 56.42 m 6. 9.06 m

EXERCISE 7.1

1. (i) Right half plane (ii) The 1st quadrant (iii) y-axis (iv) x-axis (v) 4th quadrant and negative y-axis (vi) Origin (vii) It is a line bisecting 1st and 3rd quadrant.
 (viii) The set of points lying on and right side of the line $x = 3$.
 (ix) The set of points lying above x-axis. (x) The set of points in 2nd and 4th quadrants.
 2. (i) $3\sqrt{13}$ (ii) $4\sqrt{5}$ (iii) $\sqrt{53}$ (iv) $\sqrt{113}$ 3. (i) (a) $5\sqrt{2}$ (b) $2\sqrt{29}$
 (c) $\frac{2\sqrt{109}}{3}$ (ii) (a) $\left(\frac{1}{2}, -\frac{3}{2}\right)$ (b) $(-3, 1)$ (c) $\left(-2\sqrt{5}, \frac{7}{3}\right)$ 4. (i) $(\sqrt{176}, 7)$ is at distance of 15 units from the origin. (ii) $(10, -10)$ is not a distance of 15 units from the origin.
 (iii) $(1, 15)$ is not a distance from the origin. 6. $h = 0$
 7. $h = 1$ 8. $C(0, -3)$; radius $= \sqrt{26}$ 9. $h = -10$ or $h = 6$

Exercise 7.2

1. (i) $m = 1$, $\alpha = 45^\circ$ (ii) $m = -9$, $\alpha = 96^\circ 20'$ (iii) $m = \infty$, $\alpha = 90^\circ$ 3. (i) $k = -11$
 (ii) $k = \frac{23}{2}$ 5. (a) lines are neither parallel nor perpendicular.
 (b) lines are neither parallel nor perpendicular. 6. (a) $y + 9 = 0$ (b) $x + 5 = 0$
 (c) $7x - y + 47 = 0$ (d) $y + 3 = 0$ (e) $x + 8 = 0$ (f) $x - 7y - 16 = 0$
 (g) $5x + y + 7 = 0$ (h) $4x - 3y + 12 = 0$ (i) $4x + y + 36 = 0$
 7. $4x + 2y - 37 = 0$ 8. $2x - 3y - 10 = 0$ 9. $24x + y - 259 = 0$
 10. (a) (i) $y = \frac{1}{2}x + \frac{11}{4}$ (ii) $\frac{x}{-11} + \frac{y}{11} = 1$ (iii) $x \cos(116.57^\circ) + y \sin(116.57^\circ) = \frac{11}{2\sqrt{5}}$

(b) (i) $y = \frac{-4}{7}x + \frac{2}{7}$ (ii) $\frac{x}{\frac{1}{2}} + \frac{y}{\frac{2}{7}} = 1$ (iii) $x \cos(60.26^\circ) + y \sin(60.26^\circ) = \frac{2}{\sqrt{65}}$

(c) (i) $y = \frac{8}{15}x - \frac{1}{5}$ (ii) $\frac{x}{\frac{3}{8}} + \frac{y}{-\frac{1}{5}} = 1$ (iii) $x \cos(298.07^\circ) + y \sin(298.07^\circ) = \frac{3}{17}$

11. (a) Parallel (b) Perpendicular (c) neither parallel nor perpendicular.

12. $2x - 7y + 57 = 0$ 13. $x + y + 3 = 0$

Exercise 7.3

1. $\sqrt{85} \approx 9.22$ km 2. (10, 5) 3. $\sqrt{61} \approx 7.81$ m 4. $\sqrt{89} \approx 9.43$ km
5. (6, 11) 6. (5, 7) 7. $4\sqrt{29} \approx 21.5$ units 8. 26 units 9. $10\sqrt{5} \approx 22.4$ units
10. Perimeter = 20 units 11. 16 units

REVIEW EXERCISE 7

1. (i) c (ii) a (iii) b (iv) a (v) b (vi) a (vii) b (viii) a (ix) c (x) d
2. $5\sqrt{2}$ 3. $\left(-1, \frac{1}{2}\right)$ 4. $\frac{4}{3}$ 5. $y = 2x + 1$ 6. $\frac{2}{3}$ 7. $\sqrt{97} \approx 9.85$ units
8. (6, 5) 9. $\frac{3}{2}, 4\sqrt{13} \approx 14.4$ units 10. (a) $y = -3x + 2$ (b) $y - 2 = -3(x - 1)$
(c) $\frac{y-2}{-7-2} = \frac{x-1}{4-1}$ (d) $\frac{y}{2} + \frac{x}{\frac{2}{3}} = 1$ (e) $\frac{y}{\sqrt{10}} + \frac{3x}{\sqrt{10}} = \frac{2}{\sqrt{10}}$ (f) $x \cos(-71.56^\circ) + y \sin(-71.56^\circ) = \frac{2}{\sqrt{10}}$

EXERCISE 8

1. (i) a (ii) d (iii) c (iv) a (v) b (vi) a (vii) c (viii) b (ix) c (x) b

EXERCISE 9.1

1. Similar 3. $m\overline{DF} = 10$ cm, $m\overline{EF} = 8$ cm 4. (i) $x = 3$ cm (ii) $x = 2.25$ cm (iii) $x = 2.19$ cm
5. 10 cm 6. 7.11 m 7. $x = 10\frac{2}{3}$ cm, $y = 8$ cm, $z = 13\frac{1}{3}$ cm 8. $m\overline{CE} = 1.5$ cm 9. $\frac{18\sqrt{2}}{5}$

EXERCISE 9.2

1. (i) 1:9 (ii) 9:16 (iii) 4:49 (iv) 64:81 (v) 36:25 2. (i) 86.4 cm²
(ii) 106.67 cm² (iii) 7.03125 cm² (iv) 150 cm² (v) 12.6 cm 3. (a) 100 cm²
(b) 64 cm² 4. $5\frac{5}{9}$ cm² 5. 1024 cm² 6. $\frac{4}{5}$ 7. 22.5 cm² 8. 289 cm²

EXERCISE 9.3

1. $\frac{27}{64}$ 2. $\frac{2}{3}$ 3. (i) $\frac{4}{5}$ (ii) $\frac{16}{25}$ 4. (i) 648 cm³ (ii) 4 cm³ (iii) 2744 cm³ (iv) 8 cm 5. (i) 42.67 m²
(ii) 810 cm³ 6. (i) 90 m² (ii) 1250 m³

EXERCISE 9.4

1. (i) 1440° (ii) 120° (iii) 72° (iv) 9 sides 2. 42.42 cm² 3. $m\angle ABC = 110^\circ$

- $m\angle BCD = 70^\circ$, $m\angle CDA = 110^\circ$ 4. The shape can tessellate, with interior angles summing to 360° .
 5. 600 reflections needed to cover the square. 6. 1623.8 cm^2 , 190 cm 7. 180 tiles
 8. 35 gallons 9. 6 litres 10. 4.5 m^2

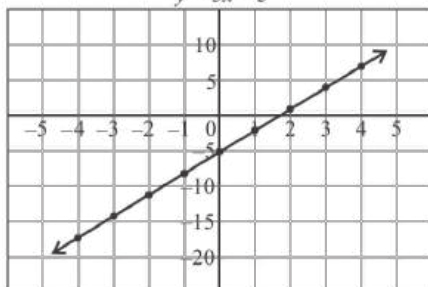
REVIEW EXERCISE 9

1. (i) a (ii) b (iii) b (iv) d (v) c (vi) d (vii) c (viii) a (ix) b (x) d
 3. 4:1, 8:1 4. (a) 1:100 (b) 1:1000 (c) 1:10 (d) 1:1 5. 1.69 litres, 4 litres
 6. 125 millilitres, 216 millilitres 7. (a) 1:50 (b) 1:125000 (c) 3 cm (d) 7500 cm^2
 8. (a) 12:13 (b) 1728:2197 10. 6.69 m^2

EXERCISE 10.1

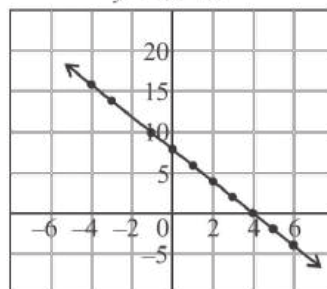
1. (i)

$$y = 3x - 5$$



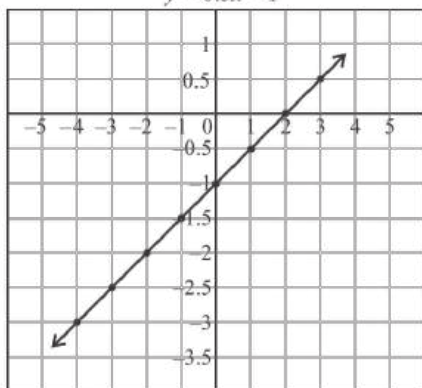
(ii)

$$y = -2x + 8$$



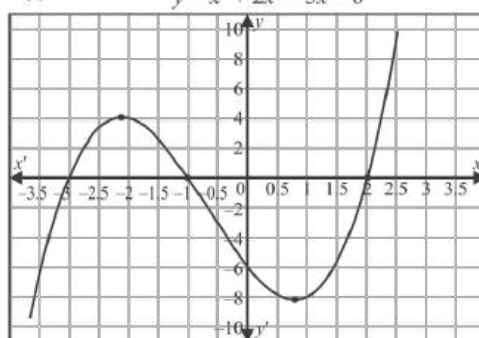
(iii)

$$y = 0.5x - 1$$



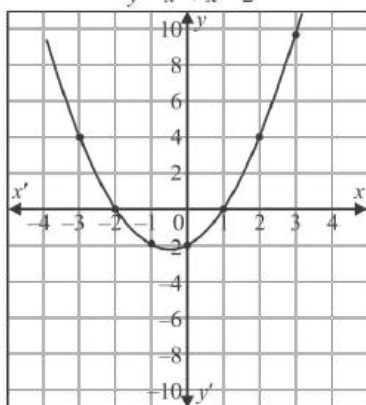
2. (i)

$$y = x^3 + 2x^2 - 5x - 6$$



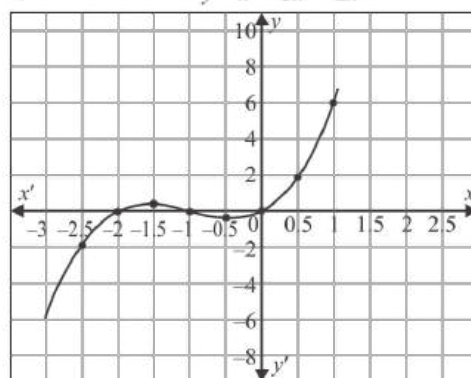
(ii)

$$y = x^2 + x - 2$$

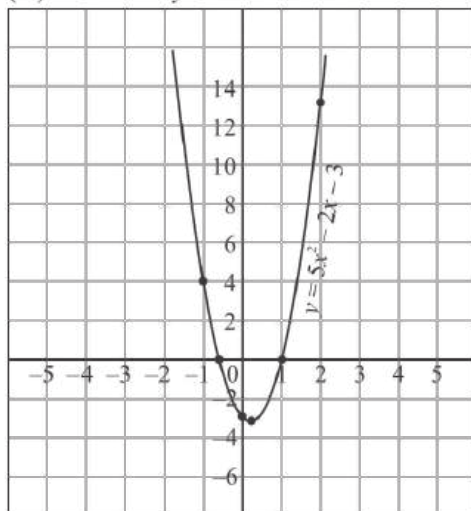


(iii)

$$y = x^3 + 3x^2 + 2x$$

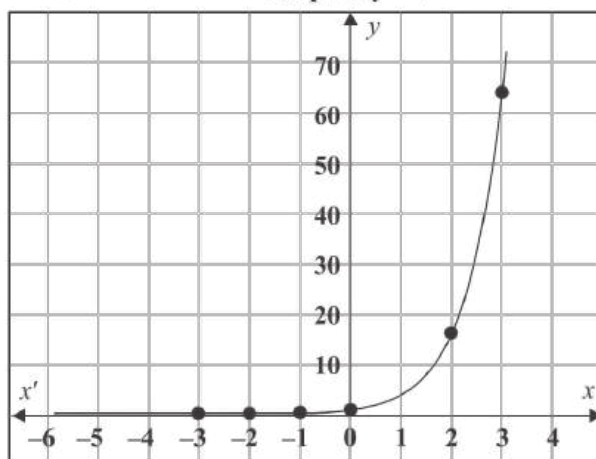


(iv) $y = 5x^2 - 2x - 3$



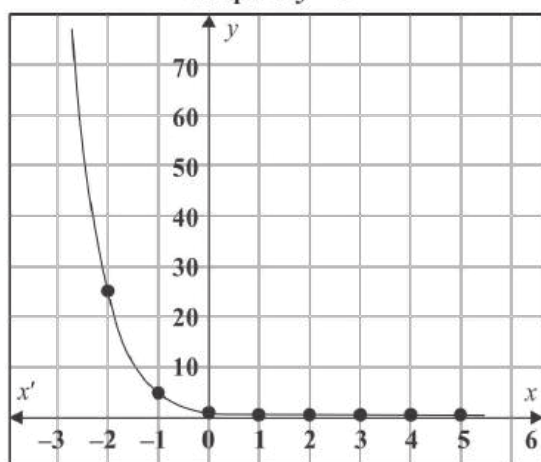
3. (i)

Graph of $y = 4^x$



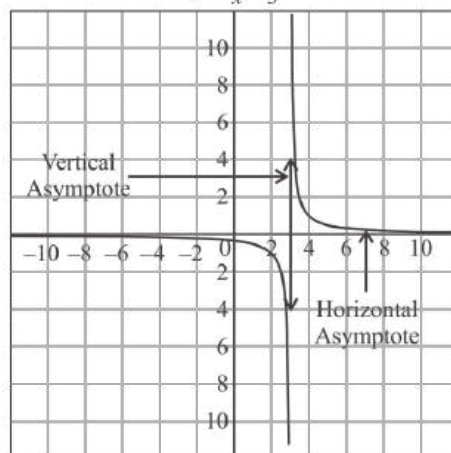
(ii)

Graph of $y = 5^{-x}$



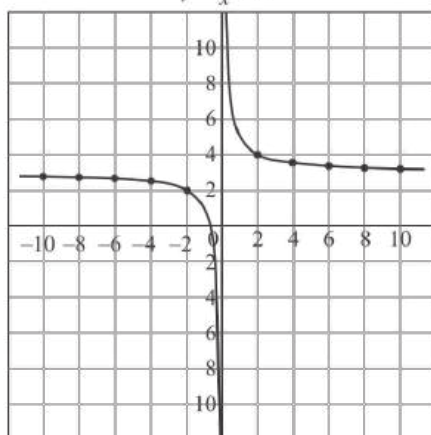
(iii)

$y = \frac{1}{x-3}$



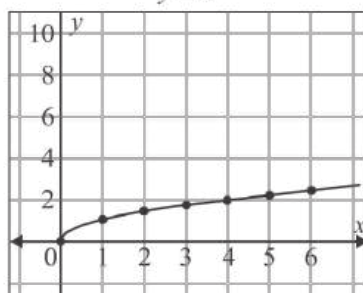
(iv)

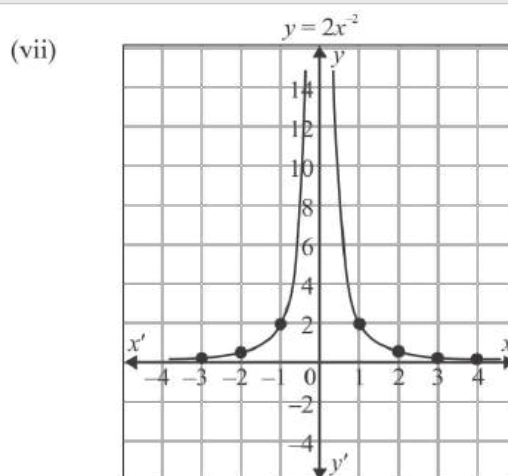
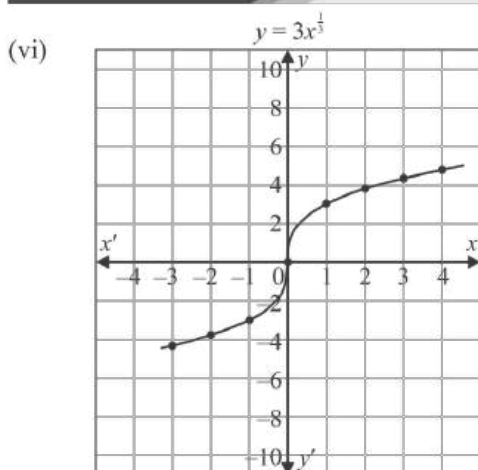
$y = \frac{2}{x} + 3$



(v)

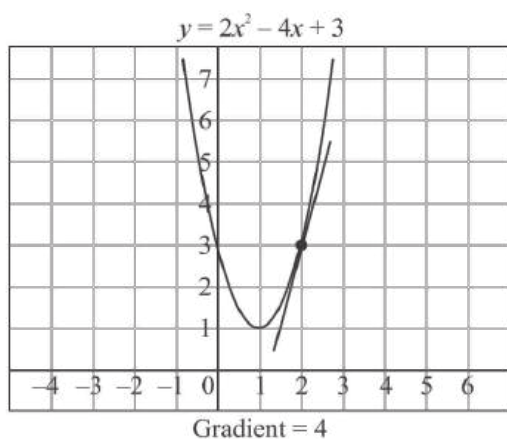
$y = x^{\frac{1}{2}}$



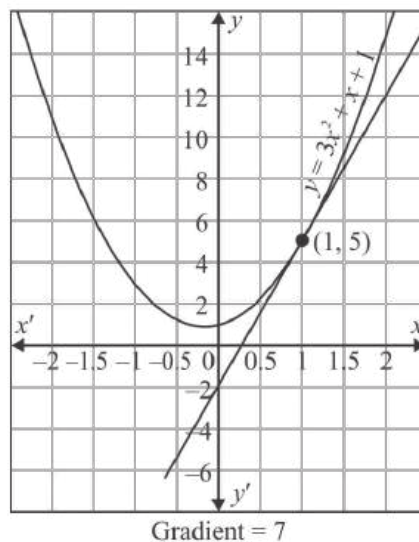


EXERCISE 10.2

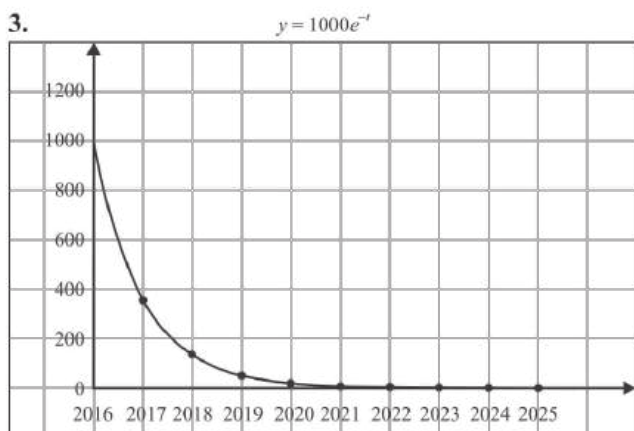
1.



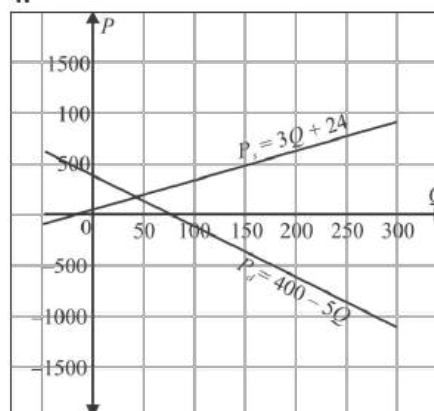
2.



3.

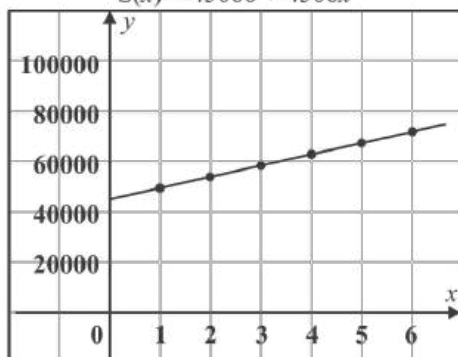


4.

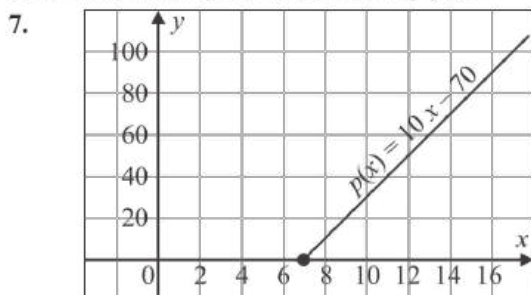


- (b) From the graph, students' strength in 2019 is approximately 50, and in 2023 approximately 1.

5. $S(x) = 45000 + 4500x$

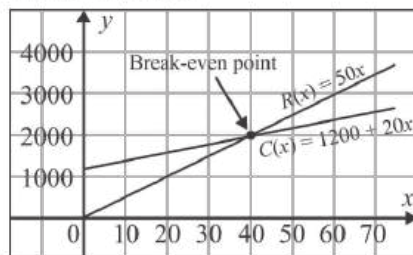


Shahid's salary increases linearly with years of service and rises by Rs. 4500 for every year.

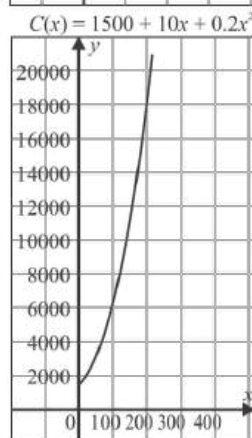


Profit for 500 newspapers = Rs. 4930

6. (a) $x = 40$ bags
(b) Profit = Rs. 6300
(c)



8.

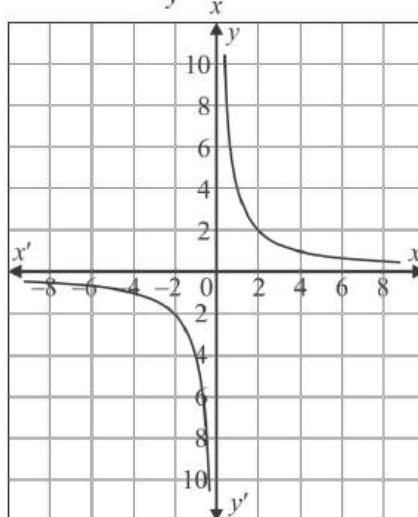
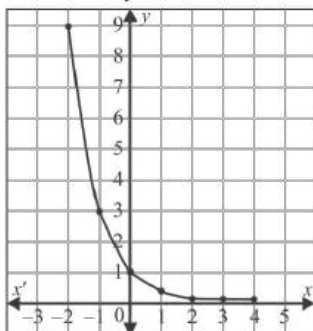


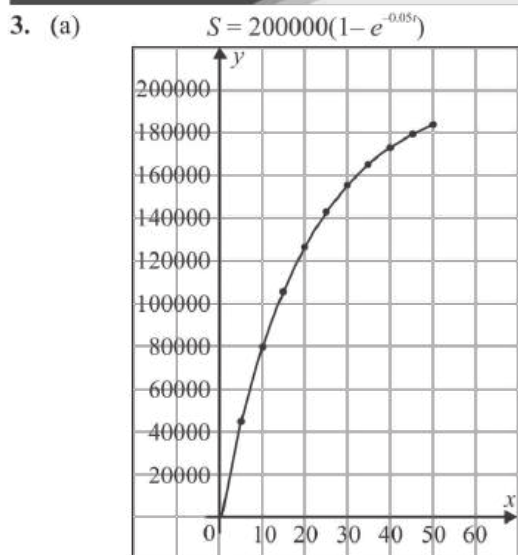
Cost of 200 shirts = Rs. 11500

REVIEW EXERCISE 10

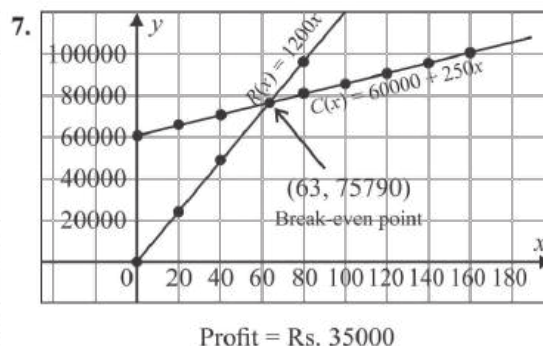
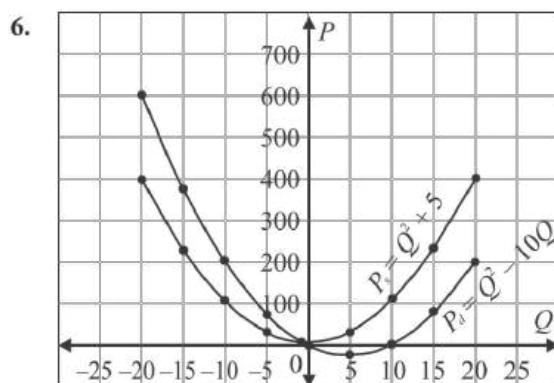
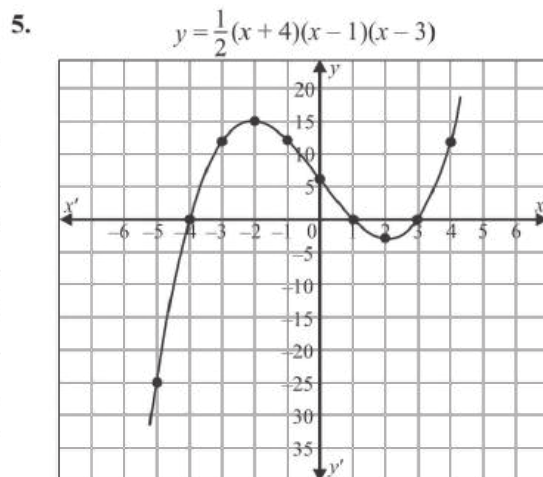
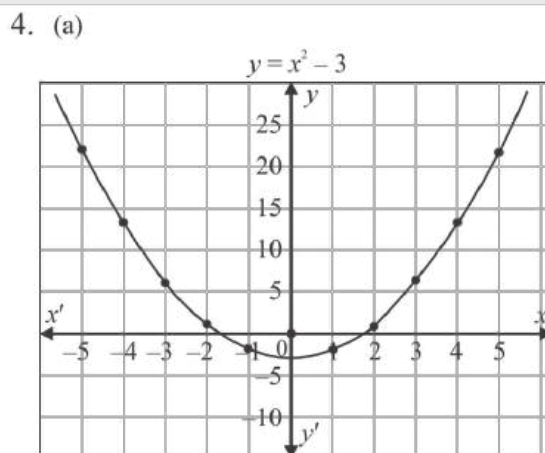
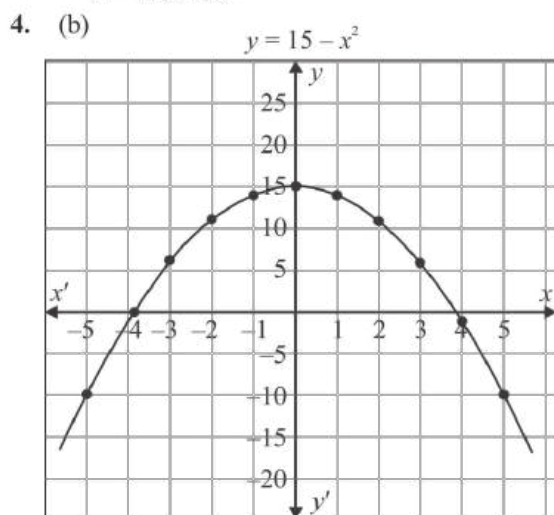
1. (i) d (ii) c (iii) c (iv) a (v) a (vi) b (vii) a (viii) d (ix) b (x) b

2. (i) $y = 3^{-x}$ (ii) $y = \frac{2}{x}$





(b) For $t = 5$, $S = 44239.84$ and for $t = 35$, $S = 165245.2$



REVIEW EXERCISE 11

1. (i) b (ii) a (iii) c (iv) a (v) a (vi) a (vii) b (viii) d (ix) c (x) c

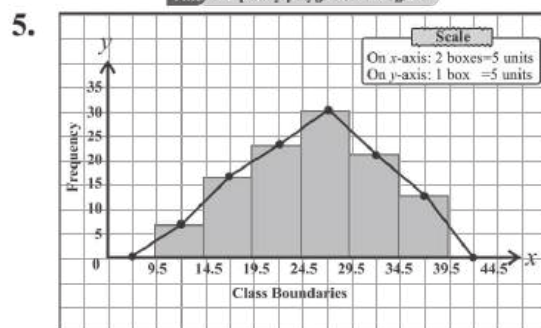
EXERCISE 12.1

1. (i) 53 (ii) 39 (iii) 36 (iv) 6 and 15 (v) 5 (vi) (24 – 28) (vii) 44 (viii) 44

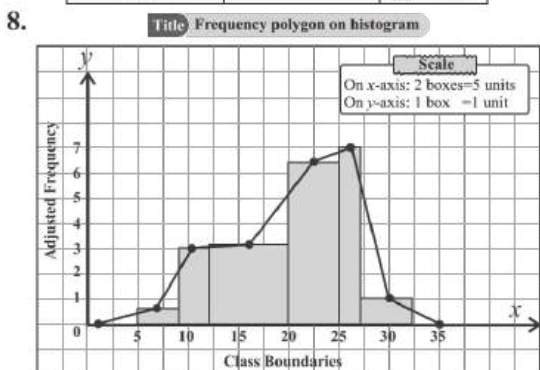
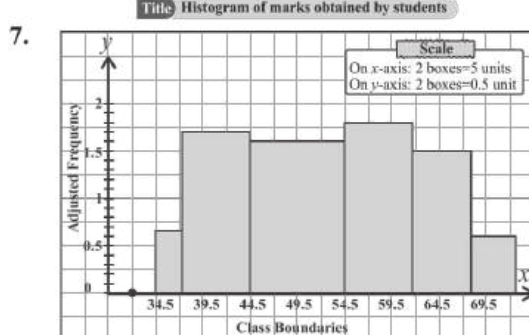
Class limits	Tally marks	f
144 – 146		4
147 – 149		3
150 – 152		7
153 – 155		5
156 – 158		4
159 – 161		4
162 – 164		1
165 – 167		2
Total		$\Sigma f = 30$

Class limits	Tally marks	f
15 – 19		2
20 – 24		3
25 – 29		5
30 – 34		10
35 – 39		6
40 – 44		4
Total		$\Sigma f = 30$

Class limits	Tally marks	f
33 – 38		1
39 – 44		6
45 – 50		15
51 – 56		4
57 – 62		6
Total		$\Sigma f = 30$



No. of heads	Tally marks	f
0		5
1		7
2		9
3		14
4		9
5		6
Total		$\Sigma f = 50$



EXERCISE 12.2

1. (i) 16.67 (ii) $\bar{X} = 0$ (iii) $\bar{X} = 14.04$
 (iv) $\bar{X} = 14.57$ 2. Median height = 56.5 inches
 3. (i) $\bar{X} = 92.1$ (ii) $X = 90$ (iii) $\hat{X} = 90$ and 95
 4. (i) $\Sigma f = 84$, $\Sigma fX = 2223$, $\bar{X} = 26.46$
 (ii) Median = 26.64, $c.f. = 9, 27, 62, 79, 84$
 5. Mode = 17.44

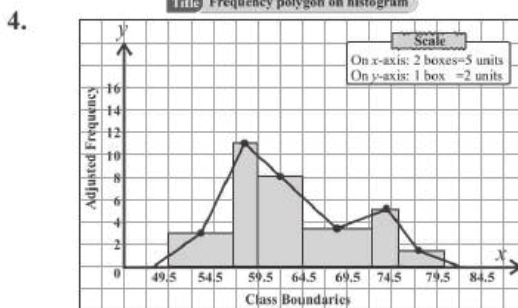
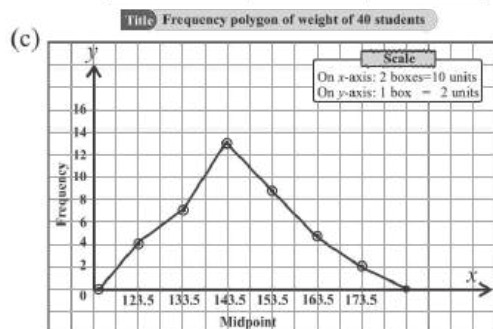
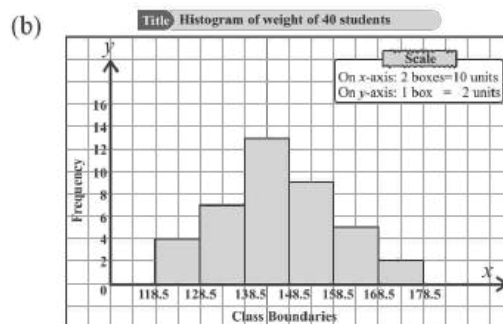
6. $\bar{X}$ = Rs. 437, $\tilde{X}$ = Rs. 437, $\hat{X}$ = Rs. 425, Rs. 435, Rs. 450 7. Σx = 3600
 8. Mean = 4.20, Median = 4, No mode 9. Mode > Median > Mean
 10. Median = 15, Mode = 15, Mean = 15.2, 160 > 156.5 > 154.33
 11. Median = 16.11, Mode = 17.25, Mean = 15.70
 12. 266 years, 11 months and 10 days, average age of 19 boys = 13 years, 3 months and 4 days approx.
 13. (i) $\bar{X}$ = 190 (ii) $\bar{X}$ = 710 (iii) $\bar{X}$ = 40 (iv) $\bar{X}$ = 123
 14. $\bar{X}_{\text{(Haris)}} = 70$, $\bar{X}_{\text{(Maham)}} = 58.6$, $\bar{X}_{\text{(Minal)}} = 40$, Haris will get awarded amount.
 15. $\bar{X}$ = 21.17 16. $\bar{X}$ = 54.13 17. $\bar{X}_w$ = Rs 120.74 18. $\bar{X}_w$ = Rs 20.25 (in thousands)
 19. Average budget = 6.6 (million) 20. $\bar{X}_w$ = 76.9 marks

REVIEW EXERCISE 12

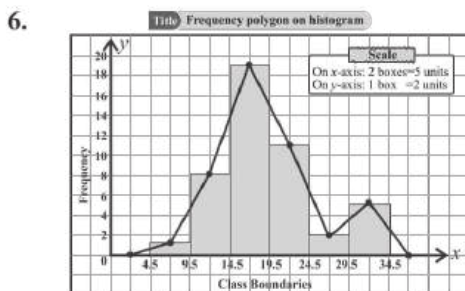
1. (i) b (ii) a (iii) d (iv) a (v) d (vi) c (vii) c (viii) d (ix) b (x) a

3. (a)

Class limits	Tally marks	f
119 – 128		4
129 – 138		7
139 – 148		13
149 – 158		9
159 – 168		5
169 – 178		2
Total		$\Sigma f = 40$



5. (i) 44 (ii) 19.5, 24.5, 29.5, 34.5, 39.5, 44.5 (iii) 22, 27, 32, 37, 42, 47 (iv) 5



7. Rs. 473.81
 8. The average of funds allocation in each sector is Rs. 10,000
 9. 80 marks 10. 108 kg
 11. Median = 6, Mode = 6
 12. Mean = 918.09, Median = 940.46, Mode = 958.33

EXERCISE 13.1

1. $\frac{2}{3}$ 2. (i) $\frac{11}{12}$ (ii) $\frac{11}{36}$ (iii) $\frac{1}{9}$ (iv) $\frac{1}{6}$ 3. (i) $\frac{4}{11}$ (ii) $\frac{7}{11}$ (iii) $\frac{1}{11}$ (iv) $\frac{2}{11}$ (v) $\frac{9}{11}$ (vi) $\frac{9}{11}$
 4. $P(\text{getting 3 or 4}) = \frac{1}{3}$, $P(\text{not getting 3 or 4}) = \frac{2}{3}$ 5. (i) $\frac{1}{30}$ (ii) $\frac{1}{5}$ (iii) $\frac{11}{30}$ (iv) $\frac{14}{15}$ (v) $\frac{13}{15}$
 6. 0.15 7. (i) $\frac{1}{4}$ (ii) $\frac{1}{6}$ (iii) $\frac{1}{6}$ (iv) $\frac{11}{12}$ (v) $\frac{5}{6}$ 8. (i) $\frac{1}{13}$ (ii) $\frac{11}{13}$ 9. (i) $\frac{1}{13}$ (ii) $\frac{3}{4}$

EXERCISE 13.2

1.	No. of death	f	$r.f.$
	0	60	$\frac{30}{147}$
	1	50	$\frac{25}{147}$
	2	87	$\frac{29}{98}$
	3	40	$\frac{20}{147}$
	4	32	$\frac{16}{147}$
	5	15	$\frac{5}{98}$
	6	10	$\frac{5}{147}$
	Total	$\Sigma f = 294$	

2.	No. of defective per sample	f	$r.f.$
	0	120	$\frac{4}{25}$
	1	140	$\frac{14}{75}$
	2	94	$\frac{47}{375}$
	3	85	$\frac{17}{150}$
	4	105	$\frac{21}{150}$
	5	50	$\frac{1}{15}$
	6	40	$\frac{4}{75}$
	7	66	$\frac{33}{150}$
8	50	$\frac{1}{15}$	
Total	$\Sigma f = 750$		

3.	X	f	$r.f.$
	0	10	$\frac{1}{10}$
	1	23	$\frac{23}{100}$
	2	15	$\frac{3}{20}$
	3	25	$\frac{1}{4}$
	4	18	$\frac{9}{50}$
	5	09	$\frac{9}{100}$
	Total	$\Sigma f = 100$	

4. (i) 37% (ii) 19% (iii) fresh juice (iv) biryani 5. $83.33 \approx 83$ 6. Rs. 60

7.	X	0	1	2	3	4	5	6
	$P(X)$	0.11	0.21	0.17	0.18	0.09	0.17	0.07
	Expected Frequency	22	42	34	36	18	34	14

8. 80 times

REVIEW EXERCISE 13

1. (i) c (ii) b (iii) c (iv) a (v) a (vi) c (vii) b (viii) c (ix) d (x) b 3. (i) $\frac{5}{23}$
 (ii) $\frac{10}{23}$ (iii) $\frac{8}{23}$ (iv) $\frac{13}{23}$ (v) $\frac{18}{23}$ 4. (i) $\frac{1}{8}$ (ii) $\frac{1}{2}$ (iii) $\frac{1}{2}$ (iv) $\frac{5}{8}$ 5. (i) $\frac{1}{13}$ (ii) $\frac{25}{26}$

6.	No. of tails	0	1	2	3	4	5	6	Total
	f	110	90	105	80	76	123	16	$\Sigma f = 600$
	Relative Frequency	$\frac{11}{60}$	$\frac{3}{20}$	$\frac{7}{40}$	$\frac{2}{15}$	$\frac{19}{150}$	$\frac{41}{200}$	$\frac{2}{75}$	

7. Relative frequency = $\frac{17}{25} = 0.68$

Expected frequency of non-defective items = 17

Glossary

Antilogarithm: An antilogarithm is the inverse operation of a logarithm.

Axiom: An axiom is a mathematical statement that we believe to be true without any evidence or requiring any proof.

Biconditional $p \leftrightarrow q$: The statement $p \rightarrow q \wedge q \rightarrow p$ is shortly written as $p \leftrightarrow q$ and is called the biconditional or equivalence.

Binary Relation: Any subset of $A \times B$ is called a binary relation, or simply a relation, from A to B .

Centroid: The point of concurrency of the medians of a triangle is called centroid of the triangle.

Characteristic: The characteristic is the integral part of the logarithm. It tells us how big or small the number is.

Circular Measure (Radian): It is defined as, “the angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle”.

Circumcenter: The point of concurrency of perpendicular bisector of the sides of a triangle is called circumcenter.

Common Logarithm: The common logarithm is the logarithm with a base of 10. It is written as $\log_{10}$ or simply as $\log$ (when no base is mentioned, it is usually assumed to be base 10).

Conditionals related with a given conditional: Let p and q be the statements and $p \rightarrow q$ be a given conditional, then

- (i) $q \rightarrow p$ is called the **converse** of $p \rightarrow q$;
- (ii) $\sim p \rightarrow \sim q$ is called the **inverse** of $p \rightarrow q$;
- (iii) $\sim q \rightarrow \sim p$ is called the **contrapositive** of $p \rightarrow q$.

Conjecture: A conjecture is a mathematical statement or hypothesis that is believed to be true based on observations but has not yet been proved.

Conjunction: The conjunction of two statements p and q is symbolically written as $p \wedge q$ (p and q). A conjunction is considered to be true only if both statements are true.

Deductive Proof: Deductive reasoning is a way of drawing conclusions from premises believed to be true. If the premises are true, then the conclusion must also be true.

Degree: A degree ($^{\circ}$) is a unit of measurement of angles. It represents $\left(\frac{1}{360}\right)^{\text{th}}$ of a full rotation around a point.

Disjunction: The disjunction of p and q is symbolically written as $p \vee q$ (p or q). The disjunction $p \vee q$ is considered to be true when at least one of the statements is true. It is false when both of them are false.

Domain: The set of the first elements of the ordered pairs forming a relation is called its domain.

Event: The set of results of an experiment is called an event

Expected Frequency: Expected frequency is a measure that estimate how often an event should be occurred depended on probability.

Experiment: The process which generates results e.g., tossing a coin, rolling a dice, etc. is called an experiment.

Favourable Outcome: An outcome which represents how many times we expect the things to be happened.

Feasible region: A region which is restricted to the first quadrant is referred to as a feasible region for the set of given constraints.

Feasible solution: Each point of the feasible region is called a feasible solution of the system of linear inequalities (or for the set of a given constraints).

Frequency Polygon: A frequency polygon is a closed geometrical figure used to display a frequency distribution graphically.

Implication or conditional: A compound statement of the form if p then q ($p \rightarrow q$) also written as p implies q is called a conditional or an implication. p is called the **antecedent** or **hypothesis** and q is called the **consequent** or the **conclusion**.

Incentre: The point of concurrency of the angle bisectors of a triangle is called incentre of the triangle.

Linear Equation: An equation of the form $ax + b = 0$ where ' a ' and ' b ' are constants, $a \neq 0$ and ' x ' is a variable, is called a linear equation in one variable.

Linear Functions: A linear function is a polynomial function of degree 1.

Loci: A locus (plural loci) is a set of points that follow a given rule. In geometry, loci are often used to define the positions of points relative to one another or to other geometric figures.

Logarithm of a Real Number: The logarithm of x to the base b is y , means that when b is raised to the power y , it equals x . The relationship between logarithmic form and exponential form is given as $\log_b(x) = y \Leftrightarrow b^y = x$ where $b > 0$, $x > 0$ and $b \neq 1$.

Logic: Logic is a systematic method of reasoning that enables one to interpret the meanings of statements, examine their truth, and deduce new information from existing.

Mantissa: The mantissa is the decimal part of the logarithm. It represents the "fractional" component and is always positive.

Measures of Location (Central Tendency): The measure that gives the centre of the data is called measure of central tendency.

Natural Logarithm: The natural logarithm is the logarithm with base e , where e is a mathematical constant approximately equal to 2.71828.

Negation: If p is any statement, its negation is denoted by $\sim p$, read 'not p '. It follows from this definition that if p is true, $\sim p$ is false, and if p is false, $\sim p$ is true.

Non-negative constraints: The variables used in the system of linear inequalities relating to the problems of everyday life are non-negative and are called non-negative constraints.

Non-Terminating and Recurring Decimal Numbers: The decimal numbers with repeating a pattern of digits after the decimal point are called non-terminating and recurring decimal numbers.

Objective function: A function which is to be maximized or minimized is called an objective function.

Optimal solution: The feasible solution which maximizes or minimizes the objective function is called the optimal solution.

Orthocentre: The point of concurrency of the altitudes of the triangle is called orthocentre of the triangle.

Outcomes: The results of an experiment are called outcomes e.g., the possible outcomes of tossing a coin are head or tail.

Point of concurrency: A point of concurrency is the single point where three or more lines, rays or line segments intersect or meet in a geometric figure.

Problem constraints: The system of linear inequalities involved in the problem concerned is called problem constraints.

Range: The set of the second elements of the ordered pairs forming a relation is called its range

Relative Frequency: Relative frequency is an estimated probability of an event occurring when an experiment is repeated a fixed number of times.

Sample Space: The set of all possible outcomes of an experiment is called sample space.

Scientific Notation: A number in scientific notation is written as: $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{Z}$. Here “a” is called the coefficient or base number.

Similar Solids: Two solids are said to be similar if they have same shape but possibly different sizes. Two solids are similar if lengths of the corresponding sides are proportional.

Similarity of Polygons: Similar figures have same shape but not necessarily of same size.

Slope or Gradient of a Line: The measure of steepness (ratio of rise to the run) is termed as slope or gradient of the inclined path.

Square Root of an Algebraic Expression: The square root of an algebraic expression refers to a value that, when multiplied by itself, gives the original expression.

Statement: A sentence or mathematical expression which may be true or false but not both is called a statement.

Terminating Decimal Numbers: A decimal number with a finite number of digits after the decimal point is called a terminating decimal number.

Tessellation: A tessellation is a pattern of shapes that fit together perfectly, without any gaps or overlaps, covering a plane.

Theorem: A theorem is a mathematical statement that has been proved true based on previously known facts.

Triangle Inequality Theorem: The sum of the measure of any two sides of a triangle is always greater than the measure of the third side.

Symbols / Notations

Symbols	Stands for
$=$	is equal to
$\neq$	is not equal to
$\in$	belongs to/element of
$\notin$	not belongs to/not element of
$\wedge$	logical and
$\vee$	logical or
$\cup$	union
$\cap$	intersection
$>$	is greater than
$<$	is less than
$\leq$	is less than or equal to
$\geq$	is greater than or equal to
$\nless$	is not greater than
$\nless$	is not less than
$ $	such that
$\subseteq$	subset
$\not\subseteq$	not a subset
$\subset$	proper subset
$\supseteq$	superset
$\not\supseteq$	not a superset
$\emptyset$ or $\{ \}$	empty set
$\therefore$	therefore/so
$\because$	since
$\approx$	is approximately equal to
$\sim$	is similar to
$\Rightarrow$	implies that
$\Leftrightarrow$	if and only if
$ x $	absolute value of x
$\sqrt{\quad}$	square root

Symbols	Stands for
$\forall$	for all
π	pi
e	euler constant
$^{\circ}\text{C}$	degree celsius
$^{\circ}\text{F}$	degree fahrenheit
$\log$	logarithm
$\ln$	natural logarithm
$\overline{AB}$	line segment AB
$m\overline{AB}$	measure of line segment AB
$\overrightarrow{AB}$	ray AB
$\overleftrightarrow{AB}$	line AB
$\angle ABC$	angle ABC
$m\angle ABC$	measure of angle ABC
$\triangle ABC$	triangle ABC
$ \overline{AB} $	length of $\overline{AB}$
$\widehat{AB}$	arc AB
$\parallel$	is parallel to
$\nparallel$	is not parallel to
$\perp$	is perpendicular to
$\rightarrow$	if . . . then or implies
θ	theta
ϕ	phi
α	alpha
$^{\circ}$	degree
$/$	telly mark
$\overline{X}$	arithmetic mean
$\overline{X}_w$	weighted mean
$\tilde{X}$	median
$\hat{X}$	mode

Logarithms

	0	1	2	3	4	5	6	7	8	9	Mean Difference								
											1	2	3	4	5	6	7	8	9
10	0000	0043	0086	0128	0170						4	9	13	17	21	26	30	34	38
						0212	0253	0294	0334	0374	4	8	12	16	20	24	28	32	37
11	0414	0453	0492	0531	0569						4	8	12	15	19	23	27	31	35
						0607	0645	0682	0719	0755	4	7	11	15	19	22	26	30	33
12	0792	0828	0864	0899	0934	0969					3	7	11	14	18	21	25	28	32
							1004	1038	1072	1106	3	7	10	14	17	20	24	27	31
13	1139	1173	1206	1239	1271						3	7	10	13	16	20	23	26	30
						1303	1335	1367	1399	1430	3	7	10	12	16	19	22	25	29
14	1461	1492	1523	1553							3	6	9	12	15	18	21	24	28
					1584	1614	1644	1673	1703	1732	3	6	9	12	15	17	20	23	26
15	1761	1790	1818	1847	1875	1903					3	6	9	11	14	17	20	23	26
							1931	1959	1987	2014	3	5	8	11	14	16	19	22	25
16	2041	2068	2095	2122	2148						3	5	8	11	14	16	19	22	24
					2175	2201	2227	2253	2279		3	5	8	10	13	15	18	21	23
17	2304	2330	2355	2380	2405	2430					3	5	8	10	13	15	18	20	23
							2455	2480	2504	2529	2	5	7	10	12	15	17	19	22
18	2553	2577	2601	2625	2648						2	5	7	9	12	14	16	19	21
						2672	2695	2718	2742	2765	2	5	7	9	11	14	16	18	21
19	2788	2810	2833	2856	2878						2	4	7	9	11	13	16	18	20
						2900	2923	2945	2967	2989	2	4	6	8	11	13	15	17	19
20	3010	3032	3054	3075	3096	3118	3139	3160	3181	3201	2	4	6	8	11	13	15	17	19
21	3222	3243	3263	3284	3304	3324	3345	3365	3385	3404	2	4	6	8	10	12	14	16	18
22	3424	3444	3464	3483	3502	3522	3541	3560	3579	3598	2	4	6	8	10	12	14	15	17
23	3617	3636	3655	3674	3692	3711	3729	3747	3766	3784	2	4	6	7	9	11	13	15	17
24	3802	3820	3838	3856	3874	3892	3909	3927	3945	3962	2	4	5	7	9	11	12	14	16
25	3979	3997	4014	4031	4048	4065	4082	4099	4116	4133	2	3	5	7	9	10	12	14	15
26	4150	4166	4183	4200	4216	4232	4249	4265	4281	4298	2	3	5	7	8	10	11	13	15
27	4314	4330	4346	4362	4378	4393	4409	4425	4440	4456	2	3	5	6	8	9	11	13	14
28	4472	4487	4502	4518	4533	4548	4564	4579	4594	4609	2	3	5	6	8	9	11	12	14
29	4624	4639	4654	4669	4683	4698	4713	4728	4742	4757	1	3	4	6	7	9	10	12	13
30	4771	4786	4800	4814	4829	4843	4857	4871	4886	4900	1	3	4	6	7	9	10	11	13
31	4914	4928	4942	4955	4969	4983	4997	5011	5024	5038	1	3	4	6	7	8	10	11	12
32	5051	5065	5079	5092	5105	5119	5132	5145	5159	5172	1	3	4	5	7	8	9	11	12
33	5185	5198	5211	5224	5237	5250	5263	5276	5289	5302	1	3	4	5	6	8	9	10	12
34	5315	5328	5340	5353	5366	5378	5391	5403	5416	5428	1	3	4	5	6	8	9	10	11
35	5441	5453	5465	5478	5490	5502	5514	5527	5539	5551	1	2	4	5	6	7	9	10	11
36	5563	5575	5587	5599	5611	5623	5635	5647	5658	5670	1	2	4	5	6	7	8	10	11
37	5682	5694	5705	5717	5729	5740	5752	5763	5775	5786	1	2	3	5	6	7	8	9	10
38	5798	5809	5821	5832	5843	5855	5866	5877	5888	5899	1	2	3	5	6	7	8	9	10
39	5911	5922	5933	5944	5955	5966	5977	5988	5999	6010	1	2	3	4	5	7	8	9	10
40	6021	6031	6042	6053	6064	6075	6085	6096	6107	6117	1	2	3	4	5	6	8	9	10
41	6128	6138	6149	6160	6170	6180	6191	6201	6212	6222	1	2	3	4	5	6	7	8	9
42	6232	6243	6253	6263	6274	6284	6294	6304	6314	6325	1	2	3	4	5	6	7	8	9
43	6335	6345	6355	6365	6375	6385	6395	6405	6415	6425	1	2	3	4	5	6	7	8	9
44	6435	6444	6454	6464	6474	6484	6493	6503	6513	6522	1	2	3	4	5	6	7	8	9
45	6532	6542	6551	6561	6571	6580	6590	6599	6609	6618	1	2	3	4	5	6	7	8	9
46	6628	6637	6646	6656	6665	6675	6684	6693	6702	6712	1	2	3	4	5	6	7	7	8
47	6721	6730	6739	6749	6758	6767	6776	6785	6794	6803	1	2	3	4	5	5	6	7	8
48	6812	6821	6830	6839	6848	6857	6866	6875	6884	6893	1	2	3	4	4	5	6	7	8
49	6902	6911	6920	6928	6937	6946	6955	6964	6972	6981	1	2	3	4	4	5	6	7	8

	0	1	2	3	4	5	6	7	8	9	Mean Difference								
											1	2	3	4	5	6	7	8	9
50	6990	6998	7007	7016	7024	7033	7042	7050	7059	7067	1	2	3	3	4	5	6	7	8
51	7076	7084	7093	7101	7110	7118	7126	7135	7143	7152	1	2	3	3	4	5	6	7	8
52	7160	7168	7177	7185	7193	7202	7210	7218	7226	7235	1	2	2	3	4	5	6	7	7
53	7243	7251	7259	7267	7275	7284	7292	7300	7308	7316	1	2	2	3	4	5	6	6	7
54	7324	7332	7340	7348	7356	7364	7372	7380	7388	7396	1	2	2	3	4	5	6	6	7
55	7404	7412	7419	7427	7435	7443	7451	7459	7466	7474	1	2	2	3	4	5	5	6	7
56	7482	7490	7497	7505	7513	7520	7528	7536	7543	7551	1	2	2	3	4	5	5	6	7
57	7559	7566	7574	7582	7589	7597	7604	7612	7619	7627	1	2	2	3	4	5	5	6	7
58	7634	7642	7649	7657	7664	7672	7679	7686	7694	7701	1	1	2	3	4	4	5	6	7
59	7709	7716	7723	7731	7738	7745	7752	7760	7767	7774	1	1	2	3	4	4	5	6	7
60	7782	7789	7796	7803	7810	7818	7825	7832	7839	7846	1	1	2	3	4	4	5	6	6
61	7853	7860	7868	7875	7882	7889	7896	7903	7910	7917	1	1	2	3	4	4	5	6	6
62	7924	7931	7938	7945	7952	7959	7966	7973	7980	7987	1	1	2	3	3	4	5	6	6
63	7993	8000	8007	8014	8021	8028	8035	8041	8048	8055	1	1	2	3	3	4	5	5	6
64	8062	8069	8075	8082	8089	8096	8102	8109	8116	8122	1	1	2	3	3	4	5	5	6
65	8129	8136	8142	8149	8156	8162	8169	8176	8182	8189	1	1	2	3	3	4	5	5	6
66	8195	8202	8209	8215	8222	8228	8235	8241	8248	8254	1	1	2	3	3	4	5	5	6
67	8261	8267	8274	8280	8287	8293	8299	8306	8312	8319	1	1	2	3	3	4	5	5	6
68	8325	8331	8338	8344	8351	8357	8363	8370	8376	8382	1	1	2	3	3	4	4	5	6
69	8388	8395	8401	8407	8414	8420	8426	8432	8439	8445	1	1	2	2	3	4	4	5	6
70	8451	8457	8463	8470	8476	8482	8488	8494	8500	8506	1	1	2	2	3	4	4	5	6
71	8513	8519	8525	8531	8537	8543	8549	8555	8561	8567	1	1	2	2	3	4	4	5	5
72	8573	8579	8585	8591	8597	8603	8609	8615	8621	8627	1	1	2	2	3	4	4	5	5
73	8633	8639	8645	8651	8657	8663	8669	8675	8681	8686	1	1	2	2	3	4	4	5	5
74	8692	8698	8704	8710	8716	8722	8727	8733	8739	8745	1	1	2	2	3	4	4	5	5
75	8751	8756	8762	8768	8774	8779	8785	8791	8797	8802	1	1	2	2	3	3	4	5	5
76	8808	8814	8820	8825	8831	8837	8842	8848	8854	8859	1	1	2	2	3	3	4	5	5
77	8865	8871	8876	8882	8887	8893	8899	8904	8910	8915	1	1	2	2	3	3	4	4	5
78	8921	8927	8932	8938	8943	8949	8954	8960	8965	8971	1	1	2	2	3	3	4	4	5
79	8976	8982	8987	8993	8998	9004	9009	9015	9020	9025	1	1	2	2	3	3	4	4	5
80	9031	9036	9042	9047	9053	9058	9063	9069	9074	9079	1	1	2	2	3	3	4	4	5
81	9085	9090	9096	9101	9106	9112	9117	9122	9128	9133	1	1	2	2	3	3	4	4	5
82	9138	9143	9149	9154	9159	9165	9170	9175	9180	9186	1	1	2	2	3	3	4	4	5
83	9191	9196	9201	9206	9212	9217	9222	9227	9232	9238	1	1	2	2	3	3	4	4	5
84	9243	9248	9253	9258	9263	9269	9274	9279	9284	9289	1	1	2	2	3	3	4	4	5
85	9294	9299	9304	9309	9315	9320	9325	9330	9335	9340	1	1	2	2	3	3	4	4	5
86	9345	9350	9355	9360	9365	9370	9375	9380	9385	9390	1	1	2	2	3	3	4	4	5
87	9395	9400	9405	9410	9415	9420	9425	9430	9435	9440	0	1	1	2	2	3	3	4	4
88	9445	9450	9455	9460	9465	9469	9474	9479	9484	9489	0	1	1	2	2	3	3	4	4
89	9494	9499	9504	9509	9513	9518	9523	9528	9533	9538	0	1	1	2	2	3	3	4	4
90	9542	9547	9552	9557	9562	9566	9571	9576	9581	9586	0	1	1	2	2	3	3	4	4
91	9590	9595	9600	9605	9609	9614	9619	9624	9628	9633	0	1	1	2	2	3	3	4	4
92	9638	9643	9647	9652	9657	9661	9666	9671	9675	9680	0	1	1	2	2	3	3	4	4
93	9685	9689	9694	9699	9703	9708	9713	9717	9722	9727	0	1	1	2	2	3	3	4	4
94	9731	9736	9741	9745	9750	9754	9759	9763	9768	9773	0	1	1	2	2	3	3	4	4
95	9777	9782	9786	9791	9795	9800	9805	9809	9814	9818	0	1	1	2	2	3	3	4	4
96	9823	9827	9832	9836	9841	9845	9850	9854	9859	9863	0	1	1	2	2	3	3	4	4
97	9868	9872	9877	9881	9886	9890	9894	9899	9903	9908	0	1	1	2	2	3	3	4	4
98	9912	9917	9921	9926	9930	9934	9939	9943	9948	9952	0	1	1	2	2	3	3	4	4
99	9956	9961	9965	9969	9974	9978	9983	9987	9991	9996	0	1	1	2	2	3	3	3	4

Antilogarithms

											Mean Difference								
	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
.00	1000	1002	1005	1007	1009	1012	1014	1016	1019	1021	0	0	1	1	1	1	2	2	2
.01	1023	1026	1028	1030	1033	1035	1038	1040	1042	1045	0	0	1	1	1	1	2	2	2
.02	1047	1050	1052	1054	1057	1059	1062	1064	1067	1069	0	0	1	1	1	1	2	2	2
.03	1072	1074	1076	1079	1081	1084	1086	1089	1091	1094	0	0	1	1	1	1	2	2	2
.04	1096	1099	1102	1104	1107	1109	1112	1114	1117	1119	0	1	1	1	1	2	2	2	2
.05	1122	1125	1127	1130	1132	1135	1138	1140	1143	1146	0	1	1	1	1	2	2	2	2
.06	1148	1151	1153	1156	1159	1161	1164	1167	1169	1172	0	1	1	1	1	2	2	2	2
.07	1175	1178	1180	1183	1186	1189	1191	1194	1197	1199	0	1	1	1	1	2	2	2	2
.08	1202	1205	1208	1211	1213	1216	1219	1222	1225	1227	0	1	1	1	1	2	2	2	3
.09	1230	1233	1236	1239	1242	1245	1247	1250	1253	1256	0	1	1	1	1	2	2	2	3
.10	1259	1262	1265	1268	1271	1274	1276	1279	1282	1285	0	1	1	1	1	2	2	2	3
.11	1288	1291	1294	1297	1300	1303	1306	1309	1312	1315	0	1	1	1	2	2	2	2	3
.12	1318	1321	1324	1327	1330	1334	1337	1340	1343	1346	0	1	1	1	2	2	2	2	3
.13	1349	1352	1355	1358	1361	1365	1368	1371	1374	1377	0	1	1	1	2	2	2	3	3
.14	1380	1384	1387	1390	1393	1396	1400	1403	1406	1409	0	1	1	1	2	2	2	3	3
.15	1413	1416	1419	1422	1426	1429	1432	1435	1439	1442	0	1	1	1	2	2	2	3	3
.16	1445	1449	1452	1455	1459	1462	1466	1469	1472	1476	0	1	1	1	2	2	2	3	3
.17	1479	1483	1486	1489	1493	1496	1500	1503	1507	1510	0	1	1	1	2	2	2	3	3
.18	1514	1517	1521	1524	1528	1531	1535	1538	1542	1545	0	1	1	1	2	2	2	3	3
.19	1549	1552	1556	1560	1563	1567	1570	1574	1578	1581	0	1	1	1	2	2	3	3	3
.20	1585	1589	1592	1596	1600	1603	1607	1611	1614	1618	0	1	1	1	2	2	3	3	3
.21	1622	1626	1629	1633	1637	1641	1644	1648	1652	1656	0	1	1	1	2	2	3	3	3
.22	1660	1663	1667	1671	1675	1679	1683	1687	1690	1694	0	1	1	1	2	2	3	3	3
.23	1698	1702	1706	1710	1714	1718	1722	1726	1730	1734	0	1	1	1	2	2	3	3	4
.24	1738	1742	1746	1750	1754	1758	1762	1766	1770	1774	0	1	1	1	2	2	3	3	4
.25	1778	1782	1786	1791	1795	1799	1803	1807	1811	1816	0	1	1	1	2	2	3	3	4
.26	1820	1824	1828	1832	1837	1841	1845	1849	1854	1858	0	1	1	1	2	2	3	3	4
.27	1862	1866	1871	1875	1879	1884	1888	1892	1897	1901	0	1	1	1	2	2	3	3	4
.28	1905	1910	1914	1919	1923	1928	1932	1936	1941	1945	0	1	1	1	2	2	3	3	4
.29	1950	1954	1959	1963	1968	1972	1977	1982	1986	1991	0	1	1	1	2	2	3	3	4
.30	1995	2000	2004	2009	2014	2018	2023	2028	2032	2037	0	1	1	1	2	2	3	3	4
.31	2042	2046	2051	2056	2061	2065	2070	2075	2080	2084	0	1	1	1	2	2	3	3	4
.32	2089	2094	2099	2104	2109	2113	2118	2123	2128	2133	0	1	1	1	2	2	3	3	4
.33	2138	2143	2148	2153	2158	2163	2168	2173	2178	2183	0	1	1	1	2	2	3	3	4
.34	2188	2193	2198	2203	2208	2213	2218	2223	2228	2234	1	1	2	2	3	3	4	4	5
.35	2239	2244	2249	2254	2259	2265	2270	2275	2280	2286	1	1	2	2	3	3	4	4	5
.36	2291	2296	2301	2307	2312	2317	2323	2328	2333	2339	1	1	2	2	3	3	4	4	5
.37	2344	2350	2355	2360	2366	2371	2377	2382	2388	2393	1	1	2	2	3	3	4	4	5
.38	2399	2404	2410	2415	2421	2427	2432	2438	2443	2449	1	1	2	2	3	3	4	4	5
.39	2455	2460	2466	2472	2477	2483	2489	2495	2500	2506	1	1	2	2	3	3	4	5	5
.40	2512	2518	2523	2529	2535	2541	2547	2553	2559	2564	1	1	2	2	3	4	4	5	5
.41	2570	2576	2582	2588	2594	2600	2606	2612	2618	2624	1	1	2	2	3	4	4	5	5
.42	2630	2636	2642	2649	2655	2661	2667	2673	2679	2685	1	1	2	2	3	4	4	5	6
.43	2692	2698	2704	2710	2716	2723	2729	2735	2742	2748	1	1	2	3	3	4	4	5	6
.44	2754	2761	2767	2773	2780	2786	2793	2799	2805	2812	1	1	2	3	3	4	4	5	6
.45	2818	2825	2831	2838	2844	2851	2858	2864	2871	2877	1	1	2	3	3	4	5	5	6
.46	2884	2891	2897	2904	2911	2917	2924	2931	2938	2944	1	1	2	3	3	4	5	5	6
.47	2951	2958	2965	2972	2979	2985	2992	2999	3006	3013	1	1	2	3	3	4	5	5	6
.48	3020	3027	3034	3041	3048	3055	3062	3069	3076	3083	1	1	2	3	4	4	5	6	6
.49	3090	3097	3105	3112	3119	3126	3133	3141	3148	3155	1	1	2	3	4	4	5	6	6

	0	1	2	3	4	5	6	7	8	9	Mean Difference								
											1	2	3	4	5	6	7	8	9
.50	3162	3170	3177	3184	3192	3199	3206	3214	3221	3228	1	1	2	3	4	4	5	6	7
.51	3236	3243	3251	3258	3266	3273	3281	3289	3296	3304	1	2	2	3	4	5	5	6	7
.52	3311	3319	3327	3334	3342	3350	3357	3365	3373	3381	1	2	2	3	4	5	5	6	7
.53	3388	3396	3404	3412	3420	3428	3436	3443	3451	3459	1	2	2	3	4	5	6	6	7
.54	3467	3475	3483	3491	3499	3508	3516	3524	3532	3540	1	2	2	3	4	5	6	6	7
.55	3548	3556	3565	3573	3581	3589	3597	3606	3614	3622	1	2	2	3	4	5	6	7	7
.56	3631	3639	3648	3656	3664	3673	3681	3690	3698	3707	1	2	3	3	4	5	6	7	8
.57	3715	3724	3733	3741	3750	3758	3767	3776	3784	3793	1	2	3	3	4	5	6	7	8
.58	3802	3811	3819	3828	3837	3846	3855	3864	3873	3882	1	2	3	4	4	5	6	7	8
.59	3890	3899	3908	3917	3926	3936	3945	3954	3963	3972	1	2	3	4	5	5	6	7	8
.60	3981	3990	3999	4009	4018	4027	4036	4046	4055	4064	1	2	3	4	5	6	6	7	8
.61	4074	4083	4093	4102	4111	4121	4130	4140	4150	4159	1	2	3	4	5	6	7	8	9
.62	4169	4178	4188	4198	4207	4217	4227	4236	4246	4256	1	2	3	4	5	6	7	8	9
.63	4266	4276	4285	4295	4305	4315	4325	4335	4345	4355	1	2	3	4	5	6	7	8	9
.64	4365	4375	4385	4395	4406	4416	4426	4436	4446	4457	1	2	3	4	5	6	7	8	9
.65	4467	4477	4487	4498	4508	4519	4529	4539	4550	4560	1	2	3	4	5	6	7	8	9
.66	4571	4581	4592	4603	4613	4624	4634	4645	4656	4667	1	2	3	4	5	6	7	9	10
.67	4677	4688	4699	4710	4721	4732	4742	4753	4764	4775	1	2	3	4	5	7	8	9	10
.68	4786	4797	4808	4819	4831	4842	4853	4864	4875	4887	1	2	3	4	6	7	8	9	10
.69	4898	4909	4920	4932	4943	4955	4966	4977	4989	5000	1	2	3	5	6	7	8	9	10
.70	5012	5023	5035	5047	5058	5070	5082	5093	5105	5117	1	2	4	5	6	7	8	9	11
.71	5129	5140	5152	5164	5176	5188	5200	5212	5224	5236	1	2	4	5	6	7	8	10	11
.72	5248	5260	5272	5284	5297	5309	5321	5333	5346	5358	1	2	4	5	6	7	9	10	11
.73	5370	5383	5395	5408	5420	5433	5445	5458	5470	5483	1	3	4	5	6	8	9	10	11
.74	5495	5508	5521	5534	5546	5559	5572	5585	5598	5610	1	3	4	5	6	8	9	10	12
.75	5623	5636	5649	5662	5675	5689	5702	5715	5728	5741	1	3	4	5	7	8	9	10	12
.76	5754	5768	5781	5794	5808	5821	5834	5848	5861	5875	1	3	4	5	7	8	9	11	12
.77	5888	5902	5916	5929	5943	5957	5970	5984	5998	6012	1	3	4	5	7	8	10	11	12
.78	6026	6039	6053	6067	6081	6095	6109	6124	6138	6152	1	3	4	6	7	8	10	11	13
.79	6166	6180	6194	6209	6223	6237	6252	6266	6281	6295	1	3	4	6	7	9	10	11	13
.80	6310	6324	6339	6353	6368	6383	6397	6412	6427	6442	1	3	4	6	7	9	10	12	13
.81	6457	6471	6486	6501	6516	6531	6546	6561	6577	6592	2	3	5	6	8	9	11	12	14
.82	6607	6622	6637	6653	6668	6683	6699	6714	6730	6745	2	3	5	6	8	9	11	12	14
.83	6761	6776	6792	6808	6823	6839	6855	6871	6887	6902	2	3	5	6	8	9	11	13	14
.84	6918	6934	6950	6966	6982	6998	7015	7031	7047	7063	2	3	5	6	8	10	11	13	15
.85	7079	7096	7112	7129	7145	7161	7178	7194	7211	7228	2	3	5	7	8	10	12	13	15
.86	7244	7261	7278	7295	7311	7328	7345	7362	7379	7396	2	3	5	7	8	10	12	13	15
.87	7413	7430	7447	7464	7482	7499	7516	7534	7551	7568	2	3	5	7	9	10	12	14	16
.88	7586	7603	7621	7638	7656	7674	7691	7709	7727	7745	2	4	5	7	9	11	12	14	16
.89	7762	7780	7798	7816	7834	7852	7870	7889	7907	7925	2	4	5	7	9	11	13	14	16
.90	7943	7962	7980	7998	8017	8035	8054	8072	8091	8110	2	4	6	7	9	11	13	15	17
.91	8128	8147	8166	8185	8204	8222	8241	8260	8279	8299	2	4	6	8	9	11	13	15	17
.92	8318	8337	8356	8375	8395	8414	8433	8453	8472	8492	2	4	6	8	10	12	14	15	17
.93	8511	8531	8551	8570	8590	8610	8630	8650	8670	8690	2	4	6	8	10	12	14	16	18
.94	8710	8730	8750	8770	8790	8810	8831	8851	8872	8892	2	4	6	8	10	12	14	16	18
.95	8913	8933	8954	8974	8995	9016	9036	9057	9078	9099	2	4	6	8	10	12	15	17	19
.96	9120	9141	9162	9183	9204	9226	9247	9268	9290	9311	2	4	6	8	11	13	15	17	19
.97	9333	9354	9376	9397	9419	9441	9462	9484	9506	9528	2	4	7	9	11	13	15	17	20
.98	9550	9572	9594	9616	9638	9661	9683	9705	9727	9750	2	4	7	9	11	13	16	18	20
.99	9772	9795	9817	9840	9863	9886	9908	9931	9954	9977	2	5	7	9	11	14	16	18	20